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A State-Space Representation of Coupled Linear Multivariate PDEs and Stability Analysis using SDP

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Physical processes evolving in both time and space are often modeled using Partial Differential Equations (PDEs). Recently, it has been shown how stability analysis and control of coupled PDEs in a single spatial variable can be more conveniently performed using an equivalent Partial Integral Equation (PIE) representation. The construction of this PIE representation is based on an analytic expression for the inverse of the univariate spatial differential operator, ∂_s^d, on the domain defined by boundary conditions. In this paper, we show how this univariate representation may be extended inductively to multiple spatial variables by representing the domain as the intersection of lifted univariate domains. However, this extension is complicated by the need to ensure consistency of the domains. Specifically, for PDEs defined on a hyper-rectangle, invertibility of the multivariate differential operator, $D^\alpha = \partial_{s_1}^{\alpha_1} \cdots \partial_{s_N}^{\alpha_N}$, requires the boundary values along each hyper-surface to be consistent at the corners. Furthermore, construction of the inverse of D^α then requires a multivariate algebra of Volterra-type operators. To resolve these challenges, we show that if each univariate domain is well-posed, then there exists a readily verified consistency condition which is necessary and sufficient for existence of an inverse to the multivariate differential operator, D^α, on the PDE domain. Furthermore, we show that this inverse is an element of a $*$-algebra of Partial Integral (PI) operators defined by polynomial semi-separable kernels. Based on this operator algebra, we show that the evolution of any suitably well-posed linear multivariate PDE may be described by a PIE of the form $\partial_t \mathcal{T}\mathbf{v} = \mathcal{A}\mathbf{v}$, where $\mathcal{T}, \mathcal{A}$ are elements of the PI algebra. A convex computational test for PDE stability is then proposed using a positive matrix parameterization of positive PI operators, and software (PIETOOLS) is provided which automates the process of representation and stability analysis of such PDEs. This software is used to analyze stability of 2D heat, wave, and plate equations, obtaining accurate bounds on the rate of decay.

Keywords: Distributed Parameter Systems; Stability Analysis; Semidefinite Programming.

1. Introduction

Partial Differential Equations (PDEs) are frequently used to model physical phenomena such as fluid flow (e.g. Navier-Stokes), vibrations in flexible structures (e.g. Kirchoff-Love plates), and population growth (e.g. Fisher's equation). Such PDE models provide a simplified representation of the phenomena which can then be used for simulation, stability analysis, or controller synthesis. However, the utility of such models is predicated on the availability of methods for simulation, analysis, and control of PDEs. While there exist well-defined state-space and transfer function based algorithms for simulation, stability analysis and control of Ordinary Differential Equations (ODEs), applicable for any suitably well-posed problem, these methods do not readily generalize to PDEs. In particular, development of similar universal tools for simulation, analysis and control of PDEs is complicated by the fact that the state of the PDE is infinite-dimensional, and is further required to satisfy a set of boundary conditions. Moreover, PDEs can involve an arbitrary number of spatial variables, and tools developed for univariate PDEs may not readily extend to a multivariate setting. As such, most work on analysis and control of

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(multivariate) PDEs is ad hoc, tailoring results to a limited class of PDEs and boundary conditions. An overview of such methods is as follows.

1.1. *Methods of Analysis and Control of Multivariate PDEs*

Semi-group theory is a mathematical framework for analysis of PDEs, with associated notions of solution and well-posedness being defined in terms of equivalence of a strongly continuous semigroup [6, 8]. In this framework, operator inequality conditions may be proposed for analysis and control purposes [25, 26, 33]. However, because the semigroup framework does not provide explicit methods for finding a solution or solving operator inequalities, PDE problems expressed in this framework are typically discretized through projection of the state onto a finite set of basis functions. Discretization of the dynamics is called "early-lumping"[1] and discretization of operator inequalities is termed "late-lumping"[2]. However, in each case, care must be taken in selection of a finite-dimensional basis to ensure compatibility with boundary conditions. Furthermore, inferring properties of the PDE from properties of the discretization requires extensive, ad hoc, a posteriori analysis.

Another PDE framework is the backstepping methodology [49], often used to design stabilizing boundary feedback controllers. The discretization step in this case lies in numerically solving certain Volterra type integral equations for reconstruction of the backstepping state transformation. The resulting kernel may then be used to define an invertible state transformation and stabilizing controller gain [24, 44]. Defining the backstepping transformation for PDEs with multiple spatial variables requires an extension of the Volterra framework, and integration of function-valued boundary conditions. Consequently, such multivariate methods typically require use of spatial invariance or symmetry of solutions to reduce the dimensionality of the problem [47, 48, 51].

Finally, spectral discretization by wave number can be used in combination with transfer-function approaches to perform energy gain analysis. This method has been applied to linearized fluid flow in multiple spatial variables [19, 20, 27], yielding decomposition into streamwise and spanwise components. This yields an operator-valued transfer function which is then discretized.

Methods which avoid discretization entirely are often based on construction of a suitable metric – i.e. a Lyapunov functional. The challenge in this case is parameterization of quadratic Lyapunov functionals and verifying negativity of the derivative along solutions to the PDE (often through the use of Linear Matrix Inequalities (LMIs)). In most cases, the Lyapunov functional is chosen to be the L_2 norm of the state, so that negativity of the derivative of this function may be established using inequalities such as Poincare, Wirtinger, Jensen, etc. – together with integration by parts. However, restrictions on the structure of the Lyapunov functional and the need for ad hoc analysis limits the scope of such methods and implies conservatism in the resulting stability conditions. This approach has been applied to design observers for the 2D Navier-Stokes equations [22, 53, 55] and controllers for the Kuramoto-Sivashinsky equation [21]. Works which extend these approaches to Lyapunov functions defined by weighted L_2 norms using Sum-of-Squares [38] for parabolic PDEs with multiple spatial variables include [34, 36] (linear PDEs) and [2, 46] (nonlinear PDEs). Applications to fluid flow include [3, 52]. Moment-based dual versions of these results can be found in [17, 23, 32]. In all cases, however, the need for ad hoc analysis and the use of restricted classes of Lyapunov functionals limits accuracy of the results [10].

While this brief survey on methods for analysis and control of multivariate PDEs is clearly incomplete, in each case we find that the need for ad hoc analysis (necessitated by the structure imposed on the solution by the boundary conditions) limits the accuracy, scope and reliability of the method. To

[1] Examples of early-lumping can be found in [13, 28, 29] as well as [7, 12, 15].
[2] Examples of late-lumping can be found in [35, 39].

address this problem, we will examine the question of whether there exists some more fundamental state space representation of PDEs which allows for the design of universal tools for simulation, analysis and control of PDEs, where such tools would be applicable to any PDE expressed in terms of this representation.

1.2. *The PIE Representation for Stability Analysis and Control of 1D PDEs*

The need for universal simulation, analysis and control methods for PDEs has motivated the development of a state-space framework for representation of these systems. Specifically, for PDEs in a single spatial variable, we have the Partial Integral Equation (PIE) representation [42]. This PIE representation defines the fundamental (or PIE) state as the highest-order spatial derivative (∂_s^d) of the PDE state. This fundamental state lies in the Hilbert space L_2^n and does not admit boundary conditions or Sobolev regularity constraints. Because the PIE state is unconstrained, establishing equivalence of PDE and PIE requires invertibility of $\partial_s^d : \mathcal{D} \to L_2^n$ on the domain of the PDE, $\mathcal{D}$, as specified by the boundary conditions and regularity constraints.

For univariate coupled PDEs, necessary and sufficient conditions for existence of an inverse to $\partial_s^d : \mathcal{D} \to L_2^n$ have been proposed (e.g. Thm. 3.1 in Sec. XIV.3 of [14]) and result in an analytic construction of this inverse as defined by a Volterra type integral operator. Specifically, it has been shown that this integral operator is of the form

$$\big(\mathcal{T}\mathbf{v}\big)(s) = \int_a^s \boldsymbol{T}_1(s,\theta)\mathbf{v}(\theta)\,d\theta + \int_s^b \boldsymbol{T}_2(s,\theta)\mathbf{v}(\theta)\,d\theta,$$

where $\boldsymbol{T}_1, \boldsymbol{T}_2$ are polynomials and can be computed analytically using the matrices which parameterize the boundary conditions associated with $\mathcal{D}$. Linear integral operators of this form (denoted Partial Integral (PI) operators) are bounded and form a $*$-algebra, where $*$ denotes the L_2-adjoint. This algebraic structure is significant in that it implies that the evolution of the fundamental state may be represented as a PIE of the form $\partial_t \mathcal{T}\mathbf{v} = \mathcal{A}\mathbf{v}$ where $\mathcal{A}$ is likewise a PI operator albeit with inclusion of multiplier operator (inclusion of polynomial multipliers preserves the $*$-algebraic structure).

Once a PIE representation has been established, there exists an array of algorithmic tools for simulation, analysis, control and estimation, most of which can be found in the PIETOOLS software package and associated documentation [43] (see also literature on PIE methods in a single spatial variable).

When PDEs involve multiple spatial variables, establishing an equivalent PIE representation raises new challenges. In this case, the fundamental state is similarly defined as the highest-order spatial derivative of the PDE state, e.g. $\mathbf{v} = \partial_x^d \partial_y^d \mathbf{u}$, and the goal (for a given PDE domain $\mathcal{D}$) is to construct an inverse of $\partial_x^d \partial_y^d : \mathcal{D} \to L_2^n$. The challenge lies in the boundary conditions which restrict the PDE state $\mathbf{u} \in \mathcal{D}$. Specifically, if the domain is an N-dimensional hyper-rectangle, the boundary conditions constrain the values of the PDE state on the $2N$ surfaces of the hyper-rectangle, each of which is itself an $(N-1)$-dimensional hyper-rectangle. Furthermore, unlike in the 1D case, the boundary values of the multivariate PDE state are coupled in ways which create new questions of well-posedness. For example, in the simplest case, the surfaces of the hyper-rectangle intersect at the corners and hence boundary conditions must be consistent at these corner values (e.g. the boundary conditions must not imply $\mathbf{u}(x,0) \neq \mathbf{u}(0,y)$ at $(x,y) = (0,0)$). This question of consistency was explored in [18], wherein a consistent basis for boundary conditions was proposed and it was shown that projection of given boundary conditions onto this basis was able to resolve the question of consistency. The disadvantage

of this approach, however, is the need for projection of boundary conditions onto the given basis and the need for inversion of multivariate integral operators in order to obtain the PIE representation.

In contrast to [18], this paper considers a restricted set of boundary conditions defined as the intersection of lifted 1D domains – e.g. $\mathcal{D} := \mathcal{D}_x \cap \mathcal{D}_y$ where $\mathcal{D}_y := \{\mathbf{u} \mid A_y\mathbf{u}(x,0) + B_y\mathbf{u}(x,1) = 0\}$ and $\mathcal{D}_x := \{\mathbf{u} \mid A_x\mathbf{u}(0,y) + B_x\mathbf{u}(1,y) = 0\}$. This approach includes many commonly used boundary conditions, and has the additional advantages that the parameters of the boundary conditions may be expressed in a more intuitive manner; and it avoids the need for inversion of multivariate operators. Furthermore, this approach greatly simplifies the PIE construction in that it allows one to define the map from PIE state to PDE state[3] inductively using a composition of 1D mappings – i.e. $\mathcal{T}_x\partial_x^d = I$ and $\mathcal{T}_y\partial_y^d = I$ implies $\mathcal{T}_y\mathcal{T}_x\partial_x^d\partial_y^d = I$. This reduces the need for complicated multivariate notation. The difficulty, however, lies in establishing that also $\mathrm{Im}(\mathcal{T}_y\mathcal{T}_x) \subseteq \mathcal{D}_x \cap \mathcal{D}_y$. In particular, although $\mathrm{Im}(\mathcal{T}_y\mathcal{T}_x) \subseteq \mathcal{D}_y$ clearly follows from $\mathrm{Im}(\mathcal{T}_y) \subseteq \mathcal{D}_y$, we show in Section 4 that $\mathrm{Im}(\mathcal{T}_y\mathcal{T}_x) \subseteq \mathcal{D}_x$ only if $\mathcal{T}_x$ and $\mathcal{T}_y$ commute[4]. The goal of Section 4, then, is to derive consistency conditions on $\mathcal{D}_x$ and $\mathcal{D}_y$ which are necessary and sufficient for the existence of an inverse of $\partial_x^d\partial_y^d$ which defines a one-to-one map from $\mathcal{T} : L_2 \to \mathcal{D}_x \cap \mathcal{D}_y$. We show that this consistency condition implies that $\mathcal{T}_x$ and $\mathcal{T}_y$ commute – allowing the associated inverse to be defined inductively. This inverse is then used to construct the PIE representation of a class of admissible, linear, multivariate PDEs. Note that while this explanation focuses on the 2D case, the result is presented for arbitrary spatial dimension.

Having obtained the main technical contribution, we then consider construction of stability conditions for the PDE. Specifically, in Section 5, induction is used to define a class of multivariate PI operators, which parameterize the PIE representation of multivariate PDEs. Using these operators, the same techniques for stability analysis of univariate PDEs can then be used to analyze stability of multivariate PDEs: parameterizing a Lyapunov functional by a PI operator, and testing decay of this functional along solutions to the PDE as a linear operator inequality on PI operator variables. To solve this operator inequality, in Section 6, a class of positive semidefinite PI operators $\mathcal{P} \succeq 0$ is parameterized by positive semidefinite matrices $P \succeq 0$ as $\mathcal{P} = \mathcal{Z}^*P\mathcal{Z}$ – defining a monomial operator $\mathcal{Z}$ using induction – allowing stability to be tested using SemiDefinite Programming (SDP). A more detailed overview of the organization of the paper is provided in Section 3.

2. Notation

Let $\mathbb{N}$ denote the natural numbers and $\mathbb{N}_0 := \mathbb{N} \cup \{0\}$. For $i, j \in \mathbb{N}_0$, we denote a set of indices as $\{i : j\} := \{i, i+1, \ldots, j\}$. For $N \in \mathbb{N}$ and multi-indices $\alpha, \beta \in \mathbb{N}_0^N$, we write $\alpha \leq \beta$ if $\alpha_i \leq \beta_i$ for all $i \in \{1, \ldots, n\}$. For scalar $\lambda \in \mathbb{R}$ and vector $\vec{b} \in \mathbb{R}^n$, $\vec{v} = \lambda \cdot \vec{b}$ implies $v_i = \lambda b_i$. We denote $\vec{1} := (1, \ldots, 1) \in \mathbb{N}^N$, and, more generally, $\vec{k} = k \cdot \vec{1} \in \mathbb{N}_0^N$ for $k \in \mathbb{N}_0$.

The spatial domains will be hyper-rectangular which are represented as $\Omega := \prod_{i=1}^N [a_i, b_i] = [a_1, b_1] \times \cdots \times [a_N, b_N]$, where $[a_i, b_i] \subset \mathbb{R}$. Denote by $\mathbb{R}^n[s]$ the space of $\mathbb{R}^n$-valued polynomials in variables s. Let $L_2^n[\Omega]$ denote the Hilbert space of (equivalence classes of) square-integrable functions on Ω, with standard inner product $\langle \mathbf{u}, \mathbf{v} \rangle_{L_2} := \int_\Omega \mathbf{u}(s)^T\mathbf{v}(s)ds$. Let $\mathcal{L}(L_2^n, L_2^m)$ denote the space of bounded linear operators from L_2^n to L_2^m, with induced operator norm $\|.\|_{\mathrm{op}}$, and let $\mathcal{L}(L_2^n) := \mathcal{L}(L_2^n, L_2^n)$. For a suitably

[3] where the map from PIE state to PDE state is defined as the inverse of $\partial_x^d\partial_y^d : \mathcal{D} \to L_2$.

[4] The need to check consistency of the boundary conditions a priori can be avoided using the projection-based approach in [18].

differentiable function $\mathbf{u} \in L_2^n[\Omega]$, we will use the shorthand notation

$$\partial_z^k \mathbf{u} := \frac{\partial^k}{\partial z^k}\mathbf{u}, \qquad\qquad D^\alpha \mathbf{u} := \partial_{s_1}^{\alpha_1} \cdots \partial_{s_N}^{\alpha_N} \mathbf{u},$$

where $k \in \mathbb{N}_0$ and $\alpha \in \mathbb{N}_0^N$. For sufficiently regular $\mathbf{u} \in L_2^n[\Omega]$, we denote limit values in one argument as $\mathbf{u}(\theta_i) := \mathbf{u}(s)|_{s_i=\theta_i} := \mathbf{u}(s_1,\dots,s_{i-1},\theta_i,s_{i+1},\dots,s_N)$.

For $\boldsymbol{K} \in L_2^{m\times n}[\Omega]$, define the multiplier operator $\mathrm{M}[\boldsymbol{K}] : L_2^n[\Omega] \to L_2^m[\Omega]$ by

$$(\mathrm{M}[\boldsymbol{K}]\mathbf{u})(s) := \boldsymbol{K}(s)\mathbf{u}(s).$$

We write $I_n \in \mathbb{R}^{n\times n}$ for the $n \times n$ identity matrix, and $0_{m\times n} \in \mathbb{R}^{m\times n}$ for a matrix of all zeros. For $A \in \mathbb{R}^{m\times n}$, we let $[A]_{i,j}$ denote the element in row i and column j of A.

2.1. *Sobolev Space with Bounded Mixed Derivatives*

Traditionally, solutions to PDEs in multiple spatial dimensions are allowed to lie in the Sobolev Hilbert spaces W_2^d – implying $D^\alpha \mathbf{u}(t) \in L_2$ for all $\alpha \geq \vec{0}$ such that $\|\alpha\|_1 := \sum_{i=1}^N \alpha_i \leq d$. The disadvantage of this approach is that solutions must be defined in the weak sense and that the boundary conditions need to be defined using trace operators. More recently, however, there has been a growing interest in multivariate PDE solutions which are required to exist in the more restrictive spaces, requiring $D^\alpha \mathbf{u}(t) \in L_2$ for all α such that $\|\alpha\|_\infty := \max_{i=1}^N \alpha_i \leq d$. More generally, for $\delta \in \mathbb{N}_0^N$, we define the Sobolev space with bounded mixed derivatives up to order δ on Ω as

$$S_2^{\delta,n}[\Omega] := \left\{\mathbf{u} \in L_2^n[\Omega] \mid D^\alpha \mathbf{u} \in L_2^n[\Omega],\ \forall \alpha \in \mathbb{N}_0^N : \alpha \leq \delta\right\}.$$

Spaces of this form are referred to as Sobolev spaces with dominating mixed smoothness, and have the advantage that if $\delta_i > 0$ for all i, then S_2^δ can be embedded in a suitable space of Hölder continuous functions [1] – implying weak and classical solutions are equivalent and that the boundary values of such functions are well-defined without the need for trace operators. In particular, for every $\alpha \leq \delta$ and $i \in \{1:N\}$, if $\alpha_i < \delta_i$, then the restrictions $D^\alpha \mathbf{u}|_{s_i=a_i}$ and $D^\alpha \mathbf{u}|_{s_i=b_i}$ of any $\mathbf{u} \in S_2^\delta[\Omega]$ are well-defined. Moreover, using S_2^δ has the added advantage that we may decompose $S_2^\delta[\Omega] = S_2^{\delta_1\cdot \mathrm{e}_1}[\Omega] \cap \cdots \cap S_2^{\delta_N\cdot \mathrm{e}_N}[\Omega]$, where $\mathrm{e}_i \in \mathbb{R}^N$ is the ith standard Euclidean basis vector.

Considering then a PDE domain $\mathcal{D} := \bigcap_{i=1}^N \mathcal{D}_i \subseteq S_2^\delta$, where $\mathcal{D}_i \subseteq S_2^{\delta_i\cdot \mathrm{e}_i}[\Omega]$ for each i, this also allows the inverse of $D^\delta : \mathcal{D} \to L_2[\Omega]$ to be constructed inductively using the inverses of the individual $\partial_{s_i}^{\delta_i} : \mathcal{D}_i \to L_2[\Omega]$, as will be shown in Section 4.

Note that for $\Omega := \prod_{i=1}^N [a_i,b_i]$ and univariate domain, $\mathcal{D} \subset S_2^d[a_i,b_i]$, we denote the ith lifting of that domain as $\hat{\mathcal{D}}[i] \subset S_2^{d\cdot \mathrm{e}_i}[\Omega]$, so that

$$\hat{\mathcal{D}}[i] := \{\mathbf{u} \in S_2^{d\cdot \mathrm{e}_i,n}[\Omega] \ \mid\ \mathbf{u}(s_1,\dots,s_{i-1},\bullet,s_{i+1},\dots,s_N) \in \mathcal{D}\ \ \forall s_j \in [a_j,b_j],\ j \neq i\ \},$$

where for $\mathbf{u} \in S_2^{d\cdot \mathrm{e}_i,n}$ and $s_j \in [a_j,b_j]$, $j \neq i$, $\mathbf{v} = \mathbf{u}(s_1,\dots,s_{i-1},\bullet,s_{i+1},\dots,s_N)$ means that $\mathbf{v}(s_i) = \mathbf{u}(s)$ for all $s_i \in [a_i,b_i]$. Thus for univariate domains $\mathcal{D}_i$, through some abuse of notation, $\mathcal{D} = \bigcap_{i=1}^N \mathcal{D}_i$ will denote $\mathcal{D} = \bigcap_{i=1}^N \hat{\mathcal{D}}_i[i]$.

3. Problem Definition and Organization: PIE Representation and Stability Analysis of *N*D PDEs

Consider the problem of representation and stability analysis of a class of autonomous linear PDEs of the form

$$\partial_t \mathbf{u}(t,s) = \sum_{0\leq\alpha\leq\delta} \boldsymbol{A}_\alpha(s) D^\alpha \mathbf{u}(t,s), \qquad t\geq 0,\ s\in\Omega, \tag{3.1}$$

parameterized by polynomials $\boldsymbol{A}_\alpha \in \mathbb{R}^{n\times n}[s]$ and defined on the hyper-rectangle $\Omega := \prod_{i=1}^N [a_i,b_i]$. We suppose that the domain of the PDE may be decomposed as $\mathbf{u}(t)\in\mathcal{D} := \mathcal{D}_1\cap\cdots\cap\mathcal{D}_N$, where for each i,

$$\mathcal{D}_i := \left\{ \mathbf{u}\in S_2^{\delta_i\cdot e_{i,n}}[\Omega] \;\middle|\; \sum_{k=0}^{\delta_i-1}\left[B^i_{j,k}(\partial_{s_i}^k\mathbf{u})|_{s_i=a_i} + C^i_{j,k}(\partial_{s_i}^k\mathbf{u})|_{s_i=b_i}\right]=0,\ \ \forall j\in\{0:\delta_i-1\}\right\}, \tag{3.2}$$

with matrix parameters $B^i_{j,k}, C^i_{j,k}\in\mathbb{R}^{n\times n}$. This parameterization of the domain is quite general but implies no coupling between spatial variables at orthogonal boundary surfaces – i.e. it is does not permit a constraint of the form $\mathbf{u}(x,0)=\mathbf{u}(0,y)$. Motivation for this restriction will be discussed in the following subsection.

The parameterization of PDEs in (3.1) includes most commonly used PDEs such as heat, wave and plate equations. For example, it includes 2nd-order 3D PDEs of the form[5]

$$\mathbf{u}_t = A_0\mathbf{u} + A_1\begin{bmatrix}\mathbf{u}_x\\ \mathbf{u}_y\\ \mathbf{u}_z\end{bmatrix} + A_2\begin{bmatrix}\mathbf{u}_{xx}\\ \mathbf{u}_{yy}\\ \mathbf{u}_{zz}\end{bmatrix},$$

with boundary conditions parameterized as

$$\mathcal{D} := \left\{ \mathbf{u} \;\middle|\; \overbrace{B_x\begin{bmatrix}\mathbf{u}(0,y,z)\\ \mathbf{u}_x(0,y,z)\\ \mathbf{u}(1,y,z)\\ \mathbf{u}_x(1,y,z)\end{bmatrix}=0}^{\mathcal{D}_1},\quad \overbrace{B_y\begin{bmatrix}\mathbf{u}(x,0,z)\\ \mathbf{u}_y(x,0,z)\\ \mathbf{u}(x,1,z)\\ \mathbf{u}_y(x,1,z)\end{bmatrix}=0}^{\mathcal{D}_2},\quad \overbrace{B_z\begin{bmatrix}\mathbf{u}(x,y,0)\\ \mathbf{u}_z(x,y,0)\\ \mathbf{u}(x,y,1)\\ \mathbf{u}_z(x,y,1)\end{bmatrix}=0}^{\mathcal{D}_3}\right\}.$$

This formulation includes heat and wave[6] equations with Dirichlet, Neumann, Robin, and mixed type boundary conditions.

Considering the class of PDEs in (3.1), our goal is to construct an equivalent PIE representation, exploiting the decomposition of the PDE domain as $\mathcal{D}=\bigcap_{i=1}^N\mathcal{D}_i$.

3.1. *PIE Representation and Fundamental State*

Having proposed a class of PDEs in Eqn. (3.1), with domain $\mathcal{D}=\bigcap_{i=1}^N\mathcal{D}_i$, where $\mathcal{D}_i$ are specified as in Eqn. (3.2), we now seek to represent the solutions to such PDEs using evolution equations of the form

$$\partial_t\mathcal{T}\mathbf{v}(t) = \mathcal{A}\mathbf{v}(t), \qquad \mathbf{v}(t)\in L_2^n[\Omega],\ t\geq 0, \tag{3.3}$$

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[5] We may convert to the form in Eqn. (3.2) using $\boldsymbol{A}_{(0,0,0)}=A_0$, $[\boldsymbol{A}_{(1,0,0)},\boldsymbol{A}_{(0,1,0)},\boldsymbol{A}_{(0,0,1)}]=A_1$ and $[\boldsymbol{A}_{(2,0,0)},\boldsymbol{A}_{(0,2,0)},\boldsymbol{A}_{(0,0,2)}]=A_2$.

[6] Let $A_0=\begin{bmatrix}0&0\\1&0\end{bmatrix}$ and $A_2=\begin{bmatrix}0&1&0&1&0&1\\0&0&0&0&0&0\end{bmatrix}$ and use state $\mathbf{u}(t)=\begin{bmatrix}u(t)\\u_t(t)\end{bmatrix}$.

where $\mathcal{T},\mathcal{A}$ are integral operators of the form

$$\mathcal{T}=\prod_{i=1}^{N}\mathcal{T}_i,\qquad (\mathcal{T}_i\mathbf{v})(s):=\int_{a_i}^{s_i}\boldsymbol{R}_1(s_i,\theta_i)\mathbf{v}(\theta_i)d\theta_i+\int_{s_i}^{b_i}\boldsymbol{R}_2(s_i,\theta_i)\mathbf{v}(\theta_i)d\theta_i, \tag{3.4}$$

and

$$\mathcal{A}=\sum_{j=1}^{M}\prod_{i=1}^{N}\mathcal{A}_{i,j},\qquad (\mathcal{A}_{i,j}\mathbf{v})(s):=\boldsymbol{Q}_0(s_i)\mathbf{v}(s)+\int_{a_i}^{s_i}\boldsymbol{Q}_1(s_i,\theta_i)\mathbf{v}(\theta_i)d\theta_i+\int_{s_i}^{b_i}\boldsymbol{Q}_2(s_i,\theta_i)\mathbf{v}(\theta_i)d\theta_i,$$

for polynomials $\boldsymbol{R}_k$ and $\boldsymbol{Q}_k$. Operators of this form belong to the algebra of Partial Integral (PI) operators defined formally in Defn. 6 and evolution equations of the form in (3.3) are referred to as Partial Integral Equations (PIEs). In such a representation, the map from PDE to PIE solution is given by $\mathbf{v}(t)=D^{\boldsymbol{\delta}}\mathbf{u}(t)$ and the map from PIE to PDE solution is given by $\mathbf{u}(t)=\mathcal{T}\mathbf{v}(t)$.

3.2. *Equivalence of PDE and PIE – Overview of Section 4*

While the map from PDE to PIE state, $D^{\boldsymbol{\delta}}:\mathcal{D}\to L_2^n$, is specified by the order $\boldsymbol{\delta}$ of the PDE in (3.1), construction of the map from PIE to PDE state, $\mathcal{T}:=(D^{\boldsymbol{\delta}})^{-1}:L_2^n\to\mathcal{D}$, is non-trivial. Indeed, inversion of $D^{\boldsymbol{\delta}}$ on the domain $\mathcal{D}$ is the essential step[7] in constructing the PIE representation — establishing a bijection between L_2^n and $\mathcal{D}$. We perform this inversion in Section 4, exploiting the decomposition of the PDE domain as $\mathcal{D}:=\mathcal{D}_1\cap\cdots\cap\mathcal{D}_N$ to inductively define $(D^{\boldsymbol{\delta}})^{-1}=(\partial_{s_N}^{\delta_N})^{-1}\cdots(\partial_{s_1}^{\delta_1})^{-1}$. In particular, in Subsection 4.1, we first recall that if each set $\mathcal{D}_i$ is defined by well-posed boundary conditions of the form in (3.2), then $\partial_{s_i}^{\delta_i}:\mathcal{D}_i\to L_2^n$ is invertible, and the inverse $\mathcal{T}_i:L_2^n\to\mathcal{D}_i$ is as in (3.4), where the formulae for $\boldsymbol{R}_1,\boldsymbol{R}_2$ are given in [14]. It is then relatively easy to show through induction that $\mathcal{T}:=\mathcal{T}_N\cdots\mathcal{T}_1$ is both a left- and right-inverse of $D^{\boldsymbol{\delta}}:=\partial_{s_1}^{\delta_1}\cdots\partial_{s_N}^{\delta_N}$ – implying $D^{\boldsymbol{\delta}}\mathcal{T}=I_n:L_2^n\to L_2^n$ and $\mathcal{T}D^{\boldsymbol{\delta}}=I_n:\mathcal{D}\to\mathcal{D}$. This implies that if $\mathbf{u}(t)$ satisfies the PDE, then $\mathbf{v}(t)=D^{\boldsymbol{\delta}}\mathbf{u}(t)$ satisfies the PIE in Eqn. (3.3). Thus if the PIE is stable, the PDE will likewise be stable. However, the converse statement requires one additional step – showing that every solution to the PIE also offers a solution to the PDE.

In order to show that every solution of the PIE defines a solution to the PDE, we require $\mathcal{T}:L_2^n\to\mathcal{D}$ – i.e. for any $\mathbf{v}(t)\in L_2$, $\mathcal{T}\mathbf{v}(t)$ satisfies the boundary conditions defining $\mathcal{D}$. While this property always holds when $\mathbf{v}(t,s)$ is scalar-valued, the extension to vector-valued states requires an additional restriction on the set of admissible boundary conditions. Specifically, in Subsection 4.2, we show that even in the 2D case, $\mathcal{D}=\mathcal{D}_1\cap\mathcal{D}_2$, there exist domains $\mathcal{D}_1,\mathcal{D}_2$ (each well-posed in the 1D sense) and $\mathbf{v}\in L_2^n[[0,1]^2]$ such that $\mathcal{T}\mathbf{v}\notin\mathcal{D}$. This problem manifests as inconsistency of $\mathcal{D}$ at corners (i.e. $(x,y)=(0,0)$) of the domain where $\mathcal{D}_1$ and $\mathcal{D}_2$ intersect. So, for example, if $\mathbf{u}\in\mathcal{D}_1$ implies $\mathbf{u}(0,x)=0$, this clearly requires $\mathbf{u}(0,0)=0$ – a restriction which may be incompatible with $\mathcal{D}_2$. Such carryover from one domain to another occurs because the domains have not been properly decoupled through projection onto a consistent basis as proposed in [18]. In this paper, however, we tolerate this carryover in order to avoid explicit construction of such a basis — instead proposing a necessary and sufficient condition on the parameters $B_{j,k}^i$ and $C_{j,k}^i$ which define $\mathcal{D}_i$ in (3.2) for these domains to be consistent (Defn. 2). This consistency condition then guarantees that $\mathcal{T}:L_2^n\to\mathcal{D}$ – implying that any solution to the PIE is a solution to the PDE. This establishes that stability of the PIE is equivalent to stability of the PDE.

Having established the main result, in Section 6, we propose a simple Lyapunov condition for stability of the PIE (and hence PDE) in terms of a set of linear PI operator inequalities. These operator

[7] Given $\mathcal{T}$, the operator $\mathcal{A}$ may be constructed as $\mathcal{A}=\sum_{\alpha\leq\delta}\mathrm{M}[\boldsymbol{A}_\alpha]D^\alpha\mathcal{T}$.

inequalities can then be tested using semidefinite programming via software such as PIETOOLS and a brief description of such tools are described in Section 7 with application to several numerical examples.

4. An Equivalent PIE Representation of Linear Multivariate PDEs

In [42], it was shown that for a univariate subspace $\mathcal{D} \subseteq S_2^d[a,b]$, if $\mathcal{D}$ is defined by a set of suitably admissible linear boundary conditions, then we can define a left- and right-inverse $\mathcal{T} : L_2[a,b] \to \mathcal{D}$ to the differential operator $\partial_s^d : \mathcal{D} \to L_2[a,b]$, establishing a bijection between $\mathcal{D}$ and $L_2[a,b]$. This $\mathcal{T}$ is an integral operator of the form $(\mathcal{T}\mathbf{v})(s) = \int_a^b \boldsymbol{k}(s,\theta)\mathbf{v}(\theta)d\theta$, with *polynomial semi-separable* kernel $\boldsymbol{k}$, so that

$$\boldsymbol{k}(s,\theta) := \begin{cases} \boldsymbol{g}(s,\theta), & \theta \leq s, \\ \boldsymbol{h}(s,\theta), & \theta > s, \end{cases} \qquad s,\theta \in [a,b],$$

where $\boldsymbol{g}$ and $\boldsymbol{h}$ are polynomial. Using this relation, it can then be shown that any univariate PDE of the form

$$\partial_t \mathbf{u}(t,s) = \sum_{k=0}^{d} \boldsymbol{A}_k(s)\partial_s^k \mathbf{u}(t,s), \qquad \mathbf{u}(t) \in \mathcal{D},$$

can be equivalently represented as a PIE of the form

$$\partial_t \mathcal{T}\mathbf{v}(t) = \mathcal{A}\mathbf{v}(t), \qquad \mathbf{v}(t) \in L_2[a,b],$$

using the state transformation $\mathbf{v}(t) = \partial_s^d \mathbf{u}(t)$ and $\mathbf{u}(t) = \mathcal{T}\mathbf{v}(t)$.

As discussed in Subsection 3.2, our approach to representation of multivariate PDEs is based on inductive application of this result when the domain of the multivariate PDE may be decomposed as $\mathcal{D} = \mathcal{D}_1 \cap \cdots \cap \mathcal{D}_N$, where each $\mathcal{D}_i \subseteq S_2^{\delta_i}[\Omega]$ is defined by boundary conditions only with respect to the ith spatial variable (as in (3.2)). When these domains are consistent, the map $\mathcal{T} : L_2^n[\Omega] \to \mathcal{D}$ may then be constructed as $\mathcal{T} := \mathcal{T}_1 \cdots \mathcal{T}_N$, where $\mathcal{T}_i : L_2^n[\Omega] \to \mathcal{D}_i$ are as defined above. To illustrate, consider the special case of Dirichlet and Neumann boundary conditions (which always define consistent domains). Specifically, define the following 1D domains as

$$\begin{aligned} \mathcal{D}_{\mathrm{D}} &:= \{\mathbf{u} \in S_2^{2,n}[a_i,b_i] \;\mid\; \mathbf{u}(a_i) = \mathbf{u}(b_i) = 0\}, \\ \mathcal{D}_{\mathrm{N1}} &:= \{\mathbf{u} \in S_2^{2,n}[a_i,b_i] \;\mid\; \mathbf{u}_s(a_i) = \mathbf{u}(b_i) = 0\}, \\ \mathcal{D}_{\mathrm{N2}} &:= \{\mathbf{u} \in S_2^{2,n}[a_i,b_i] \;\mid\; \mathbf{u}(a_i) = \mathbf{u}_s(b_i) = 0\}. \end{aligned} \tag{4.1}$$

These 1D domains may be trivially embedded in a multivariate space using the lifted 1D domains, denoted $\hat{\mathcal{D}}[i]$ and $\mathcal{D} = \bigcap_{i=1}^N \mathcal{D}_i := \bigcap_{i=1}^N \hat{\mathcal{D}}_i[i]$ as defined in Subsec. 2.1. Then we have the following result (a special case of the main result in Thm. 9).

Corollary 1 *Let $\mathcal{D} = \mathcal{D}_1 \cap \cdots \cap \mathcal{D}_N$. Then $\mathbf{u} = \mathcal{T}D^{(2,\cdots,2)}\mathbf{u}$, $\mathbf{v} = D^{(2,\cdots,2)}\mathcal{T}\mathbf{v}$ and $\mathcal{T}\mathbf{v} \in \mathcal{D}$ for all $\mathbf{v} \in L_2^n$ and $\mathbf{u} \in \mathcal{D}$, where $(\mathcal{T}\mathbf{v})(s) = \int_\Omega \boldsymbol{G}(s,\theta)\mathbf{v}(\theta)d\theta$ with $\boldsymbol{G}(s,\theta) = \prod_{i=1}^N \boldsymbol{G}_i(s_i,\theta_i)$ and*

$$\boldsymbol{G}_i(s,\theta) = \begin{cases} \boldsymbol{h}_i(s,\theta) + (s-\theta), & \theta \leq s, \\ \boldsymbol{h}_i(s,\theta), & \theta > s, \end{cases} \quad \textit{where} \quad \boldsymbol{h}_i(s,\theta) = \begin{cases} \frac{-(s-a_i)(b_i-\theta)}{b_i-a_i}, & \mathcal{D}_i = \mathcal{D}_{\mathrm{D}}, \\ a_i - s, & \mathcal{D}_i = \mathcal{D}_{\mathrm{N1}}, \\ s - b_i, & \mathcal{D}_i = \mathcal{D}_{\mathrm{N2}}. \end{cases}$$

Moreover, $\mathbf{u}(t) \in \mathcal{D}$ *satisfies the PDE* $\partial_t \mathbf{u}(t) = \nabla^2 \mathbf{u}(t) = \sum_{i=1}^N \partial_{s_i}^2 \mathbf{u}(t)$ *if and only if* $\mathbf{v} = D^{(2,\cdots,2)}\mathbf{u} \in L_2^n$ *satisfies the PIE* $\partial_t \mathcal{T}\mathbf{v}(t) = \mathcal{A}\mathbf{v}(t)$*, where* $\mathcal{A}\mathbf{v}(s) = \sum_{j=1}^N \prod_{i\neq j}(\int_{[a_i,b_i]} \boldsymbol{G}_i(s_i,\theta_i))\mathbf{v}(\theta)d\theta$.

Note: For $\mathcal{T}$ as in Cor. 1, $\mathcal{T} = \mathcal{T}_1 \cdots \mathcal{T}_N$, where $(\mathcal{T}_i\mathbf{v})(s) := \int_{a_i}^{b_i} \boldsymbol{G}_i(s_i,\theta)\mathbf{u}(s_1,\cdots,s_{i-1},\theta,s_{i+1},\cdots,s_N)d\theta$.

To establish results of this form, in Subsection 4.1, we recall the construction of $\mathcal{T}_i$ and the PIE representation in the 1D case. Consistency of the domains, $\mathcal{D}_i$ will be addressed in Subsection 4.2. Finally, construction of $\mathcal{T} = \mathcal{T}_1 \cdots \mathcal{T}_N$ and the PIE representation in *N*D is presented in Subsection 4.3.

4.1. *The PIE Representation of 1D PDEs*

Consider construction of a PIE representation of a 1D PDE defined as

$$\partial_t \mathbf{u}(t,s) = \sum_{k=0}^{d} \boldsymbol{A}_k(s)\partial_s^k \mathbf{u}(t,s), \qquad \mathbf{u}(t) \in \mathcal{D},\ s \in [a,b],$$

where $\mathcal{D}$ is the domain of the PDE, defined by a suitable set of linear boundary conditions. The crucial step in constructing the PIE representation is that of inverting the operator $\partial_s^d : \mathcal{D} \to L_2^n[a,b]$, which is only possible if the boundary conditions defining $\mathcal{D}$ are admissible. We recall the following notion of an admissible PDE domain in 1D.

Definition 1 *We say that* $\mathcal{D} \subset S_2^{d,n}[a,b]$ *is* ***admissible*** *if*

$$\mathcal{D} := \left\{ \mathbf{u} \in S_2^{d,n}[a,b] \;\middle|\; \sum_{k=0}^{d-1} \Big[B_{j,k}(\partial_s^k \mathbf{u})(a) + C_{j,k}(\partial_s^k \mathbf{u})(b) \Big] = 0,\ \ \forall j \in \{0 : d-1\} \right\}. \tag{4.2}$$

with $\big(H_a + H_b \boldsymbol{Q}(b-a)\big)$ *invertible, where* $H_a, H_b \in \mathbb{R}^{nd\times nd}$ *are defined by* $[H_a]_{j,k} = B_{j,k}$, $[H_b]_{j,k} = C_{j,k}$, *and where*

$$\boldsymbol{Q}(z) := e^{Uz} = \begin{bmatrix} I_n & zI_n & \cdots & \frac{z^{d-1}}{(d-1)!}I_n \\ 0_n & I_n & \cdots & \frac{z^{d-2}}{(d-2)!}I_n \\ \vdots & \vdots & \ddots & \vdots \\ 0_n & 0_n & \cdots & I_n \end{bmatrix}, \qquad U = \begin{bmatrix} 0_n & I_n & & \\ & \ddots & \ddots & \\ & & 0_n & I_n \\ & & & 0_n \end{bmatrix}. \tag{4.3}$$

Note that many common boundary conditions (Dirichlet, Neumann, Robin) define admissible PDE domains by this definition. Given such an admissible domain, the following result, a corollary of Thm. 3.1 in Sec. XIV.3 of [14], proves that $\partial_s^d : \mathcal{D} \to L_2^n[a,b]$ is invertible, offering an explicit expression of the inverse $\mathcal{T} : L_2[a,b] \to \mathcal{D}$, as well as for the composition $\mathcal{T}$ with the differential operator $\mathcal{H}$.

Corollary 2 *For given* $[a,b] \subset \mathbb{R}$, $d \in \mathbb{N}_0$ *and* $\boldsymbol{H}_k \in \mathbb{R}^{n\times n}[s]$, *let* $\mathcal{D} \subseteq S_2^{d,n}[a,b]$ *be admissible as per Defn. 1 with* H_a, H_b *as defined therein. Let* $\mathcal{H} : \mathcal{D} \to L_2^n$ *be defined as* $(\mathcal{H}\mathbf{u})(s) := \sum_{k=0}^d \boldsymbol{H}_k(s-a)\partial_s^k \mathbf{u}(s)$. *Then for any* $\mathbf{u} \in \mathcal{D}$ *we have that* $\mathcal{T}\partial_s^d \mathbf{u} = \mathbf{u}$ *and* $\mathcal{H}\mathbf{u} = \mathcal{A}\partial_s^d \mathbf{u}$, *and for any* $\mathbf{v} \in L_2^n[a,b]$ *we have that*

$\mathcal{T}\mathbf{v} \in \mathcal{D}$, $\partial_s^d \mathcal{T}\mathbf{v} = \mathbf{v}$, *and* $\mathcal{H}\mathcal{T}\mathbf{v} = \mathcal{A}\mathbf{v}$, *where for* $d > 0$,

$$(\mathcal{T}\mathbf{v})(s) = \int_a^b \boldsymbol{G}_T(s,\theta)\mathbf{v}(\theta)d\theta, \qquad \boldsymbol{G}_T(s,\theta) = \begin{cases} \mathbf{e}_1(s-a)^T(I_n - K)\mathbf{e}_d(b-\theta) & \theta \le s, \\ -\mathbf{e}_1(s-a)^T K\mathbf{e}_d(b-\theta) & s < \theta, \end{cases}$$

$$(\mathcal{A}\mathbf{v})(s) = \boldsymbol{H}_d(s-a)\mathbf{v}(s) + \int_a^b \boldsymbol{G}_A(s,\theta)\mathbf{v}(\theta)d\theta, \quad \boldsymbol{G}_A(s,\theta) = \begin{cases} \mathbf{c}_A(s-a)^T(I_n - K)\mathbf{e}_d(b-\theta) & \theta \le s, \\ -\mathbf{c}_A(s-a)^T K\mathbf{e}_d(b-\theta) & s < \theta, \end{cases}$$

where $K := (H_a + H_b\boldsymbol{Q}(b-a))^{-1}H_b$ *with* $\boldsymbol{Q}$ *as defined in Eqn. (4.3) and*

$$\mathbf{e}_1(z) = \begin{bmatrix} 1 \\ z \\ \frac{1}{2}z^2 \\ \vdots \\ \frac{z^{d-1}}{(d-1)!} \end{bmatrix} \otimes I_n, \qquad \mathbf{e}_d(z) = \begin{bmatrix} \frac{z^{d-1}}{(d-1)!} \\ \vdots \\ \frac{1}{2}z^2 \\ z \\ 1 \end{bmatrix} \otimes I_n, \qquad \mathbf{c}_A(z) = \begin{bmatrix} \boldsymbol{H}_0(z) \\ \boldsymbol{H}_0(z)z + \boldsymbol{H}_1(z) \\ \vdots \\ \sum_{k=0}^{d-1} \boldsymbol{H}_k(z)\frac{z^{d-k-1}}{(d-k-1)!} \end{bmatrix}.$$

Additionally, for $d = 0$, *the identities hold with* $\mathcal{T} = I_n$ *and* $\mathcal{A} = \mathrm{M}[\boldsymbol{H}_0]$.

Cor. 2 shows that, for $\mathcal{D}$ admissible, we can define the inverse to the differential operator $\partial_s^d : \mathcal{D} \to L_2$ as an integral operator $\mathcal{T} : L_2 \to \mathcal{D}$. Moreover, for a PDE of the form $\partial_t \mathbf{u}(t) = \mathcal{H}\mathbf{u}(t)$, we can define an equivalent PIE representation as $\partial_t \mathcal{T}\mathbf{v}(t) = \mathcal{A}\mathbf{v}(t)$, using fundamental state $\mathbf{v}(t) = \partial_s^d \mathbf{u}(t)$. The following corollary adapts this result to multivariate space, $\partial_{s_i}^{\delta_i} : \hat{\mathcal{D}}[i] \to L_2[\Omega]$, using the lifted operator $\mathcal{T}_i : L_2[\Omega] \to \hat{\mathcal{D}}[i]$.

Corollary 3 *For any* $i \in \{1:N\}$, *let* $\mathcal{D}_i := \hat{\mathcal{D}}[i] \subseteq S_2^{\delta_i \cdot e_i, n}[\Omega]$ *for* $\mathcal{D}$ *of the form in (4.2) with* $d = \delta_i$ *and* $[a,b] = [a_i, b_i]$. *Define associated* $\mathcal{T}, \mathcal{A}$, *and* $\mathcal{H}$ *as in Cor. 2. If we define the extensions of these operators as*

$$\begin{aligned} &\mathcal{T}_i : L_2^n[\Omega] \to S_2^{\delta_i \cdot \mathrm{e}_i, n}[\Omega] \qquad \textit{where} \qquad (\mathcal{T}_i\mathbf{v})(s) := \big(\mathcal{T}\mathbf{v}(s_1, \cdots, s_{i-1}, \bullet, s_{i+1}, \cdots, s_N)\big)(s_i), \\ &\mathcal{A}_i : L_2^n[\Omega] \to L_2^n[\Omega] \qquad \textit{where} \qquad (\mathcal{A}_i\mathbf{v})(s) := \big(\mathcal{A}\mathbf{v}(s_1, \cdots, s_{i-1}, \bullet, s_{i+1}, \cdots, s_N)\big)(s_i), \\ &\mathcal{H}_i : \mathcal{D}_i \to L_2^n[\Omega] \qquad \textit{where} \qquad (\mathcal{H}_i\mathbf{v})(s) := \big(\mathcal{H}\mathbf{v}(s_1, \cdots, s_{i-1}, \bullet, s_{i+1}, \cdots, s_N)\big)(s_i), \end{aligned}$$

then for all $\mathbf{u} \in \mathcal{D}_i$ *we have that* $\mathcal{T}_i \partial_{s_i}^{\delta_i} \mathbf{u} = \mathbf{u}$ *and* $\mathcal{A}_i \partial_{s_i}^{\delta_i} \mathbf{u} = \mathcal{H}_i \mathbf{u}$, *and for all* $\mathbf{v} \in L_2^n[\Omega]$ *we have* $\mathcal{T}_i\mathbf{v} \in \mathcal{D}_i$, $\partial_{s_i}^{\delta_i}\mathcal{T}_i\mathbf{v} = \mathbf{v}$ *and* $\mathcal{H}_i\mathcal{T}_i\mathbf{v} = \mathcal{A}_i\mathbf{v}$.

Proof For any $\mathbf{u} \in \mathcal{D}_i$ and $s_j \in [a_j, b_j]$ for $j \neq i$, we have $\mathbf{u}(s_1, \cdots, s_{i-1}, \bullet, s_{i+1}, \cdots, s_N) \in \mathcal{D}$. By Cor. 2, it follows that

$$\mathcal{T}_i \partial_{s_i}^{\delta_i} \mathbf{u}(s) = (\mathcal{T}\partial_{s_i}^{\delta_i} \mathbf{u}(s_1, \cdots, s_{i-1}, \bullet, s_{i+1}, \cdots, s_N))(s_i) = \mathbf{u}(s),$$

and similarly $\mathcal{A}_i \partial_{s_i}^{\delta_i} \mathbf{u} = \mathcal{H}_i \mathbf{u}$. Conversely, for any $\mathbf{v} \in L_2^n[\Omega]$, we have $\mathbf{v}(s_1, \cdots, s_{i-1}, \bullet, s_{i+1}, \cdots, s_N) \in L_2^n[a_i, b_i]$ for almost every $s_j \in [a_j, b_j]$ for $j \neq i$, and therefore by Cor. 2,

$$\partial_{s_i}^{\delta_i} \mathcal{T}_i \mathbf{v}(s) = \partial_{s_i}^{\delta_i} (\mathcal{T}_i \mathbf{v}(s_1, \cdots, s_{i-1}, \bullet, s_{i+1}, \cdots, s_N))(s_i) = \mathbf{v}(s),$$

and similarly $\mathcal{H}_i\mathcal{T}_i\mathbf{v} = \mathcal{A}_i\mathbf{v}$. Finally, by Cor. 2, also $(\mathcal{T}\mathbf{v})(s_1, \cdots, s_{i-1}, \bullet, s_{i+1}, \cdots, s_N) \in \mathcal{D} \subseteq S_2^{\delta_i, n}[a_i, b_i]$, and therefore $\mathcal{T}_i\mathbf{v} \in \mathcal{D}_i$. $\square$

By Cor. 3, if $\mathcal{D}_i$ is defined by admissible boundary conditions, then we can define an inverse to $\partial_{s_i} : \mathcal{D}_i \to L_2^n$ as a lifted, univariate integral operator $\mathcal{T}_i : L_2^n \to \mathcal{D}_i$. Using this result, the following corollary shows that $\mathcal{T} := \mathcal{T}_N \cdots \mathcal{T}_1$ defines a left- and right-inverse of D^δ when the domain of D^δ is extended to the range of $\mathcal{T}$.

Corollary 4 *For $\mathcal{D}_i \subseteq S_2^{\delta_i \cdot \mathrm{e}_i, n}$ and associated $\mathcal{T}_i : L_2^n \to \mathcal{D}_i$ as in Cor. 3, let $\mathcal{T} := \mathcal{T}_N \cdots \mathcal{T}_1$ and $\mathcal{D} := \mathcal{D}_1 \cap \cdots \cap \mathcal{D}_N$. Then $\mathcal{T} D^\delta \mathbf{u} = \mathbf{u}$ for any $\mathbf{u} \in \mathcal{D}$, and $D^\delta \mathcal{T} \mathbf{v} = \mathbf{v}$ for any $\mathbf{v} \in L_2^n$.*

Proof The result follows from the fact that $\mathcal{T}_i$ defines both a left- and right-inverse to $\partial_{s_i}^{\delta_i} : \mathcal{D}_i \to L_2^n$. A formal proof is given in Cor. 21 in Appx. A. □

Note that Cor. 4 does not necessarily imply that $\mathcal{T}$ is a right-inverse to D^δ in the standard sense unless $\mathcal{T} : L_2^n \to \mathcal{D}$. Indeed, such a right-inverse exists only if D^δ is surjective, a condition which requires consistency of the domains $\mathcal{D}_i$ as described in the following subsection.

4.2. *Consistency of PDE Domains*

In this subsection, we see that if $\mathcal{D}_i$ are admissible in the sense of Defn. 1, and $\mathcal{T}_i$ are as defined in Cor. 3, then we may define a necessary and sufficient condition for invertibility of $D^\delta := \prod_i \partial_{s_i}^{\delta_i}$ on $\mathcal{D} = \bigcap_i \mathcal{D}_i$ and show that such an inverse may be constructed as $\mathcal{T} := \prod_i \mathcal{T}_i$. Since we have already shown in Cor. 4 that this $\mathcal{T}$ is a left- and right-inverse of D^δ on the range of $\mathcal{T}$, the only challenge is now to show that $D^\delta : \mathcal{D} \to L_2$ is surjective or, equivalently, that the image of $\mathcal{T}$ is $\mathcal{D}$. To illustrate this challenge, let us consider an example where D^δ is not surjective and hence the image of $\mathcal{T}$ is not $\mathcal{D}$. This example illustrates a case where the domains $\mathcal{D}_i$ are not "consistent" at the corners of the spatial domain.

Example 1 *To illustrate the problem of inconsistent PDE domains, consider the 2D domain $\mathcal{D} = \mathcal{D}_1 \cap \mathcal{D}_2$, where*

$$\mathcal{D}_1 := \left\{ \begin{bmatrix} \mathbf{u}_1 \\ \mathbf{u}_2 \end{bmatrix} \in S_2^{(1,0),2}[[0,1]^2] \;\middle|\; \begin{array}{l} \mathbf{u}_1(0,y) = 0 \\ \mathbf{u}_2(0,y) = \mathbf{u}_1(1,y) \end{array} \right\},$$
$$\mathcal{D}_2 := \left\{ \begin{bmatrix} \mathbf{u}_1 \\ \mathbf{u}_2 \end{bmatrix} \in S_2^{(0,1),2}[[0,1]^2] \;\middle|\; \begin{array}{l} \mathbf{u}_1(x,0) = \mathbf{u}_2(x,1) \\ \mathbf{u}_2(x,0) = 0 \end{array} \right\}.$$

Both of these sets are admissible in the sense of Defn. 1, and using Cor. 3, we can express $\mathbf{u} = \mathcal{T}_1 \partial_x \mathbf{u} = \mathcal{T}_2 \partial_y \mathbf{u}$ for all $\mathbf{u} \in \mathcal{D}$, where

$$\big(\mathcal{T}_1 \mathbf{v}\big)(x,y) := \int_0^x \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \mathbf{v}(\theta,y) d\theta + \int_x^1 \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \mathbf{v}(\theta,y) d\theta,$$
$$\big(\mathcal{T}_2 \mathbf{v}\big)(x,y) := \int_0^y \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \mathbf{v}(x,\eta) d\eta + \int_y^1 \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \mathbf{v}(x,\eta) d\eta.$$

Defining then $\mathcal{T} = \mathcal{T}_2 \mathcal{T}_1$, as per Cor. 4, we have $\partial_x \partial_y \mathcal{T} \mathbf{v} = \mathbf{v}$ and $\mathcal{T} \partial_x \partial_y \mathbf{u} = \mathbf{u}$ for all $\mathbf{v} \in L_2^2$ and $\mathbf{u} \in \mathcal{D}$. However, if we define $\mathbf{v}_1 = \mathbf{v}_2 = 1$, then $\mathbf{v} \in L_2^2[[0,1]^2]$, and $\mathbf{u} = \mathcal{T}\mathbf{v}$ implies $\mathbf{u}_1 = 1 + x + xy$ and $\mathbf{u}_2 = y + xy$. Clearly, $\partial_x \partial_y \mathcal{T} \mathbf{v} = \mathbf{v}$ and $\mathcal{T} \partial_x \partial_y \mathbf{u} = \mathbf{u}$, but $\mathbf{u} = \mathcal{T}\mathbf{v} \notin \mathcal{D}$ since $\mathbf{u}_1(0,y) \neq 0$. The problem, here, is that the domains $\mathcal{D}_1$ and $\mathcal{D}_2$ are not consistent. In particular, any $\mathbf{u} \in \mathcal{D}_1$ must satisfy $\mathbf{u}_1(0,0) = 0$,

whereas any $\mathbf{u} \in \mathcal{D}_2$ *must satisfy* $\mathbf{u}_1(0,0) = \mathbf{u}_2(0,1)$. *Although it is possible for both of these constraints to be satisfied simultaneously, this does impose another constraint,* $\mathbf{u}_2(0,1) = 0$, *which is not implied by either* $\mathbf{u} \in \mathcal{D}_1$ *or* $\mathbf{u} \in \mathcal{D}_2$ *individually. Similarly, any* $\mathbf{u} \in \mathcal{D}$ *must also satisfy* $\mathbf{u}_2(1,1) = 0$, *as well as* $\mathbf{u}_1(1,0) = \mathbf{u}_1(1,1) = 0$ *– all constraints that need not be satisfied for* $\mathbf{u} \in \mathcal{D}_1$ *or* $\mathbf{u} \in \mathcal{D}_2$. *As a result of these auxiliary constraints, any* $\mathbf{u} \in \mathcal{D}$ *must satisfy* $\int_0^1 \int_0^1 \partial_x \partial_y \mathbf{u}(x,y) dy dx = \mathbf{u}(1,1) - \mathbf{u}(0,1) - \mathbf{u}(1,0) + \mathbf{u}(0,0) = 0$, *meaning that* $\partial_x \partial_y : \mathcal{D} \to L_2^2[[0,1]^2]$ *is not surjective onto* L_2, *and thus not right-invertible.*

This example illustrates a 2D case where the domains $\mathcal{D}_i$ impose different constraints at the corners of the spatial domain. To avoid such inconsistencies, we propose the following constraint on the $\mathcal{D}_i$ in terms of the corresponding parameters $B^i_{j,k}, C^i_{j,k}$.

Definition 2 (Consistent Boundary Conditions) *For each* $i \in \{1:N\}$, *let* $\{[a_i,b_i], \delta_i, B^i_{j,k}, C^i_{j,k}\}$ *define* $\mathcal{D}_i := \hat{\mathcal{D}}[i]$, *where* $\mathcal{D} \subseteq S_2^{\delta_i,n}[a_i,b_i]$ *is of the form in (4.2) and admissible as per Defn. 1. Define* $[H^i_a]_{j,k} := B^i_{j,k}$, $[H^i_b]_{j,k} := C^i_{j,k}$ *and associated* $K^i := \left(H^i_a + H^i_b \boldsymbol{Q}(b_i - a_i)\right)^{-1} H^i_b \in \mathbb{R}^{n\delta_i \times n\delta_i}$ *for* $\boldsymbol{Q}$ *as in Defn. 1, and decompose the matrix* K^i *as*

$$K^i = \begin{bmatrix} K^i_{1,1} & \cdots & K^i_{1,\delta_i} \\ \vdots & \ddots & \vdots \\ K^i_{\delta_i,1} & \cdots & K^i_{\delta_i,\delta_i} \end{bmatrix}, \qquad \text{where} \qquad K^i_{k,\ell} \in \mathbb{R}^{n\times n},\ \forall k,\ell \in \{1:\delta_i\}. \tag{4.4}$$

We say that the $\mathcal{D}_i$ *are* ***consistent*** *if, for every* $i,j \in \{1:N\}$, *we have* $K^i_{k,p} K^j_{\ell,q} = K^j_{\ell,q} K^i_{k,p}$ *for all* $k,p \in \{1:\delta_i\}$ *and* $\ell,q \in \{1:\delta_j\}$.

Note: *For* $\delta_i = \delta_j = 1$, *commutability of* K^i *and* K^j *is equivalent to consistency. In any other case, however, commutability of* K^i *and* K^j *is neither necessary nor sufficient for consistency.*

We now apply this consistency definition to Example 1.

Example 2 *To illustrate the consistency condition from Defn. 2, consider again the 2D domain* $\mathcal{D} = \mathcal{D}_1 \cap \mathcal{D}_2$ *from Ex. 1. In this case,* $\delta_1 = \delta_2 = 1$, $n = 2$, *and* $\mathcal{D}_1$ *is defined by* $B^1 = I_2$ *and* $C^1 = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$, *and* $\mathcal{D}_2$ *is defined by* $B^2 = I_2$ *and* $C^2 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$. *The corresponding matrices* K^i *in Defn. 2 are given by* $K^1 = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$ *and* $K^2 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, *which do not commute. Thus,* $\mathcal{D}_1$ *and* $\mathcal{D}_2$ *are not consistent.*

Given this definition of consistency, the following lemma shows that $D^\delta : \bigcap_{i=1}^N \mathcal{D}_i \to L_2^n$ is right-invertible only if the domains $\mathcal{D}_i$ are consistent in the sense of Defn. 2. For simplicity, we consider only 2 spatial variables, extending this result to ND in Lem. 7.

Lemma 5 *Let* $\mathcal{D}_1$, $\mathcal{D}_2$ *(of the form in Eqn. (4.2) with* $d = \delta_1$ *and* $d = \delta_2$, *respectively) be admissible as per Defn. 1, and define* $\mathcal{D} = \mathcal{D}_1 \cap \mathcal{D}_2$. *If there exists an operator* $\mathcal{T} : L_2^n \to \mathcal{D}$ *such that* $\partial_x^{\delta_1} \partial_y^{\delta_2} \mathcal{T}\mathbf{v} = \mathbf{v}$ *for all* $\mathbf{v} \in L_2^n$, *then* $\mathcal{D}_1$ *and* $\mathcal{D}_2$ *are consistent.*

Proof We provide an outline of the proof here, referring to Lem. 22 in Appx. A for a full proof. To reduce notation, we assume $\delta_1 = \delta_2 = d$, $a_1 = a_2 = 0$, $b_1 = b_2 = 1$ and define $\Omega = [0,1]^2$.

Suppose there exists an operator $\mathcal{T} : L_2^n[[0,1]^2] \to \mathcal{D}$ such that $\partial_x^d \partial_y^d \mathcal{T}\mathbf{v} = \mathbf{v}$ for all $\mathbf{v} \in L_2^n[[0,1]^2]$. For any $\mathbf{v} \in L_2^n[\Omega]$, let $\mathbf{u} = \mathcal{T}\mathbf{v}$. Then, $\mathbf{v} = \partial_x^d \partial_y^d \mathbf{u}$ and $\mathbf{u} \in \mathcal{D} = \mathcal{D}_1 \cap \mathcal{D}_2$. It follows that also $\partial_y^d \mathbf{u} \in \mathcal{D}_1$ and $\partial_x^d \mathbf{u} \in \mathcal{D}_2$ (see also Lem. 20 in Appx. A). Using the fact that $\mathbf{u} \in \mathcal{D}_2$, and $\partial_y^d \mathbf{u} \in \mathcal{D}_1$, we can then show that at the corner $(x,y) = (0,0)$, each lower-order derivative of $\mathbf{u}$ can be expressed as

$$\big(\partial_x^{k-1}\partial_y^{\ell-1}\mathbf{u}\big)(0,0) = \int_0^1 \int_0^1 \big(K^2_{(\ell,:)}\mathbf{e}_d(1-y)\big)\big(K^1_{(k,:)}\mathbf{e}_d(1-x)\big)\mathbf{v}(x,y)\,dy\,dx, \qquad \forall k,\ell \in \{1:d\},$$

where $\mathbf{e}_d$ is as in Cor. 2, and where $K^1_{(k,:)} := \begin{bmatrix} K^1_{k,1} & \cdots & K^1_{k,\delta_1} \end{bmatrix}$. On the other hand, using the fact that $\mathbf{u} \in \mathcal{D}_1$ and $\partial_x^d \mathbf{u} \in \mathcal{D}_2$, we can similarly express

$$\big(\partial_x^{k-1}\partial_y^{\ell-1}\mathbf{u}\big)(0,0) = \int_0^1 \int_0^1 \big(K^1_{(k,:)}\mathbf{e}_d(1-x)\big)\big(K^2_{(\ell,:)}\mathbf{e}_d(1-y)\big)\mathbf{v}(x,y)\,dy\,dx, \qquad \forall k,\ell \in \{1:d\}.$$

Since $\mathbf{u} \in \mathcal{D} \subseteq S_2^{(d,d),n}[\Omega]$ and $k,\ell \le d$, the derivative $\partial_x^{k-1}\partial_y^{\ell-1}\mathbf{u}$ is continuous, and therefore the two expressions for $(\partial_x^{k-1}\partial_y^{\ell-1}\mathbf{u})(0,0)$ must be identical for all $\mathbf{v} \in L_2^n[[0,1]^2]$. It follows that

$$\big(K^2_{(\ell,:)}\mathbf{e}_d\big)\big(K^1_{(k,:)}\mathbf{e}_d\big) - \big(K^1_{(k,:)}\mathbf{e}_d\big)\big(K^2_{(\ell,:)}\mathbf{e}_d\big) = 0 \quad \in L_2^n\big[[0,1]^2\big].$$

Here, since $K^1_{(k,:)}\mathbf{e}_d$ and $K^2_{(\ell,:)}\mathbf{e}_d$ are matrix-valued polynomials with coefficients defined by $K^1_{k,p}$ and $K^2_{\ell,q}$, respectively, this implies that $K^1_{k,p}K^2_{\ell,q} = K^2_{\ell,q}K^1_{k,p}$ for all $k,p,\ell,q \in \{1:d\}$. □

While Lem. 5 only shows that consistency of the $\mathcal{D}_i$ is necessary for $\partial_{s_i}^{\delta_i}\partial_{s_j}^{\delta_j} : \mathcal{D}_i \cap \mathcal{D}_j \to L_2^n$ to be right-invertible, this condition is also sufficient. To establish this result, we first show that if $\mathcal{D}_i$ are consistent, then the associated $\mathcal{T}_i$ commute – i.e. $\mathcal{T}_i\mathcal{T}_j = \mathcal{T}_j\mathcal{T}_i$.

Lemma 6 *For each $i \in \{1:N\}$, let $\mathcal{D}_i$ and associated $\mathcal{T}_i : L_2^n \to \mathcal{D}_i$ be as in Cor. 3. Then, for any $i,j \in \{1:N\}$, $\mathcal{T}_i\mathcal{T}_j\mathbf{v} = \mathcal{T}_j\mathcal{T}_i\mathbf{v}$ for all $\mathbf{v} \in L_2$ if and only $\mathcal{D}_i$ and $\mathcal{D}_j$ are consistent in the sense of Defn. 2.*

Proof A formal proof is given in Lem. 24 in Appx. A. This proof uses the definition

$$(\mathcal{T}_i\mathbf{v})(s) := -\int_{a_i}^{b_i} \Big[\mathbf{e}_1(s_i - a_i)^T K^i \mathbf{e}_{\delta_i}(b_i - \theta_i)\,\mathbf{v}(s)|_{s_i=\theta_i}\Big]\,d\theta_i + \int_{a_i}^{s_i} \frac{(s_i-\theta_i)^{\delta_i-1}}{(\delta_i-1)!}\mathbf{v}(s)|_{s_i=\theta_i}\,d\theta_i,$$

for $\mathbf{v} \in L_2[\Omega]$, and where $\mathbf{e}_1, \mathbf{e}_{\delta_i}$ are as in Cor. 2 (letting $d = \delta_i$). From this definition, it follows that $\mathcal{T}_i$ and $\mathcal{T}_j$ commute if and only if $\mathbf{e}_1(s_i-a_i)^T K^i \mathbf{e}_{\delta_i}(b_i-\theta_i)$ and $\mathbf{e}_1(s_j-a_j)^T K^j \mathbf{e}_{\delta_j}(b_j-\theta_j)$ commute for all $s_i,\theta_i \in [a_i,b_i]$ and $s_j,\theta_j \in [a_j,b_j]$. Since the $\mathbf{e}_1^T K^i \mathbf{e}_{\delta_i}$ are matrix-valued polynomial with coefficients $K^i_{p,\ell}$, we find that $\mathbf{e}_1(x)^T K^i \mathbf{e}_{\delta_i}(\eta)\mathbf{e}_1^T(y)K^j\mathbf{e}_{\delta_j}(\zeta) = \mathbf{e}_1^T(y)K^j\mathbf{e}_{\delta_j}(\zeta)\mathbf{e}_1(x)^T K^i\mathbf{e}_{\delta_i}(\eta)$ for all x,y,η,ζ if and only if $K^i_{k,p}K^j_{\ell,q} = K^j_{\ell,q}K^i_{k,p}$ for all k,p,ℓ,q, and thus $\mathcal{D}_i$ and $\mathcal{D}_j$ are consistent. □

Lem. 6 shows that consistency of the $\mathcal{D}_i$ is necessary and sufficient for the operators $\mathcal{T}_i : L_2^n \to \mathcal{D}_i$ and $\mathcal{T}_j : L_2^n \to \mathcal{D}_j$ for $i,j \in \{1:N\}$ to commute. Commutability then readily implies that the image of

$\mathcal{T} = \prod_i \mathcal{T}_i$ is $\mathcal{D} = \bigcap_i \mathcal{D}_i$, since for any $\mathbf{v}$, we have $\mathcal{T}\mathbf{v} = \mathcal{T}_i \prod_{j\neq i} \mathcal{T}_j \mathbf{v} \in \mathcal{D}_i$ for every i. This is formalized in the following lemma which shows that $D^\delta : \mathcal{D} \to L_2^n$ is right-invertible if and only if the domains $\mathcal{D}_i$ are consistent.

Lemma 7 *For each $i \in \{1:N\}$, let $\mathcal{D}_i \subseteq S_2^{\delta_i \cdot e_i, n}$ as in Cor. 3 be admissible as per Defn. 1, and let $\mathcal{D} := \bigcap_{i=1}^N \mathcal{D}_i$. Then $D^\delta : \mathcal{D} \to L_2^n$ is right-invertible if and only if the $\mathcal{D}_i$ are consistent as per Defn. 2. Furthermore, such a right-inverse is then given by $\mathcal{T} := \prod_{i=1}^N \mathcal{T}_i$ for $\mathcal{T}_i$ as in Cor. 3.*

Proof For sufficiency, suppose the domains $\mathcal{D}_i$ are consistent as per Defn. 2. Then, by Lem. 6, the operators $\mathcal{T}_i : L_2^n \to \mathcal{D}_i$ defined in Cor. 3 commute. Define $\mathcal{T} := \prod_{i=1}^N \mathcal{T}_i$. Since the operators $\mathcal{T}_i$ commute, it follows that $\mathcal{T}\mathbf{v} = \mathcal{T}_i \prod_{j\neq i} \mathcal{T}_j \mathbf{v} \in \mathcal{D}_i$ for all i, and therefore $\mathcal{T} : L_2^n \to \mathcal{D}$. Furthermore, by Cor. 4, we have $D^\delta \mathcal{T}\mathbf{v} = \mathbf{v}$ for all $\mathbf{v} \in L_2^n$. Thus, $\mathcal{T}$ defines a right-inverse to $D^\delta : \mathcal{D} \to L_2^n$.

For necessity, suppose that $D^\delta : \mathcal{D} \to L_2^n$ is right-invertible, so that there exists $\mathcal{T} : L_2^n \to \mathcal{D}$ such that $D^\delta \mathcal{T} = I_n$. To prove that the domains $\mathcal{D}_i$ are consistent, we first show that for every $i \neq j$, the operator $\partial_{s_i}^{\delta_i} \partial_{s_j}^{\delta_j} : \mathcal{D}_i \cap \mathcal{D}_j \to L_2^n$ is right-invertible. To this end, note that the operators $\partial_{s_i}^{\delta_i}$ commute on $\mathcal{D} \subseteq S_2^{\delta,n}$, so that $D^\delta \mathbf{u} = \partial_{s_i}^{\delta_i} \partial_{s_j}^{\delta_j} \prod_{k\notin\{i,j\}} \partial_{s_k}^{\delta_k} \mathbf{u}$ for $\mathbf{u} \in \mathcal{D}$. Here, by Lem. 20 in Appx. A, we have for every $i \neq j$ that $\prod_{k\notin\{i,j\}} \partial_{s_k}^{\delta_k} \mathbf{u} \in \mathcal{D}_i \cap \mathcal{D}_j$ for all $\mathbf{u} \in \mathcal{D}$. Defining then $\tilde{\mathcal{T}}_{i,j} := (\prod_{k\notin\{i,j\}} \partial_{s_k}^{\delta_k}) \mathcal{T}$, it follows from the fact that $\mathcal{T} : L_2^n \to \mathcal{D}$ that $\tilde{\mathcal{T}}_{i,j} : L_2^n \to \mathcal{D}_i \cap \mathcal{D}_j$. Furthermore, $\tilde{\mathcal{T}}_{i,j}$ also satisfies $\partial_{s_i}^{\delta_i} \partial_{s_j}^{\delta_j} \tilde{\mathcal{T}}_{i,j} \mathbf{v} = \partial_{s_i}^{\delta_i} \partial_{s_j}^{\delta_j} \left[\prod_{k\notin\{i,j\}} \partial_{s_k}^{\delta_k} \mathcal{T} \right] \mathbf{v} = D^\delta \mathcal{T}\mathbf{v} = \mathbf{v}$ for all $\mathbf{v} \in L_2^n$, and thus defines a right-inverse to $\partial_{s_i}^{\delta_i} \partial_{s_j}^{\delta_j} : \mathcal{D}_i \cap \mathcal{D}_j \to L_2^n$. Since this holds for all $i \neq j$, it follows by Lem. 5 that the domains $\mathcal{D}_i$ are consistent. □

Lem. 7 shows that if the $\mathcal{D}_i$ are admissible as per Defn. 1 and we define $\mathcal{D} := \bigcap_i \mathcal{D}_i$, then consistency of $\mathcal{D}_i$ is necessary and sufficient for $D^\delta : \mathcal{D} \to L_2^n$ to be right-invertible, with right-inverse $\mathcal{T} := \prod_{i=1}^N \mathcal{T}_i$. Furthermore recall that Cor. 4 establishes that $\mathcal{T}$ is a left inverse of D^δ — implying that $D^\delta : \mathcal{D} \to L_2$ is invertible if and only if the domains $\mathcal{D}_i$ are consistent. This implies a bijection between the domain $\mathcal{D}$ and the Hilbert space L_2. In the following subsection, we will apply this bijection to the state of a class of linear coupled multivariate PDEs in order to construct an equivalent evolution equation on the Hilbert space L_2, where this evolution equation is not constrained by boundary conditions.

4.3. *A PIE Representation of Multivariate PDEs*

Having established necessary and sufficient conditions for $D^\delta : \bigcap_i \mathcal{D}_i \to L_2^n$ to be invertible, and having obtained an explicit representation of this inverse in terms of the composition of integral operators, we now apply this result to a class of linear PDEs in order to construct an alternative representation of the evolution of the the state of such PDEs. Specifically, we consider a class of PDEs defined by states $\mathbf{u}(t) \in \mathcal{D} := \bigcap_{i=1}^N \mathcal{D}_i \subseteq S_2^{\delta,n}$, and map such states to an associated *fundamental state* defined as $\mathbf{v}(t) = D^\delta \mathbf{u}(t) \in L_2^n$. We then define an evolution equation for $\mathbf{v}(t)$, any solution of which maps back to a solution of the PDE as $\mathbf{u}(t) = \mathcal{T}\mathbf{v}(t)$.

Consider the class of linear coupled multivariate PDEs parameterized by $\boldsymbol{\delta}, \boldsymbol{A}_\alpha, \mathcal{D}_i$ as

$$\partial_t \mathbf{u}(t,s) = \sum_{\vec{0}\leq\alpha\leq\delta} \boldsymbol{A}_\alpha(s) D^\alpha \mathbf{u}(t,s), \qquad \mathbf{u}(t) \in \bigcap_{i=1}^N \mathcal{D}_i, \qquad s \in \Omega := \prod_{i=1}^N [a_i, b_i], \tag{4.5}$$

where $\boldsymbol{A}_\alpha \in \mathbb{R}[s]$, and where the $\mathcal{D}_i$ are, lifted univariate and admissible domains as per Defn. 1, and are consistent as per Defn. 2. We formally define a solution to a PDE of the form in (4.5) as follows.

Definition 3 (Solution to the PDE) *For given $\boldsymbol{A}_\alpha \in \mathbb{R}^{n\times n}[s]$ and admissible and consistent $\mathcal{D}_i$, we say that $\mathbf{u}(t) \in \mathcal{D} := \bigcap_{i=1}^N \mathcal{D}_i$ solves the PDE defined by $\{\boldsymbol{A}_\alpha, \mathcal{D}_i\}$ with initial condition $\mathbf{u}_0 \in \mathcal{D}$ if $\mathbf{u}(t)$ is Frechét differentiable, $\mathbf{u}(0) = \mathbf{u}_0$, and $\mathbf{u}(t)$ satisfies (4.5) for all $t \geq 0$.*

Having specified a class of multivariate PDEs, we now map solutions of such PDEs to solutions of an associated evolution equation on the fundamental state $\mathbf{v}(t) := D^\delta \mathbf{u}(t)$. To define this evolution, however, we need a mapping from $\mathbf{v}(t)$ to $\mathbf{u}(t)$. This is achieved in the following theorem, by applying Cor. 2 to each of the univariate differential operators in $D^\alpha = \prod \partial_{s_i}^{\alpha_i}$, and combining the results using the embedding of univariate operators in multivariate space. To state this theorem, recall that $\mathrm{M}[\boldsymbol{A}]$ is the multiplier operator with multiplier $\boldsymbol{A}$.

Theorem 8 (Multivariate Extension of Cor. 2) *For $N \in \mathbb{N}$, $\Omega = \prod_{i=1}^N [a_i, b_i]$, and $\delta \in \mathbb{N}_0^N$, let $\mathcal{D} := \bigcap_{i=1}^N \mathcal{D}_i \subseteq S_2^{\delta,n}[\Omega]$ with $\mathcal{D}_i \subseteq S_2^{\delta_i \cdot \mathrm{e}_i, n}[\Omega]$ admissible and consistent and define $\mathcal{H} := \sum_{\vec{0}\leq\alpha\leq\delta} \mathrm{M}[\boldsymbol{A}_\alpha] D^\alpha$.*

Let $\mathcal{T}_i, \mathcal{A}_{i,j}$ be defined as in Cor. 2 and extended to the multivariate space as in Cor. 3 (letting $\mathcal{T} \mapsto \mathcal{T}_i$ and $\mathcal{A} \mapsto \mathcal{A}_{i,j}$), for $d = \delta_i$, $\mathcal{D} = \mathcal{D}_i$, and $\{\boldsymbol{H}_k\}$ where $\boldsymbol{H}_k = \begin{cases} I_n, & k = j, \\ 0, & k \neq j, \end{cases}$ for each $k \in \{0 : d\}$. Then for $\mathcal{T} := \prod_{i=1}^N \mathcal{T}_i$ and $\mathcal{A} := \sum_{\vec{0}\leq\alpha\leq\delta} \big(\mathrm{M}[\boldsymbol{A}_\alpha] \prod_{i=1}^N \mathcal{A}_{i,\alpha_i}\big)$, the following hold:

1. *For all $\mathbf{u} \in \mathcal{D}$, we have $D^\delta \mathbf{u} \in L_2^n[\Omega]$, $\mathbf{u} = \mathcal{T} D^\delta \mathbf{u}$, and $\mathcal{H}\mathbf{u} = \mathcal{A} D^\delta \mathbf{u}$.*
2. *For all $\mathbf{v} \in L_2^n[\Omega]$, we have $\mathcal{T}\mathbf{v} \in \mathcal{D}$, $\mathbf{v} = D^\delta \mathcal{T}\mathbf{v}$, and $\mathcal{A}\mathbf{v} = \mathcal{H}\mathcal{T}\mathbf{v}$.*

Proof For the first statement, fix arbitrary $\mathbf{u} \in \mathcal{D} \subseteq S_2^{\delta,n}$. Then $D^\delta \mathbf{u} \in L_2^n$ by definition of $S_2^{\delta,n}$, and $\mathbf{u} = \mathcal{T} D^\delta \mathbf{u}$ by Cor. 4. Finally, by Cor. 3, we have $\mathcal{A}_{i,j} \partial_{s_i}^{\delta_i} \mathbf{u} = \partial_{s_i}^j \mathbf{u}$ for all j and i, and therefore

$$\begin{aligned}
\mathcal{A} D^\delta \mathbf{u} &= \sum_{\vec{0}\leq\alpha\leq\delta} \mathrm{M}[\boldsymbol{A}_\alpha] \left(\prod_{i=1}^N \mathcal{A}_{i,\alpha_i}\right) \left(\prod_{i=1}^N \partial_{s_i}^{\delta_i}\right) \mathbf{u} \\
&= \sum_{\vec{0}\leq\alpha\leq\delta} \mathrm{M}[\boldsymbol{A}_\alpha] \left(\prod_{i=1}^N (\mathcal{A}_{i,\alpha_i} \partial_{s_i}^{\delta_i})\right) \mathbf{u} = \sum_{\vec{0}\leq\alpha\leq\delta} \mathrm{M}[\boldsymbol{A}_\alpha] \left(\prod_{i=1}^N \partial_{s_i}^{\alpha_i}\right) \mathbf{u} = \mathcal{H}\mathbf{u},
\end{aligned}$$

where we remark that $\mathcal{A}_{i,\alpha_i}$ and $\partial_{s_j}^{\delta_j}$ commute for $i \neq j$ by Lem. 25 in Appx. A.

For the second statement, fix arbitrary $\mathbf{v} \in L_2^n$. Then, by Cor. 4 and Lem. 7, we have $D^\delta \mathcal{T}\mathbf{v} = \mathbf{v}$ and $\mathcal{T}\mathbf{v} \in \mathcal{D}$. Finally, by Cor. 2, we have $\mathcal{A}_{i,j}\mathbf{v} = \partial_{s_i}^j \mathcal{T}_i \mathbf{v}$, and therefore

$$\begin{aligned}
\mathcal{A}\mathbf{v} &= \sum_{\vec{0}\leq\alpha\leq\delta} \mathrm{M}[\boldsymbol{A}_\alpha] \left(\prod_{i=1}^N \mathcal{A}_{i,\alpha_i}\right) \mathbf{v} \\
&= \sum_{\vec{0}\leq\alpha\leq\delta} \mathrm{M}[\boldsymbol{A}_\alpha] \left(\prod_{i=1}^N \partial_{s_i}^{\alpha_i} \mathcal{T}_i\right) \mathbf{v} = \sum_{\vec{0}\leq\alpha\leq\delta} \mathrm{M}[\boldsymbol{A}_\alpha] \left(\prod_{i=1}^N \partial_{s_i}^{\alpha_i}\right) \left(\prod_{i=1}^N \mathcal{T}_i\right) \mathbf{v} = \mathcal{H}\mathcal{T}\mathbf{v},
\end{aligned}$$

where we remark that $\mathcal{T}_i$ and $\partial_{s_j}^{\alpha_j}$ commute for $i \neq j$ by Lem. 25 in Appx. A. □

We may now readily apply the results of Thm. 8 to the PDE defined in Eqn. (4.5) by defining the fundamental state as $\mathbf{v}(t) := D^{\delta}\mathbf{u}(t)$ and using the mappings $\mathbf{u}(t) = \mathcal{T}\mathbf{v}(t)$ and $\sum_{\vec{0}\leq\alpha\leq\delta} \mathrm{M}[\boldsymbol{A}_\alpha]D^{\alpha}\mathbf{u}(t) = \mathcal{A}\mathbf{v}(t)$. Doing so, we obtain an evolution equation for $\mathbf{v}(t)$ as

$$\partial_t \mathcal{T}\mathbf{v}(t) = \mathcal{A}\mathbf{v}(t), \qquad \mathbf{v}(t) \in L_2^n[\Omega]. \tag{4.6}$$

We refer to a system of this form as a Partial Integral Equation (PIE), being parameterized by integral operators $\{\mathcal{T},\mathcal{A}\}$ with polynomial semi-separable kernels. We define solutions to the PIE as follows.

Definition 4 (Solution to the PIE) *For a given initial state $\mathbf{v}_0 \in L_2^n[\Omega]$, we say that $\mathbf{v}(t) \in L_2^n$ solves the PIE defined by $\{\mathcal{T},\mathcal{A}\}$ if $\mathbf{v}(t)$ is Frechét differentiable, $\mathbf{v}(0) = \mathbf{v}_0$, and $\mathbf{v}(t)$ satisfies (4.6) for all $t \geq 0$.*

For a given PDE as in (4.5), defining $\{\mathcal{T},\mathcal{A}\}$ as in Thm. 8, the following theorem shows that any solution to the PDE can be mapped to an associated solution to the PIE (4.6), and vice versa, so that the PIE offers an equivalent representation of the PDE.

Theorem 9 *For $N \in \mathbb{N}$ and $\boldsymbol{\delta} \in \mathbb{N}_0^N$, let $\mathcal{D}_i \subseteq S_2^{\delta_i\cdot \mathrm{e}_i,n}$ be admissible and consistent, and let $\boldsymbol{A}_\alpha \in \mathbb{R}^{n\times n}[s]$, for each $i \in \{1:N\}$ and $\vec{0} \leq \alpha \leq \delta$. Define associated operators $\{\mathcal{T},\mathcal{A}\}$ as in Thm. 8. Then the following hold:*

1. *If $\mathbf{u}$ solves the PDE defined by $\{\boldsymbol{A}_\alpha, \mathcal{D}_i\}$ with initial value $\mathbf{u}_0 \in \mathcal{D} := \bigcap_{i=1}^N \mathcal{D}_i$, then $\mathbf{v} := D^{\delta}\mathbf{u}$ is a solution to the PIE defined by $\{\mathcal{T},\mathcal{A}\}$ with initial value $\mathbf{v}_0 := D^{\delta}\mathbf{u}_0$, and $\mathbf{u} = \mathcal{T}\mathbf{v}$.*
2. *If $\mathbf{v}$ solves the PIE defined by $\{\mathcal{T},\mathcal{A}\}$ with initial value $\mathbf{v}_0 \in L_2^n[\Omega]$, then $\mathbf{u} := \mathcal{T}\mathbf{v}$ is a solution to the PDE defined by $\{\boldsymbol{A}_\alpha, \mathcal{D}_i\}$ with initial value $\mathbf{u}_0 := \mathcal{T}\mathbf{v}_0$, and $\mathbf{v} = D^{\delta}\mathbf{u}$.*

Proof For the first statement, fix arbitrary $\mathbf{u}_0 \in \mathcal{D}$, and let $\mathbf{u}$ be an associated solution to the PDE (4.5). Then $\mathbf{u}(t) \in \mathcal{D} \subseteq S_2^{\delta,n}$ for $t \geq 0$. Let $\mathbf{v}(t) := D^{\delta}\mathbf{u}(t) \in L_2^n$ for all $t \geq 0$ and $\mathbf{v}_0 := D^{\delta}\mathbf{u}_0 \in L_2^n$. Clearly, then, $\mathbf{u}(0) = \mathbf{u}_0$ implies $\mathbf{v}(0) = \mathbf{v}_0$. In addition, by Thm. 8, and by definition of the operators $\{\mathcal{T},\mathcal{A}\}$, we have $\mathbf{u}(t) = \mathcal{T}D^{\delta}\mathbf{u}(t) = \mathcal{T}\mathbf{v}(t)$ and $\sum_{\vec{0}\leq\alpha\leq\delta} \boldsymbol{A}_\alpha(s)D^{\alpha}\mathbf{u}(t,s) = \mathcal{A}\mathbf{v}(t,s)$. It follows that

$$\partial_t \mathcal{T}\mathbf{v}(t,s) - \mathcal{A}\mathbf{v}(t,s) = \partial_t \mathbf{u}(t,s) - \sum_{\vec{0}\leq\alpha\leq\delta} \boldsymbol{A}_\alpha(s)D^{\alpha}\mathbf{u}(t,s). \tag{$\star$}$$

Since $\mathbf{u}(t)$ satisfies the dynamics in (4.5), it follows that $\mathbf{v}(t)$ satisfies (4.6), and the first statement holds.

For the second statement, fix arbitrary $\mathbf{v}_0 \in L_2^n[\Omega]$, and let $\mathbf{v}$ be a corresponding solution to the PIE defined by $\{\mathcal{T},\mathcal{A}\}$. Let $\mathbf{u}(t) := \mathcal{T}\mathbf{v}(t)$. Then, by Thm. 8, $\mathbf{u}(t) \in \mathcal{D}$ for all $t \geq 0$, and $\sum_{\vec{0}\leq\alpha\leq\delta} \boldsymbol{A}_\alpha(s)D^{\alpha}\mathbf{u}(t,s) = \mathcal{A}\mathbf{v}(t,s)$. By the proof of the first statement, it follows that the equality in ($\star$) holds. Since $\mathbf{v}(t)$ satisfies (4.6), $\mathbf{u}(t)$ satisfies (4.5), and thus the second statement holds as well. □

By Thm. 9, for any PDE of the form in (4.5) with admissible and consistent $\mathcal{D}_i$, we can define an equivalent representation as a PIE of the form in (4.6), with any solution $\mathbf{u}$ to the PDE admitting a solution $D^{\delta}\mathbf{u}$ to the PIE, and any solution $\mathbf{v}$ to the PIE admitting a solution $\mathcal{T}\mathbf{v}$ to the PDE. Using this equivalence, stability properties of the PDE can be inferred from properties of the associated PIE. Verification of such stability properties using the PIE representation is significantly easier, since the fundamental state $\mathbf{v}(t)$ of the PIE lies in the Hilbert space L_2^n, obviating the need for ad hoc manipulations to account for boundary conditions. Moreover, as we will show in the following section,

the operators $\mathcal{T},\mathcal{A}$ defining the PIE representation belong to a $*$-algebra of Partial Integral (PI) operators, being closed under composition and adjoint operations. These properties allow e.g. stability in the PIE representation to be tested by solving a convex optimization program, as we will show in Section 6.

Of course, Thm. 9 only holds if the domains $\mathcal{D}_i$ defining $\mathcal{D} := \bigcap_{i=1}^N \mathcal{D}_i$ are admissible and consistent. However, this assumption imposes only mild well-posedness constraints on the boundary conditions: ensuring that the differential operator $D^\delta : \mathcal{D} \to L_2^n[\Omega]$ is indeed bijective, rather than e.g. mapping to a proper subspace of $L_2^n[\Omega]$. In practice, most common boundary conditions, including Dirichlet, Robin, and mixed boundary conditions satisfy the admissibility constraints from Defn. 1. Regarding the consistency conditions from Defn. 2, we note that these conditions are trivially satisfied in the scalar-valued case, $n = 1$. More generally, for $n > 1$, the boundary conditions are also consistent if the matrices $B^i_{j,k}, C^i_{j,k}$ in Defn. 1 are all diagonal – i.e. if there is no coupling between state variables in the boundary conditions. The following corollary – a reformulation of Cor. 1 from the start of this section – explicitly constructs the operators defining the PIE representation of the *N*D heat equation, with mixed Dirichlet and Neumann boundary conditions.

Corollary 10 *Let $\mathcal{D} = \bigcap_{i=1}^N \mathcal{D}_i$. Then $\mathbf{u} = \mathcal{T}D^{(2,\dots,2)}\mathbf{u}$, $\mathbf{v} = D^{(2,\dots,2)}\mathcal{T}\mathbf{v}$ and $\mathcal{T}\mathbf{v} \in \mathcal{D}$ for all $\mathbf{v} \in L_2$ and $\mathbf{u} \in \mathcal{D}$ where $(\mathcal{T}\mathbf{v})(s) = \int_\Omega \boldsymbol{G}(s,\theta)\mathbf{v}(\theta)d\theta$ with $\boldsymbol{G}(s,\theta) = \prod_{i=1}^N \boldsymbol{G}_i(s_i,\theta_i)$ and*

$$\boldsymbol{G}_i(s,\theta) = \begin{cases} \boldsymbol{h}_i(s,\theta) + (s-\theta), & \theta \le s, \\ \boldsymbol{h}_i(s,\theta), & \theta > s, \end{cases} \qquad \text{where} \qquad \boldsymbol{h}_i(s,\theta) = \begin{cases} \frac{-(s-a_i)(b_i-\theta)}{b_i-a_i}, & \mathcal{D}_i = \mathcal{D}_{\mathrm{D}}, \\ a_i - s, & \mathcal{D}_i = \mathcal{D}_{\mathrm{N1}}, \\ s - b_i, & \mathcal{D}_i = \mathcal{D}_{\mathrm{N2}}, \end{cases}$$

for $\mathcal{D}_{\mathrm{D}}$, $\mathcal{D}_{\mathrm{N1}}$ and $\mathcal{D}_{\mathrm{N2}}$ as in (4.1), and using the notation $\bigcap_i \mathcal{D}_i := \bigcap_i \hat{\mathcal{D}}[i]$ as defined in Subsection 2.1. Moreover, $\mathbf{u}(t) \in \mathcal{D}$ satisfies the PDE $\partial_t \mathbf{u}(t) = \nabla^2 \mathbf{u}(t) = \sum_{i=1}^N \partial_{s_i}^2 \mathbf{u}(t)$ if and only if $\mathbf{v} = D^{(2,\dots,2)}\mathbf{u} \in L_2^n$ satisfies the PIE $\partial_t \mathcal{T}\mathbf{v}(t) = \mathcal{A}\mathbf{v}(t)$, where $\mathcal{A} = \sum_{j=1}^N \mathcal{A}_j$ for $(\mathcal{A}_j\mathbf{v})(s) := \prod_{i\neq j}(\int_{[a_i,b_i]} \boldsymbol{G}_i(s_i,\theta_i))\mathbf{v}(\theta)d\theta$.

Proof First, we define the parameters of the $\mathcal{D}_i$ used in (4.2) as

$$B^i_{0,0} = C^i_{1,0} = I_n, \text{ if } \mathcal{D}_i = \mathcal{D}_{\mathrm{D}}, \qquad B^i_{0,0} = C^i_{1,1} = I_n, \text{ if } \mathcal{D}_i = \mathcal{D}_{\mathrm{N1}},$$
$$B^i_{0,1} = C^i_{1,0} = I_n, \text{ if } \mathcal{D}_i = \mathcal{D}_{\mathrm{N2}},$$

with $B^i_{j,k} = C^i_{j,k} = 0_n$ for all other $j,k \in \{0,1\}$. Clearly, each domain $\mathcal{D}_i$ is admissible as per Defn. 1. Now, define $K^i = (H^i_a + H^i_b \boldsymbol{Q}(b_i - a_i))^{-1} H^i_b$ for each $i \in \{1:N\}$ as

$$K^i = \begin{bmatrix} 0_n & 0_n \\ \frac{1}{b_i-a_i} I_n & 0_n \end{bmatrix}, \quad \mathcal{D}_i = \mathcal{D}_{\mathrm{D}}, \quad K^i = \begin{bmatrix} 0_n & 0_n \\ I_n & 0_n \end{bmatrix}, \quad \mathcal{D}_i = \mathcal{D}_{\mathrm{N1}}, \quad K^i = \begin{bmatrix} I_n & 0_n \\ 0_n & 0_n \end{bmatrix}, \quad \mathcal{D}_i = \mathcal{D}_{\mathrm{N2}}.$$

Here, decomposing $K^i = \left[\begin{smallmatrix} K^i_{1,1} & K^i_{1,2} \\ K^i_{2,1} & K^i_{2,2} \end{smallmatrix}\right]$, the blocks $K^i_{k,p} \in \mathbb{R}^{n\times n}$ and $K^j_{\ell,q} \in \mathbb{R}^{n\times n}$ commute for all $i,j \in \{1:N\}$ and $k,\ell,p,q \in \{1,2\}$, and thus the $\mathcal{D}_i$ are also consistent as per Defn. 2. Constructing parameters $\boldsymbol{G}_i$ as in Lem. 2, we then find that $\boldsymbol{G}_i(s,\theta) = (\boldsymbol{h}_i(s,\theta) + (s-\theta)) \cdot I_n$ for $\theta \le s$ and $\boldsymbol{G}_i(s,\theta) = \boldsymbol{h}_i(s,\theta) \cdot I_n$ for $\theta > s$. By Thm. 8, it follows that $\mathbf{u} = (\prod_{i=1}^N \mathcal{T}_i) D^{\vec{2}}\mathbf{u} = \mathcal{T}D^{\vec{2}}\mathbf{u}$ and $\partial_{s_j}^2 \mathbf{u} = (\prod_{i\neq j} \mathcal{T}_i) D^{\vec{2}}\mathbf{u} = \mathcal{A}_j D^{\vec{2}}\mathbf{u}$ for all $\mathbf{u} \in \mathcal{D}$. Conversely, by that same theorem, $\mathbf{v} = D^{\vec{2}}\mathcal{T}\mathbf{v}$ and $\partial_{s_j}^2 \mathcal{T}\mathbf{v} = \mathcal{A}_j\mathbf{v}$ for all $\mathbf{v} \in L_2^n$. By Thm. 9, it follows that $\mathbf{u}(t) \in \mathcal{D}$ satisfies $\partial_t \mathbf{u}(t) = \sum_{i=1}^N \partial_{s_i}^2 \mathbf{u}(t)$ if and only if $\mathbf{v} = D^{\vec{2}}\mathbf{u} \in L_2^n$ satisfies $\partial_t \mathcal{T}\mathbf{v}(t) = \mathcal{A}\mathbf{v}(t)$. $\square$

Cor. 10 shows how an ND heat equation with mixed Dirichlet-Neumann boundary conditions can be expressed as a PIE, providing an explicit expression for the operators $\{\mathcal{T},\mathcal{A}\}$ defining this PIE representation. The following example applies this result to a 2D example, illustrating how the operators $\{\mathcal{T},\mathcal{A}\}$ are defined for a reaction-diffusion equation on the unit hypercube, with a particular set of Dirichlet- and Neumann-type boundary conditions.

Example 3 *To illustrate the construction of the PIE representation for a multivariate PDE, consider the 2D reaction-diffusion equation*

$$\mathbf{u}_t(t,x,y) = \mathbf{u}_{xx}(t,x,y) + \mathbf{u}_{yy}(t,x,y) + r\mathbf{u}(t,x,y), \qquad \mathbf{u}(t) \in \mathcal{D}, \quad t \geq 0, \tag{4.7}$$

where $r \in \mathbb{R}$, and where the PDE domain $\mathcal{D} = \mathcal{D}_1 \cap \mathcal{D}_2 \subseteq S_2^{(2,2)}[[0,1]^2]$ is defined by

$$\mathcal{D}_1 := \left\{\mathbf{u} \in S_2^{(2,0)}[[0,1]^2] \mid \mathbf{u}(0,y) = \mathbf{u}(1,y) = 0\right\}, \quad \mathcal{D}_2 := \left\{\mathbf{u} \in S_2^{(0,2)}[[0,1]^2] \mid \mathbf{u}(x,0) = \mathbf{u}_y(x,1) = 0\right\}.$$

Then, by Cor. 3, $\mathbf{u} \in \mathcal{D}_1$ if and only if $\mathbf{u} = \mathcal{T}_1\mathbf{v}$ for $\mathbf{v} = \mathbf{u}_{xx} \in L_2[[0,1]^2]$, and where

$$(\mathcal{T}_1\mathbf{v})(x,y) := -\int_0^x \theta(1-x)\mathbf{v}(\theta,y)d\theta - \int_x^1 x(1-\theta)\mathbf{v}(\theta,y)d\theta.$$

Similarly, $\mathbf{u} \in \mathcal{D}_2$ if and only if $\mathbf{u} = \mathcal{T}_2\mathbf{v}$ for $\mathbf{v} = \mathbf{u}_{yy} \in L_2[[0,1]^2]$, where

$$(\mathcal{T}_2\mathbf{v})(x,y) := -\int_0^y \eta\mathbf{v}(x,\eta)d\eta - \int_y^1 y\mathbf{v}(x,\eta)d\eta.$$

By Thm. 8 (or Cor. 10), it follows that for any $\mathbf{u} \in \mathcal{D}$ we have $\mathbf{u}_{yy} = \mathcal{T}_1\mathbf{u}_{xxyy}$, $\mathbf{u}_{xx} = \mathcal{T}_2\mathbf{u}_{xxyy}$, and $\mathbf{u} = \mathcal{T}\mathbf{u}_{xxyy}$, where now

$$\begin{aligned}(\mathcal{T}\mathbf{v})(x,y) := (\mathcal{T}_1(\mathcal{T}_2\mathbf{v}))(x,y) = &\int_0^x \theta(1-x)\left[\int_0^y \eta\,\mathbf{v}(\theta,\eta)\,d\eta + \int_0^y y\mathbf{v}(\theta,\eta)\,d\eta\right]d\theta \\ &+ \int_x^1 x(1-\theta)\left[\int_0^y \eta\,\mathbf{v}(\theta,\eta)\,d\eta + \int_y^1 y\mathbf{v}(\theta,\eta)\,d\eta\right]d\theta.\end{aligned}$$

Conversely, for any $\mathbf{v} \in L_2[[0,1]^2]$, we have $\mathbf{v} = \partial_x^2\mathcal{T}_1\mathbf{v}$, $\mathbf{v} = \partial_y^2\mathcal{T}_2\mathbf{v}$, and $\mathbf{v} = \partial_x^2\partial_y^2\mathcal{T}\mathbf{v}$. Moreover, by Thm. 9, $\mathbf{u}$ is a solution to the PDE (4.7) if and only if $\mathbf{v} = \partial_x^2\partial_y^2\mathbf{u}$ is a solution to the PIE

$$\partial_t\mathcal{T}\mathbf{v}(t) = \mathcal{A}\mathbf{v}(t) = \mathcal{T}_2\mathbf{v}(t) + \mathcal{T}_1\mathbf{v}(t) + r\mathcal{T}\mathbf{v}(t), \qquad \mathbf{v}(t) \in L_2[[0,1]^2],\ t \geq 0,$$

where $\mathcal{A} := \mathcal{T}_2 + \mathcal{T}_1 + r\mathcal{T}$.

Using Thm. 9, a broad class of linear ND PDEs with admissible and consistent domains can be equivalently represented as PIEs, modeling the dynamics of the highest-order spatial derivative of the PDE state. In the following section, we will show that the operators defining this PIE representation belong to an algebra of integral operators with polynomial semi-separable kernels, referred to as Partial Integral (PI) operators. This allows a stability test for PIEs to be formulated as a linear operator inequality on PI operator variables, as will be shown in the subsequent section.

5. A $*$-Algebra of Multivariate PI Operators

In the previous section, it was shown that a broad class of multivariate, linear PDEs can be equivalently represented as PIEs, taking the form $\partial_t \mathcal{T}\mathbf{v}(t) = \mathcal{A}\mathbf{v}(t)$. In this section, we show that such a representation retains many elements of the classical, matrix-parameterized, state-space representation of linear ordinary differential equations. Such state-space properties are useful from a computational perspective in that they allow for the analysis of system properties (e.g. stability) through analysis of the PIE system parameters $\mathcal{T}$ and $\mathcal{A}$.

Specifically, the essential elements of a state-space are: a notion of state; algebraic properties of the system parameters; and computational methods defined on those system parameters. As to the notion of state, we have already shown that our definition of fundamental state $\mathbf{v}(t)$ lies in a Hilbert space (unconstrained by boundary conditions or Sobolev regularity constraints) and any such state uniquely defines the forward evolution of the PIE (subject to well-posedness of the underlying PDE). In this section, then, we embed the operators $\mathcal{T}$ and $\mathcal{A}$ within a parameterized algebra of integral operators. This algebra is referred to as the class of Partial Integral (PI) operators and, as will be shown, is closed under addition, composition and adjoint operations. In the next section, we show how system properties such as stability may be inferred by properties of these system parameters and provide computational methods for verification of such properties.

Let us begin by recalling that, in constructing the operators $\mathcal{T}$ and $\mathcal{A}$ defining the PIE representation of a given multivariate PDE, an inductive approach was used, defining each multivariate operator as the composition (product) and sum of univariate operators $\mathcal{T}_i$ and $\mathcal{A}_{i,\alpha_i}$ as in

$$\mathcal{T} := \prod_{i=1}^{N} \mathcal{T}_i, \qquad \mathcal{A} := \sum_{\vec{0}\leq\alpha\leq\delta} \mathrm{M}[\boldsymbol{A}_\alpha] \prod_{i=1}^{N} \mathcal{A}_{i,\alpha_i}.$$

Here, by construction, each of the operators $\mathcal{T}_i$ and $\mathcal{A}_{i,\alpha_i}$ takes the form

$$(\mathcal{R}\mathbf{v})(s) = \boldsymbol{R}_0(s)\mathbf{v}(s) + \int_a^s \boldsymbol{R}_1(s,\theta)\mathbf{v}(\theta)d\theta + \int_s^b \boldsymbol{R}_2(s,\theta)\mathbf{v}(\theta)d\theta,$$

defined by 3 polynomial parameters, $\boldsymbol{R}_0$, $\boldsymbol{R}_1$, and $\boldsymbol{R}_2$. Such univariate operators have been referred to as 3-PI operators in, e.g. [40], and have been used to parameterize a class of 1D PIEs. Moreover it has been shown that these univariate 3-PI operators form a $*$-algebra – being closed under linear combination, composition, and adjoint operations.

In this section, we show that in the multivariate setting, the operators $\mathcal{T}$ and $\mathcal{A}$ defining the PIE representation of a multivariate PDE similarly belong to a $*$-algebra of PI operators. In particular, based on the inductive definition of $\mathcal{T}$ and $\mathcal{A}$, we define a class of *N*D PI operators as linear combinations of compositions of univariate 3-PI operators along each spatial direction. Next, we prove that this class of operators is closed under composition and adjoint, providing explicit formulae for computing the composition and adjoint. In Section 6, we will then use these algebraic properties to establish tests for stability verification of multivariate linear PDEs.

5.1. *A Parameterization of Multivariate Partial Integral Operators on* $L_2^n[\Omega]$

In this subsection, we define a parameterized class of multivariate PI operators on $L_2[\Omega]$ which includes the operators $\mathcal{T},\mathcal{A}$, as constructed in Thm. 9. In Subsection 5.2, we will then show that this class of operators is a *-algebra.

We define the class of multivariate 3-PI operators using an inductive approach, wherein multivariate PI operators on the hyper-rectangle $\Omega := \prod_{i=1}^{N}[a_i,b_i]$ are defined as the sum of product compositions of univariate PI operators on each interval $[a_i,b_i]$. To start, recall from, e.g. [42], that the class of univariate 3-PI operators acting on $L_2^n[a,b]$ is formally defined as follows.

Definition 5 (Univariate 3-PI operators, $\mathbf{\Pi}_1^{m\times n}[a,b]$) *For a given interval $[a,b]$, we say that $\mathcal{R}$ is a **univariate 3-PI operator** (denoted $\mathcal{R}\in\mathbf{\Pi}_1^{m\times n}[a,b]\subseteq\mathcal{L}(L_2^n[a,b],L_2^m[a,b])$) if there exist polynomials*

$$\mathbf{R} := \{\boldsymbol{R}_0,\boldsymbol{R}_1,\boldsymbol{R}_2\} \in \mathbb{R}^{m\times n}[s]\times\mathbb{R}^{m\times n}[s,\theta]\times\mathbb{R}^{m\times n}[s,\theta] =: \mathbf{\Gamma}_1^{m\times n}[a,b],$$

such that

$$(\mathcal{R}\mathbf{v})(s) = \boldsymbol{R}_0(s)\mathbf{v}(s) + \int_a^s \boldsymbol{R}_1(s,\theta)\mathbf{v}(\theta)d\theta + \int_s^b \boldsymbol{R}_2(s,\theta)\mathbf{v}(\theta)d\theta, \qquad s\in[a,b].$$

We denote the map from parameters $\mathbf{R}\in\mathbf{\Gamma}_1^{m\times n}[a,b]$ to the operator $\mathcal{R}\in\mathbf{\Pi}_1^{m\times n}[a,b]$ as $\mathcal{R}=\Pi_1[\mathbf{R}]$.

The class of univariate 3-PI operators includes both multiplier operators with polynomial multipliers and integral operators with polynomial semi-separable kernels on $L_2[a,b]$, where we recall that a kernel, $\boldsymbol{K}$, is polynomial semi-separable if it may be expressed as

$$\boldsymbol{K}(s,\theta) := \begin{cases} \boldsymbol{R}_1(s,\theta), & s\geq\theta, \\ \boldsymbol{R}_2(s,\theta), & s<\theta, \end{cases}$$

for polynomials $\boldsymbol{R}_1,\boldsymbol{R}_2\in\mathbb{R}^{m\times n}[s,\theta]$. The restriction to polynomial multipliers and polynomial semi-separable kernels will allow us in Section 6 to verify the properties of 3-PI operators using algorithms based on the monomial coefficients of these polynomial parameters. Note that, by inspection, the operators $\mathcal{T}$ and $\mathcal{A}$ defining the univariate PIE representation in Cor. 2 are univariate 3-PI operators.

Having defined the class of univariate 3-PI operators, we may now define the class of ND 3-PI operators as the sums of compositions of N univariate 3-PI operators, each acting on a distinct spatial dimension.

Definition 6 (ND 3-PI operators, $\mathbf{\Pi}_N[\Omega]$) *Given $N\in\mathbb{N}$ and $\Omega=\prod_{i=1}^{N}[a_i,b_i]\subseteq\mathbb{R}^N$, we say that $\mathcal{R}\in\mathcal{L}(L_2[\Omega])$ is a **ND 3-PI operator** (denoted $\mathcal{R}\in\mathbf{\Pi}_N[\Omega]\subseteq\mathcal{L}(L_2[\Omega])$) if there exist $M\in\mathbb{N}$ and $\mathcal{R}_{j,i}\in\mathbf{\Pi}_1[a_i,b_i]$ such that*

$$\mathcal{R} = \sum_{j=1}^{M}\prod_{i=1}^{N}\mathcal{R}_{j,i},$$

where each univariate $\mathcal{R}_{j,i}$ operates on variable s_i – i.e

$$(\mathcal{R}_{j,i}\mathbf{v})(s) := (\mathcal{R}_{j,i}\mathbf{v}(s_1,\cdots,s_{i-1},\bullet,s_{i+1},\cdots s_N))(s_i).$$

For convenience, we denote the map from parameters $\mathbf{R}=\{\mathbf{R}_i\}_{i=1}^N\in\prod_{i=1}^N\mathbf{\Gamma}_1[a_i,b_i] =: \mathbf{\Gamma}_N[\Omega]$ to the multivariate operator $\mathcal{R}\in\mathbf{\Pi}_N[\Omega]$ as $\Pi_N[\mathbf{R}] := \prod_{i=1}^N\Pi_1[\mathbf{R}_i]$. Using this notation, any $\mathcal{R}\in\mathbf{\Pi}_N[\Omega]$ may be represented as $\mathcal{R}=\sum_j\Pi_N[\mathbf{R}_j]$ for some $\mathbf{R}_j\in\mathbf{\Gamma}_N[\Omega]$. Finally, for matrix-valued operators, $\mathcal{R}\in\mathcal{L}(L_2^n[\Omega],L_2^m[\Omega])$, we say $\mathcal{R}$ is ND 3-PI (denoted $\mathcal{R}\in\mathbf{\Pi}_N^{m\times n}[\Omega]$) if $[\mathcal{R}]_{k,\ell}\in\mathbf{\Pi}_N[\Omega]$ for each $k\in\{1:m\}$ and $\ell\in\{1:n\}$.

Defn. 6 offers a natural parameterization of multivariate 3-PI operators, by taking sums and compositions of univaraite 3-PI operators – i.e. $\mathcal{R} = \sum_{j=1}^{M} \prod_{i=1}^{N} \Pi_1[\mathbf{R}_{i,j}]$. Note that, although the class of multivariate 3-PI operators is defined inductively, the inclusion of the summation $\sum_{j=1}^{M}$ implies that this class includes, e.g., any operator of the form

$$(\mathcal{R}\mathbf{v})(s) := \boldsymbol{M}(s)\mathbf{v}(s) + \int_{\Omega} \boldsymbol{K}(s,\theta)\mathbf{v}(\theta)d\theta,$$

for $\boldsymbol{M} \in \mathbb{R}^{m\times n}[s]$ polynomial and $\boldsymbol{K}(s,\theta)$ polynomial semi-separable, where the definition of polynomial semi-separable kernels may be extended to multivariate domains as follows.

Definition 7 *For $N \in \mathbb{N}$ and $\Omega := \prod_{i=1}^{N}[a_i,b_i]$, we say that $\boldsymbol{K} \in L_2^{m\times n}[\Omega]$ is **polynomial semi-separable** if for each $\alpha \in \{-1,1\}^N$ there exists polynomials $\boldsymbol{K}_\alpha \in \mathbb{R}^{m\times n}[s,\theta]$ such that*

$$\boldsymbol{K}(s,\theta) = \boldsymbol{K}_\alpha(s,\theta) \qquad \forall (s,\theta) \in \big\{(s,\theta) \in \Omega \,\big|\, \alpha_i \cdot (s_i - \theta_i) \leq 0,\ \forall i \in \{1:N\}\big\}.$$

We may now observe that, given a multivariate PDE as in (4.5), the operators $\mathcal{T}$ and $\mathcal{A}$ defining the associated PIE representation as per Thm. 8 are *N*D 3-PI operators.

Lemma 11 *For $N \in \mathbb{N}$ and $\delta \in \mathbb{N}_0^N$, let $\mathcal{T}$ and $\mathcal{A}$ be as defined in Thm. 8. Then $\mathcal{T},\mathcal{A} \in \boldsymbol{\Pi}_N^{n\times n}[\Omega]$.*

The proof follows directly using the inductive definition of multivariate 3-PI operators (see Lem. 27 in Appx. B for a formal proof). The following example illustrates the parameterization ($\boldsymbol{\Gamma}_N[\Omega]$) of multivariate 3-PI operators ($\boldsymbol{\Pi}_N[\Omega]$) by examining the $\mathcal{T}$ and $\mathcal{A}$ as obtained from the PIE representation of the 2D reaction-diffusion equation from Example 3.

Example 4 *Consider the reaction-diffusion equation*

$$\mathbf{u}_t(t,x,y) = \mathbf{u}_{xx}(t,x,y) + \mathbf{u}_{yy}(t,x,y) + \mathbf{u}(t,x,y), \qquad \mathbf{u}(t) \in S_2^{(2,2)}[[0,1]^2],\ t \geq 0,$$
$$\mathbf{u}(t,0,y) = \mathbf{u}(t,1,y) = 0, \qquad \mathbf{u}(t,x,0) = \mathbf{u}_y(t,x,1) = 0.$$

By Example 3, we can represent this PDE equivalently as a PIE $\partial_t \mathcal{T}\mathbf{v}(t) = \mathcal{A}\mathbf{v}(t)$, where $\mathcal{T} = \mathcal{T}_1\mathcal{T}_2$ and $\mathcal{A} = \mathcal{T}_2 + \mathcal{T}_1 + \mathcal{T}$, where

$$(\mathcal{T}_1\mathbf{v})(x,y) = -\int_0^x \theta(1-x)\mathbf{v}(\theta,y)d\theta - \int_x^1 x(1-\theta)\mathbf{v}(\theta,y)d\theta,$$
$$(\mathcal{T}_2\mathbf{v})(x,y) = -\int_0^y \eta\,\mathbf{v}(x,\eta)d\eta - \int_y^1 y\mathbf{v}(x,\eta)d\eta.$$

Then, the operators $\mathcal{T}$ and $\mathcal{A}$ are 2D 3-PI operators. In particular, define $\mathbf{I} := \{I,0,0\} \in \boldsymbol{\Gamma}_1[0,1]$ and $\mathbf{T}_i = \{0,\boldsymbol{T}_i^1,\boldsymbol{T}_i^2\} \in \boldsymbol{\Gamma}_1[0,1]$ for $i \in \{1,2\}$, where

$$\boldsymbol{T}_1^1(x,\theta) := -\theta(1-x), \qquad \boldsymbol{T}_1^2(x,\theta) := -x(1-\theta), \qquad \boldsymbol{T}_2^1(y,\eta) := -\eta, \qquad \boldsymbol{T}_2^2(y,\eta) := -y.$$

Then, defining $\mathbf{T} := \{\mathbf{T}_1,\mathbf{T}_2\} \in \boldsymbol{\Gamma}_2[[0,1]^2]$, $\mathbf{R} := \{\mathbf{T}_1,\mathbf{I}\} \in \boldsymbol{\Gamma}_2[[0,1]^2]$ and $\mathbf{Q} := \{\mathbf{I},\mathbf{T}_2\} \in \boldsymbol{\Gamma}_2[[0,1]^2]$, we have $\mathcal{T} = \Pi_2[\mathbf{T}]$, $\mathcal{T}_1 = \Pi_2[\mathbf{R}]$ and $\mathcal{T}_2 = \Pi_2[\mathbf{Q}]$, so that also $\mathcal{A} = \Pi_2[\mathbf{T}] + \Pi_2[\mathbf{R}] + \Pi_2[\mathbf{Q}]$.

Having now defined a suitable class of *N*D 3-PI operators, in the following subsection we show that this class is a $*$-algebra, providing explicit expressions for composition and adjoint in terms of the parameters of the corresponding operators.

5.2. *ND 3-PI Operators Form a $*$-Algebra*

A $*$-algebra is a vector space equipped with a multiplication and involution operation which satisfy certain properties. It is well-known that both the space of multiplier operators with bounded multipliers and that of integral operators with square-integrable kernels are $*$-algebras, using composition ($\circ$) for the multiplication operation, and the adjoint ($*$) with respect to the L_2 inner product as involution operation. Since 3-PI operators are defined by (bounded) multiplier and integral operators, we can similarly define a multiplication and involution operation on $\mathbf{\Pi}_N^{n\times n}[\Omega]$ as the composition and adjoint, respectively, so that for $\mathcal{Q},\mathcal{R}\in\mathbf{\Pi}_N^{n\times n}[\Omega]$ we define $(\mathcal{Q}\circ\mathcal{R})\mathbf{v}:=\mathcal{Q}(\mathcal{R}\mathbf{v})$ and $\mathcal{P}=\mathcal{R}^*$ if $\langle\mathcal{P}\mathbf{u},\mathbf{v}\rangle_{L_2}=\langle\mathbf{u},\mathcal{R}\mathbf{v}\rangle_{L_2}$ for all $\mathbf{u},\mathbf{v}\in L_2^n$. In order to prove that $\mathbf{\Pi}_N^{n\times n}[\Omega]$ is a $*$-algebra, then, we need only show that $\mathbf{\Pi}_N^{n\times n}$ is a vector space which is closed under composition and adjoint. To reduce notational complexity, we will prove this fact only for the scalar-valued case, $n=1$, noting that this result readily generalizes to arbitrary $n\in\mathbb{N}$ using the standard matrix product and transpose operations.

We start by showing that $\mathbf{\Pi}_N[\Omega]$ is indeed a vector space. This follows naturally from the fact that any $\mathcal{R}\in\mathbf{\Pi}_N[\Omega]$ may be expressed as a sum of product compositions of univariate 3-PI operators, $\mathcal{R}=\sum_{j=1}^{M}\prod_{i=1}^{N}\Pi_1[\mathbf{R}_{i,j}]$, where each product $\prod_{i=1}^{N}\Pi_1[\mathbf{R}_{i,j}]$ is itself an element of $\mathbf{\Pi}_N[\Omega]$. Specifically, we have the following result.

Lemma 12 *For $N\in\mathbb{N}$ and $\Omega:=\prod_{i=1}^{N}[a_i,b_i]$, if $\mathcal{Q},\mathcal{R}\in\mathbf{\Pi}_N[\Omega]$, then $\lambda\mathcal{Q}+\mu\mathcal{R}\in\mathbf{\Pi}_N[\Omega]$ for all $\lambda,\mu\in\mathbb{R}$.*

Proof The result follows immediately from the definition of 3-PI operators, and linearity of integral and multiplier operators. A full proof is given in Lem. 28 in Appx. B. □

Having established that the set $\mathbf{\Pi}_N$ is a vector space, it remains to prove that it is also closed under composition and adjoint operations. However, these properties also readily follow from the inductive definition of multivariate 3-PI operators in terms of univariate 3-PI operators – the latter of which have already been proven to form a $*$-algebra. Specifically, recall the following result from e.g. [42], proving that the composition of two univariate 3-PI operators is again a univariate 3-PI operator, and providing formulae for composition of two operators in terms of the associated polynomial parameters.

Lemma 13 *If $\mathcal{Q}\in\mathbf{\Pi}_1[a,b]$ and $\mathcal{R}\in\mathbf{\Pi}_1[a,b]$, then $\mathcal{Q}\circ\mathcal{R}\in\mathbf{\Pi}_1[a,b]$. In particular, if $\mathbf{Q}:=\{\boldsymbol{Q}_0,\boldsymbol{Q}_1,\boldsymbol{Q}_2\}\in\mathbf{\Gamma}_1[a,b]$ and $\mathbf{R}:=\{\boldsymbol{R}_0,\boldsymbol{R}_1,\boldsymbol{R}_2\}\in\mathbf{\Gamma}_1[a,b]$, then $\Pi_1[\mathbf{Q}]\circ\Pi_1[\mathbf{R}]=\Pi_1[\mathbf{Q}\circ\mathbf{R}]$, where we define $\mathbf{Q}\circ\mathbf{R}:=\{\boldsymbol{P}_0,\boldsymbol{P}_1,\boldsymbol{P}_2\}\in\mathbf{\Gamma}_1[a,b]$ with*

$$
\begin{aligned}
\boldsymbol{P}_0(s)&:=\boldsymbol{Q}_0(s)\boldsymbol{R}_0(s),\\
\boldsymbol{P}_1(s,\theta)&:=\boldsymbol{Q}_0(s)\boldsymbol{R}_1(s,\theta)+\boldsymbol{Q}_1(s,\theta)\boldsymbol{R}_0(\theta)\\
&\quad+\int_a^\theta\boldsymbol{Q}_1(s,\eta)\boldsymbol{R}_2(\eta,\theta)d\eta+\int_\theta^s\boldsymbol{Q}_1(s,\eta)\boldsymbol{R}_1(\eta,\theta)d\eta+\int_s^b\boldsymbol{Q}_2(s,\eta)\boldsymbol{R}_1(\eta,\theta)d\eta,\\
\boldsymbol{P}_2(s,\theta)&:=\boldsymbol{Q}_0(s)\boldsymbol{R}_2(s,\theta)+\boldsymbol{Q}_2(s,\theta)\boldsymbol{R}_0(\theta)\\
&\quad+\int_a^s\boldsymbol{Q}_1(s,\eta)\boldsymbol{R}_2(\eta,\theta)d\eta+\int_s^\theta\boldsymbol{Q}_2(s,\eta)\boldsymbol{R}_2(\eta,\theta)d\eta+\int_\theta^b\boldsymbol{Q}_2(s,\eta)\boldsymbol{R}_1(\eta,\theta)d\eta.
\end{aligned}
$$

Proof The result follows by explicitly taking the composition $\Pi_1[\mathbf{Q}]\circ\Pi_1[\mathbf{R}]$, and performing standard algebraic manipulations. A full proof is given in Lem. 36 in [41]. □

Lem. 13 shows that the composition of univariate 3-PI operators again defines a univariate 3-PI operator, providing an explicit expression for the parameter map $\circ : \mathbf{\Gamma}_1 \times \mathbf{\Gamma}_1 \to \mathbf{\Gamma}_1$ defining this composition. Using this expression, and the inductive definition of multivariate PI operators, we then obtain the following result for the composition of multivariate 3-PI operators.

Proposition 14 (Composition of PI Operators in *N*D) *For any $N \in \mathbb{N}$ and $\Omega = \prod_{i=1}^{N}[a_i, b_i]$, if $\mathcal{Q}, \mathcal{R} \in \mathbf{\Pi}_N[\Omega]$, then $\mathcal{Q} \circ \mathcal{R} \in \mathbf{\Pi}_N[\Omega]$. In particular, if $\mathcal{Q} = \sum_{j=1}^{M} \Pi_N[\mathbf{Q}_j]$ and $\mathcal{R} = \sum_{k=1}^{K} \Pi_N[\mathbf{R}_k]$ with $\mathbf{Q}_j = \{\mathbf{Q}_{j,i}\}_{i=1}^{N} \in \mathbf{\Gamma}_N[\Omega]$ and $\mathbf{R}_k = \{\mathbf{R}_{k,i}\}_{i=1}^{N} \in \mathbf{\Gamma}_N[\Omega]$ for each $j \in \{1 : M\}$ and $k \in \{1 : K\}$, then $\mathcal{Q} \circ \mathcal{R} = \sum_{j=1}^{M} \sum_{k=1}^{K} \Pi_N[\mathbf{Q}_j \circ \mathbf{R}_k]$, where for each $\{j,k\}$,*

$$\mathbf{Q}_j \circ \mathbf{R}_k := \{\mathbf{Q}_{j,i} \circ \mathbf{R}_{k,i}\}_{i=1}^{N} \in \mathbf{\Gamma}_N[\Omega],$$

where the univariate composition $\mathbf{Q}_{j,i} \circ \mathbf{R}_{k,i} \in \mathbf{\Gamma}_1[a_i, b_i]$ is as defined in Lem. 13.

Proof Fix arbitrary $\mathcal{Q} = \sum_{j=1}^{M} \Pi_N[\mathbf{Q}_j] \in \mathbf{\Pi}_N[\Omega]$ and $\mathcal{R} = \sum_{k=1}^{K} \Pi_N[\mathbf{R}_k] \in \mathbf{\Pi}_N[\Omega]$, where $\mathbf{Q}_j := \{\mathbf{Q}_{j,i}\}_{i=1}^{N}$ and $\mathbf{R}_k := \{\mathbf{R}_{k,i}\}_{i=1}^{N}$ with $\mathbf{Q}_{j,i}, \mathbf{R}_{k,i} \in \mathbf{\Gamma}_1[a_i, b_i]$. Then, by Lem. 13, $\Pi_1[\mathbf{Q}_{j,i}] \circ \Pi_1[\mathbf{R}_{k,i}] = \Pi_1[\mathbf{Q}_{j,i} \circ \mathbf{R}_{k,i}]$ for each i, j, k. It follows that

$$\begin{aligned}
\mathcal{Q} \circ \mathcal{R} &= \left(\sum_{j=1}^{M} \prod_{i=1}^{N} \Pi_1[\mathbf{Q}_{j,i}] \right) \circ \left(\sum_{k=1}^{K} \prod_{\ell=1}^{N} \Pi_1[\mathbf{R}_{k,\ell}] \right) \\
&= \sum_{j=1}^{M} \sum_{k=1}^{K} \left(\prod_{i=1}^{N} \Pi_1[\mathbf{Q}_{j,i}] \circ \prod_{\ell=1}^{N} \Pi_1[\mathbf{R}_{k,\ell}] \right) \\
&= \sum_{j=1}^{M} \sum_{k=1}^{K} \left(\prod_{i=1}^{N} (\Pi_1[\mathbf{Q}_{j,i}] \circ \Pi_1[\mathbf{R}_{k,i}]) \right) = \sum_{j=1}^{M} \sum_{k=1}^{K} \prod_{i=1}^{N} \Pi_1\left[\mathbf{Q}_{j,i} \circ \mathbf{R}_{k,i}\right] = \sum_{j=1}^{M} \sum_{k=1}^{K} \Pi_N\left[\mathbf{Q}_j \circ \mathbf{R}_k\right],
\end{aligned}$$

where we remark that $\prod_{i=1}^{N} \Pi_1[\mathbf{Q}_{j,i}] \circ \prod_{\ell=1}^{N} \Pi_1[\mathbf{R}_{k,\ell}] = \prod_{i=1}^{N} (\Pi_1[\mathbf{Q}_{j,i}] \circ \Pi_1[\mathbf{R}_{k,i}])$, since the operators $\Pi_1[\mathbf{Q}_{j,i}]$ and $\Pi_1[\mathbf{R}_{k,\ell}]$ act on different spatial dimensions for $i \neq \ell$ (see also Cor. 30 in Appx. B). Since $\mathbf{Q}_j \circ \mathbf{R}_k \in \mathbf{\Gamma}_N[\Omega]$ for every $j \in \{1 : M\}$ and $k \in \{1 : K\}$, we conclude that $\mathcal{Q} \circ \mathcal{R} \in \mathbf{\Pi}_N[\Omega]$. □

In order to prove that $\mathbf{\Pi}_N[\Omega]$ is a $*$-algebra, it remains only to prove that this set is also closed under the adjoint with respect to the L_2 inner product. This is relatively trivial, based on the following result for computing the adjoint of a univariate 3-PI operator from e.g. [42].

Lemma 15 *For any $[a,b] \subseteq \mathbb{R}$, if $\mathcal{R} \in \mathbf{\Pi}_1[a,b]$, then $\mathcal{R}^* \in \mathbf{\Pi}_1[a,b]$. In particular, if $\mathcal{R} = \Pi_1[\mathbf{R}]$ where $\mathbf{R} := \{\boldsymbol{R}_0, \boldsymbol{R}_1, \boldsymbol{R}_2\} \in \mathbf{\Gamma}_1[a,b]$, then $\mathcal{R}^* = \Pi_1[\mathbf{R}^*]$, where $\mathbf{R}^* := \{\boldsymbol{P}_0, \boldsymbol{P}_1, \boldsymbol{P}_2\} \in \mathbf{\Gamma}_1[a,b]$ for*

$$\boldsymbol{P}_0(s) := \boldsymbol{R}_0(s), \qquad \boldsymbol{P}_1(s,\theta) := \boldsymbol{R}_2(\theta, s), \qquad \boldsymbol{P}_2(s,\theta) := \boldsymbol{R}_1(\theta, s).$$

Proof We refer to Lem. 37 in [41] for a proof. □

Using the formulae for computing the adjoint of a univariate 3-PI operator from Lem. 15, the following result provides similar formulae for computing the adjoint of a multivariate 3-PI operator, proving that this adjoint is a multivariate 3-PI operator as well.

Lemma 16 *For $N \in \mathbb{N}$ and $\Omega := \prod_{i=1}^{N}[a_i, b_i] \subseteq \mathbb{R}^N$, if $\mathcal{R} \in \mathbf{\Pi}_N[\Omega]$, then $\mathcal{R}^* \in \mathbf{\Pi}_N[\Omega]$. In particular, if $\mathcal{R} = \sum_{j=1}^{M} \Pi_N[\mathbf{R}_j]$, where $\mathbf{R}_j := \{\mathbf{R}_{j,i}\}_{i=1}^{N} \in \mathbf{\Gamma}_N[\Omega]$, then $\mathcal{R}^* = \sum_{j=1}^{M} \Pi_N[\mathbf{R}_j^*]$, where $\mathbf{R}_j^* := \{\mathbf{R}_{j,i}^*\}_{i=1}^{N}$, where the univariate adjoint $\mathbf{R}_{j,i}^* \in \mathbf{\Gamma}_1[a_i, b_i]$ is as defined in Lem. 15.*

Proof The result follows immediately from the definition of multivariate 3-PI operators (Defn. 6), the commutative properties of (scalar-valued) univariate operators (Lem. 29), and the expression for the adjoint of univariate 3-PI operators (Lem. 15). A full proof is given in Lem. 31 in Appx. B. □

By Lem. 16, the adjoint of any multivariate 3-PI operator is again a multivariate 3-PI operator. Using this result, as well as Prop. 14 and Lem. 12, it then follows that $\mathbf{\Pi}_N[\Omega]$ in fact defines a $*$-algebra.

Proposition 17 *For any $N \in \mathbb{N}$ and $\Omega := \prod_{i=1}^{N}[a_i, b_i]$, the set $\mathbf{\Pi}_N[\Omega] \subseteq \mathcal{L}(L_2)$ as defined in Defn. 6, with multiplication defined by composition and involution defined by the adjoint with respect to the L_2 inner product, is a $*$-algebra.*

Proof To prove this result, we use the fact that $\mathcal{L}(L_2[\Omega])$ is a $*$-algebra, with multiplication defined by composition, and involution defined by the adjoint with respect to the L_2 inner product. Since $\mathbf{\Pi}_N[\Omega] \subseteq \mathcal{L}(L_2[\Omega])$, it follows that the composition and adjoint also define suitable multiplication and involution operations, respectively, on $\mathbf{\Pi}_N[\Omega]$. In order to prove that $\mathbf{\Pi}_N[\Omega]$ is a $*$-algebra, then, it suffices to prove that $\mathbf{\Pi}_N[\Omega]$ is a vector space closed under these multiplication and involution operations. Here, the fact that $\mathbf{\Pi}_N[\Omega]$ is a vector space has been shown in Lem. 12. In addition, by Prop. 14, we have for all $\mathcal{Q}, \mathcal{R} \in \mathbf{\Pi}_N[\Omega]$ that $\mathcal{Q} \circ \mathcal{R} \in \mathbf{\Pi}_N[\Omega]$, whence $\mathbf{\Pi}_N[\Omega]$ is closed under the multiplication operation defined by composition. Finally, by Lem. 16, we have for all $\mathcal{Q} \in \mathbf{\Pi}_N[\Omega]$ that $\mathcal{Q}^* \in \mathbf{\Pi}_N[\Omega]$, so that $\mathbf{\Pi}_N[\Omega]$ is also closed under the adjoint operation. We conclude that $\mathbf{\Pi}_N[\Omega]$ is a $*$-algebra. □

Prop. 17 proves that the set $\mathbf{\Pi}_N[\Omega]$ of ND 3-PI operators is a $*$-algebra, being closed under linear combinations, as well as composition and adjoint operations. This result also generalizes directly to square matrix-valued operators, using the standard formulae for matrix multiplication and transposition. In the following section, we will use the $*$-algebraic properties of ND 3-PI operators to show how stability of linear ND PDEs can be tested using convex optimization.

6. Stability Analysis in the PIE Representation using Semidefinite Programming

Having shown that any linear ND PDE as in (4.5) with admissible boundary conditions admits an equivalent PIE state-space representation of the form $\partial_t \mathcal{T}\mathbf{v} = \mathcal{A}\mathbf{v}$ where $\mathbf{v} \in L_2$, and having shown that the operators, $\mathcal{T}, \mathcal{A}$ lie in the $*$-algebra of bounded linear multivariate 3-PI operators, we now exploit this representation to obtain conditions for stability which may be tested using convex optimization. Specifically, in Subsection 6.1, we will define a suitable notion of exponential stability and show that a sufficient condition for this exponential stability is the existence of a PI operator $\mathcal{P}$ such that

$$\mathcal{P}^*\mathcal{T} = \mathcal{T}^*\mathcal{P} \succeq \varepsilon^2 \mathcal{T}^*\mathcal{T}, \qquad \mathcal{P}^*\mathcal{A} + \mathcal{A}^*\mathcal{P} \preceq -2k\mathcal{P}^*\mathcal{T},$$

for some $\varepsilon, k \geq 0$. Inequalities of this form, expressed in terms of PI operators and operator inequalities, are referred to as Linear PI Inequalities (LPIs). Then, in Subsection 6.2, we will show how to use positive matrices to parameterize positive multivariate PI operators and exploit the algebraic structure of PI operators to solve LPIs and test stability of multivariate PDEs using semidefinite programming.

6.1. *A Linear PI Inequality for Exponential Stability of PDEs*

To construct a PIE-based stability test for linear multivariate PDEs, we first define a notion of exponential stability. Specifically, we use the notion of *exponential PIE to PDE stability*, which is slightly weaker than the classical notion of exponential L_2 stability. The weaker definition is used because the stronger classical notion of exponential L_2 stability, when applied to PDEs in first-order form (i.e. $\partial_t \mathbf{u} = \mathcal{A}\mathbf{u}$), does not account for certain notions of energy storage (e.g. strain) and is not satisfied for many commonly used PDEs such as wave and beam equations (see e.g. the wave equation in Appx. C.2).

Definition 8 *A PDE defined by parameters $\{\boldsymbol{A}_\alpha, \mathcal{D}_i\}$ as in (4.5), admitting a unique solution that depends continuously on the initial state, is said to be **exponentially PIE to PDE stable** with rate $k \geq 0$ and gain $M \geq 1$ if for every initial state $\mathbf{u}_0 \in \mathcal{D} \subseteq S_2^{\delta,n}[\Omega]$, the associated solution $\mathbf{u}$ to the PDE satisfies*

$$\|\mathbf{u}(t)\|_{L_2} \leq Me^{-kt}\|D^\delta \mathbf{u}_0\|_{L_2}, \qquad \forall t \geq 0.$$

In contrast to classical notions of exponential stability, Defn. 8 defines exponential stability using a Sobolev norm on the initial PDE state, corresponding to the L_2 norm of the fundamental state, $\|D^\delta \mathbf{u}(0)\|_{L_2}$ (rather than $\|\mathbf{u}(0)\|_{L_2}$). Since the L_2 norm of the PDE state is bounded by that of the fundamental state, the classical notion of stability then implies PIE to PDE stability, but the converse is not true.

Having defined a notion of exponential PIE to PDE stability in Defn. 8, suppose now we want to verify this stability property for a first order linear PDE as defined in (4.5), with an associated PIE representation of the form

$$\partial_t \mathcal{T}\mathbf{v}(t) = \mathcal{A}\mathbf{v}(t),$$

so that for any solution, $\mathbf{u}(t)$, to the PDE, $\mathbf{v}(t) := D^\delta \mathbf{u}(t)$ is a solution to the PIE and for any solution, $\mathbf{v}(t)$, to the PIE, $\mathbf{u}(t) = \mathcal{T}\mathbf{v}(t)$ is a solution to the PDE. Our stabiltiy test, then, is based on existence of a Lyapunov functional $V : L_2^n \to \mathbb{R}$ such that $\frac{d}{dt}V(\mathbf{v}(t)) \leq -2kV(\mathbf{v}(t))$ for any $\mathbf{v}(t)$ which satisfies the PIE – implying $V(\mathbf{v}(t)) \leq e^{-2kt}V(\mathbf{v}(0))$. The distinction between L_2 and PIE to PDE stability, then, lies only in the nature of the *upper* bound on V – i.e. $V(\mathbf{v}) \leq C\|\mathbf{v}\|_{L_2}$ as opposed to $V(\mathbf{v}) \leq C\|\mathcal{T}\mathbf{v}\|_{L_2}$. The following theorem shows how existence of such a functional V can be tested by solving a linear operator inequality on PI operator variables.

Theorem 18 *For given $\{\boldsymbol{A}_\alpha, \mathcal{D}_i\}$ defining a PDE as in (4.5), let associated operators $\mathcal{T}, \mathcal{A} \in \boldsymbol{\Pi}_N^{n\times n}$ be as defined in Thm. 9. If for some $\varepsilon > 0$ and $k \geq 0$, there exists $\mathcal{P} \in \boldsymbol{\Pi}_N^{n\times n}$ such that*

$$\mathcal{P}^*\mathcal{T} = \mathcal{T}^*\mathcal{P} \succeq \varepsilon^2\mathcal{T}^*\mathcal{T}, \qquad \mathcal{P}^*\mathcal{A} + \mathcal{A}^*\mathcal{P} \preceq -2k\mathcal{P}^*\mathcal{T}, \tag{6.1}$$

then the PDE defined by $\{\boldsymbol{A}_\alpha, \mathcal{D}_i\}$ is exponentially PIE to PDE stable with rate k and gain $M := \sqrt{\|\mathcal{P}^*\mathcal{T}\|_{\text{op}}}/\varepsilon$.

Proof Let $\mathcal{P}$ be such that (6.1) is satisfied, and consider the functional $V : L_2^n[\Omega] \to \mathbb{R}$ defined by $V(\mathbf{v}) := \langle \mathbf{v}, \mathcal{P}^*\mathcal{T}\mathbf{v}\rangle_{L_2}$ for $\mathbf{v} \in L_2^n[\Omega]$. Since $\mathcal{P}^*\mathcal{T} \succeq \varepsilon^2\mathcal{T}^*\mathcal{T}$, this functional satisfies

$$\varepsilon^2\|\mathcal{T}\mathbf{v}\|_{L_2}^2 \leq V(\mathbf{v}) \leq C^2\|\mathbf{v}\|_{L_2}^2,$$

where $C := \sqrt{\|\mathcal{P}^*\mathcal{T}\|_{\text{op}}}$. Now, let $\mathbf{u}_0 \in \bigcap_{i=1}^N \mathcal{D}_i \subseteq S_2^{\delta,n}[\Omega]$ be an arbitrary initial state, and let $\mathbf{u}$ be the associated solution to the PDE defined by $\{\boldsymbol{A}_\alpha, \mathcal{D}_i\}$. Let $\mathbf{v}(t) = D^\delta \mathbf{u}(t)$ for all $t \geq 0$. Then, by Thm. 9, $\mathbf{u}(t) = \mathcal{T}\mathbf{v}(t)$ and $\partial_t \mathcal{T}\mathbf{v}(t) = \mathcal{A}\mathbf{v}(t)$ for all $t \geq 0$, with $\mathbf{v}(0) = D^\delta \mathbf{u}_0$. Taking the temporal derivative of $V(\mathbf{v}(t))$ and exploiting the algebraic properties of $\boldsymbol{\Pi}_N$, we then find

$$\begin{aligned}\frac{d}{dt}V(\mathbf{v}(t)) &= \langle \mathbf{v}(t), \partial_t \mathcal{P}^*\mathcal{T}\mathbf{v}(t)\rangle_{L_2} + \langle \partial_t \mathcal{P}^*\mathcal{T}\mathbf{v}(t), \mathbf{v}(t)\rangle_{L_2} \\ &= \langle \mathbf{v}(t), [\mathcal{P}^*\mathcal{A} + \mathcal{A}^*\mathcal{P}]\mathbf{v}(t)\rangle_{L_2} \leq -2k\langle \mathbf{v}(t), \mathcal{P}^*\mathcal{T}\mathbf{v}(t)\rangle_{L_2} = -2kV(\mathbf{v}(t)).\end{aligned}$$

By Grönwall-Bellman inequality, it follows that $V(\mathbf{v}(t)) \leq e^{-2kt}V(\mathbf{v}(0))$, and therefore, for all $t \geq 0$,

$$\|\mathbf{u}(t)\|_{L_2} = \|\mathcal{T}\mathbf{v}(t)\|_{L_2} \leq \frac{1}{\varepsilon}\sqrt{V(\mathbf{v}(t))} \leq \frac{1}{\varepsilon}e^{-kt}\sqrt{V(\mathbf{v}(0))} \leq \frac{C}{\varepsilon}e^{-kt}\|\mathbf{v}(0)\|_{L_2} = Me^{-kt}\left\|D^\delta \mathbf{u}_0\right\|_{L_2}.$$

Since this holds for arbitrary initial states $\mathbf{u}^0$, this concludes the proof. □

Thm. 18 proves that exponential PIE to PDE stability of any PDE of the form in (4.5) can be tested using the associated PIE representation, by testing for existence of a solution $\mathcal{P}$ to the LPI in (6.1). In the following subsection, we will show how such an LPI can be solved numerically using semidefinite programming, by parameterizing positive semidefinite multivariate 3-PI operators using positive semidefinite matrices.

6.2. *Solving LPIs using Semidefinite Programming*

Having provided a test for exponential PIE to PDE stability as feasibility of the LPI in (6.1), we now briefly show how sufficient conditions for feasibility of this LPI may be tested using semidefinite programming (SDP). This approach is based on the observation that for any positive semidefinite matrix, $R \succeq 0$, and any given $\mathcal{Z} \in \boldsymbol{\Pi}_N[\Omega]$, we have $\mathcal{R} := \mathcal{Z}^*\mathrm{M}[R]\mathcal{Z} \succeq 0$ (where $\mathrm{M}[R] := \Pi_1[\{R,0,0\}]$ is just standard matrix multiplication extended to the function space L_2^n). Furthermore, using the composition maps defined previously, there is a linear map from the elements of R to the coefficients of the polynomials defining $\mathcal{R}$. Thus, for a given $\mathcal{Z} \in \boldsymbol{\Pi}_N[\Omega]$, if we can find an operator $\mathcal{P}$ and matrices $Q, R \succeq 0$ such that the equalities

$$\mathcal{P}^*\mathcal{T} = \mathcal{T}^*\mathcal{P}, \qquad \mathcal{T}^*\mathcal{P} - \varepsilon^2\mathcal{T}^*\mathcal{T} = \mathcal{Z}^*\mathrm{M}[R]\mathcal{Z}, \qquad \mathcal{P}^*\mathcal{A} + \mathcal{A}^*\mathcal{P} + 2k\mathcal{P}^*\mathcal{T} = -\mathcal{Z}^*\mathrm{M}[Q]\mathcal{Z},$$

hold, then we may conclude exponential PIE to PDE stability. Such equalities may then be enforced using SDP. All that remains, therefore, is to choose a suitable $\mathcal{Z}$ so as to render the equality constraints feasible. Specifically, $\mathcal{Z}$ is selected by defining a vector of basis operators for $\boldsymbol{\Pi}_N$. This is done inductively by first defining a vector of basis operators for $\boldsymbol{\Pi}_1[a_i, b_i]$ for each i, and then constructing the multivariate Kronecker product of these bases. From [42], we recall that such a basis for $\boldsymbol{\Pi}_1[a,b]$ is constructed using the degree-d monomial basis vector, $\mathbf{z}_d$, which yields an associated basis $\mathbf{Z}_d$ of polynomial 3-PI parameters as

$$\mathbf{Z}_d := \left\{ \begin{bmatrix} \mathbf{z}_d(s) \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ \mathbf{z}_d(s,\theta) \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ \mathbf{z}_d(s,\theta) \end{bmatrix} \right\} \in \boldsymbol{\Gamma}_1^{\mu(d)\times 1}[a,b], \qquad \text{where} \quad \mu(d) := d^2 + 4d + 3, \quad (6.2)$$

which in turn defines the basis of operators $\Pi_1[\mathbf{Z}_d] \in \boldsymbol{\Pi}_1^{\mu(d)\times 1}$. Given this basis, any $\mathcal{P} \in \boldsymbol{\Pi}_1[a,b]$ (with polynomial parameters of degree at most d) can be represented as $\mathcal{P} = c^T\Pi_1[\mathbf{Z}_d]$ for a unique vector of

coefficients $c \in \mathbb{R}^{\mu(d)}$. Given this basis for the univariate operators, and defining $\mathbf{Z}_{d,i} \in \mathbf{\Gamma}_1^{\mu(d)\times 1}[a_i, b_i]$ for each $i \in \{1:N\}$ to be of the form in (6.2), a basis for $\mathbf{\Pi}_N$ is then constructed as the Kronecker product of the univariate bases as

$$\mathcal{Z}_d := \Pi_1[\mathbf{Z}_{d,1}] \otimes \cdots \otimes \Pi_1[\mathbf{Z}_{d,N}] \in \mathbf{\Pi}_N^{\mu(d)^N \times 1}[\Omega], \tag{6.3}$$

so that any $\mathcal{P} \in \mathbf{\Pi}_N[\Omega]$ (defined by parameters of degree at most d in each variable s_i) can be represented as $\mathcal{P} = c^T \mathcal{Z}_d$. Note that while $\mathcal{Z}_d$ here defines a basis for the *linear* representation of multivariate 3-PI operators, $\mathcal{Z}_d^* \mathrm{M}[R] \mathcal{Z}_d$ is a *quadratic* form. However, by the composition rules of multivariate 3-PI operators, $\mathcal{Z}_d^* \mathrm{M}[R] \mathcal{Z}_d$ will be defined by polynomial parameters of degree at most $2d+1$ in each variable. Then, based on the monomial ordering of $\mathbf{z}_d$, we can define an associated matrix A, such that for any $R \in \mathbb{R}^{\mu(d)^N \times \mu(d)^N}$, we have $\mathcal{Z}_d^* \mathrm{M}[R] \mathcal{Z}_d = \mathrm{vec}(R)^T A \mathcal{Z}_{2d+1}$, where $\mathrm{vec}(R)$ is the vector of columns of R. In this way, $\mathcal{Z}_d$ also defines a basis for the quadratic representation of $\mathbf{\Pi}_N[\Omega]$. Using this basis of *N*D 3-PI operators with polynomial parameters of bounded degree, the following corollary tightens the stability LPI in Thm. 18 to an optimization problem that can be solved using SDP.

Corollary 19 *For given $\{\boldsymbol{A}_\alpha, \mathcal{D}_i\}$ defining a linear PDE as in (4.5), let associated operators $\mathcal{T}, \mathcal{A} \in \mathbf{\Pi}^{n\times n}$ be as defined in Thm. 9. Define $\mathcal{Z}_d$ as in (6.3). If for some $d, d' \in \mathbb{N}_0$, $\varepsilon > 0$ and $k \geq 0$, there exist $P \in \mathbb{R}^{n \times n\mu(d)^N}$ and $R, Q \succeq 0$ such that, for $\mathcal{P} = \mathrm{M}[P](I_n \otimes \mathcal{Z}_d)$,*

$$\begin{aligned} \mathcal{P}^*\mathcal{T} = \mathcal{T}^*\mathcal{P} &= \varepsilon^2 \mathcal{T}^*\mathcal{T} + (I_n \otimes \mathcal{Z}_{d'})^* \mathrm{M}[R](I_n \otimes \mathcal{Z}_{d'}), \\ \mathcal{P}^*\mathcal{A} + \mathcal{A}^*\mathcal{P} + 2k\mathcal{P}^*\mathcal{T} &= -(I_n \otimes \mathcal{Z}_{d'})^* \mathrm{M}[Q](I_n \otimes \mathcal{Z}_{d'}), \end{aligned} \tag{6.4}$$

then the PDE defined by $\{\boldsymbol{A}_\alpha, \mathcal{D}_i\}$ is exponentially PIE to PDE stable with rate k and gain $M := \sqrt{\|\mathcal{P}^*\mathcal{T}\|_{\mathrm{op}}}/\varepsilon$.

Proof Suppose there exist P, Q, R satisfying the conditions of the corollary, with $\mathcal{P} = \mathrm{M}[P](I_n \otimes \mathcal{Z}_d)$. Since $R \succeq 0$, we then have $\mathcal{P}^*\mathcal{T} - \varepsilon^2 \mathcal{T}^*\mathcal{T} \succeq 0$, and therefore $\mathcal{P}^*\mathcal{T} \succeq \varepsilon^2 \mathcal{T}^*\mathcal{T}$. Furthermore, since $Q \succeq 0$, we have $\mathcal{P}^*\mathcal{A} + \mathcal{A}^*\mathcal{P} + 2k\mathcal{P}^*\mathcal{T} \preceq 0$, and thus the LPI (6.1) is feasible. By Thm. 18, it follows that the PDE defined by $\{\boldsymbol{A}_\alpha, \mathcal{D}_i\}$ is exponentially PIE to PDE stable, with rate k and gain $M := \sqrt{\|\mathcal{P}^*\mathcal{T}\|_{\mathrm{op}}}/\varepsilon$. □

Cor. 19 allows stability of a broad class of *N*D PDEs to be tested using SDP. We note, however, that the computational complexity associated with solving the resulting SDP scales with the size of the basis $\mathcal{Z}_d$. Specifically, while the length of the univariate parameter basis $\mathbf{Z}_d$ in (6.2) is proportional to d^2, the use of the Kronecker product implies the length of $\mathcal{Z}_d$ scales as d^{2N} – implying the number of decision variables in P, Q, R scale as d^{4N}. For higher spatial dimension, then, solving the resulting SDP will require either very low degree or the use of specialized large-scale SDP solvers [4, 50, 54]. As in the univariate case, however, computational complexity can be partially mitigated by reducing the number of monomials. Specifically, given a degree, d, for the Lyapunov variable $\mathcal{P}$, the corresponding degree d' in Cor. 19 should be chosen so as to avoid unused monomial terms. Meanwhile, the degree d itself should be chosen to balance accuracy and complexity – the required degree will depend strongly on the type of PDE being analyzed, the required accuracy in decay rate k, and the underlying difficulty of the stability problem. In addition, kernels may be eliminated from $\mathcal{P}$ and multipliers do not appear in the expression for $\mathcal{T}^*\mathcal{P}$. Finally, it is often sufficient to use polynomial rather than polynomial semi-separable kernels (i.e. $\mathbf{Z}_i^d = \{0, \boldsymbol{R}_{i,1}, \boldsymbol{R}_{i,1}\}$).

7. Software Implementation and Numerical Examples

Having developed a framework for representation and computational stability analysis of a broad class of linear multivariate PDEs, we now demonstrate the application of this framework: providing efficient software construction of the PIE representation and testing of LPIs; providing expressions of the PIE representation for 2D heat, wave, and plate equations; verifying accuracy of the stability test; and providing non-conservative bounds on the associated rates of exponential decay.

7.1. *The PIETOOLS Software Package*

While Thm. 9 provides analytic expressions for the conversion of a PDE to a PIE, and while such expressions often provide insight into the structure of the associated PDE and PIE, evaluating such expressions may require substantial effort on the part of the user. Therefore, to facilitate the rapid prototyping and conversion of PDE models to a PIE representation, and to construct the associated stability test, such functions have been automated in the Matlab software package PIETOOLS [43]. A brief introduction and illustrative example of the most useful features of this package are provided here.

Specifically, PIETOOLS is a MATLAB software package whose essential function is to allow for the declaration and manipulation of PI operators and PI operator variables in a syntax similar to matrices – adapting approaches from the widely used SOSTOOLS [37] and YALMIP [31] interfaces. In addition, a central feature of PIETOOLS is the command-line interface for declaration of 1D and 2D ODE-PDE systems. Given such a PDE model, the software allows for: conversion to PIE; numerical simulation; stability analysis; optimal controller synthesis, etc.

The PIETOOLS command-line interface begins by declaration of PDE state variables using `pde_var`, which creates objects which can be added (`+`), differentiated (`diff`) and evaluated at spatial positions (`subs`) in order to define the evolution equations and boundary conditions which constitute a PDE model. Using `convert`, such a PDE model may then be converted to a PIE model, which defines the associated PI operators $\{\mathcal{T},\mathcal{A}\}$. For example, to declare the following heat equation

$$\begin{aligned} &\mathbf{u}_t(t,x,y) = \mathbf{u}_{xx}(t,x,y) + \mathbf{u}_{yy}(t,x,y), && (x,y) \in [0,1]^2, \\ &\mathbf{u}(t,0,y) = \mathbf{u}(t,1,y) = 0, \qquad \mathbf{u}(t,x,0) = \mathbf{u}_y(t,x,1) = 0, && t \geq 0, \end{aligned} \tag{7.1}$$

and construct the associated PIE representation, we can use the following code

```
>> pvar t x y
>> dom = [0,1; 0,1];
>> u = pde_var('state',[x;y],dom);
>> PDE = [diff(u,t)==diff(u,x,2)+diff(u,y,2);
          subs(u,x,0)==0;      subs(u,x,1)==0;
          subs(u,y,0)==0;      subs(diff(u,y),y,1)==0];
>> PIE = convert(PDE,'pie');
>> Top = PIE.T;     Aop = PIE.A;
```

Here, the `convert` function checks the declared `PDE` for the highest order of the derivative of the PDE state taken along each spatial direction – in this case finding a 2nd-order derivative along both the x- and y-direction ($\mathbf{u}_{xx}$ and $\mathbf{u}_{yy}$) – and selects the fundamental state as the corresponding highest-order mixed derivative – in this case $\mathbf{v} = D^{(2,2)}\mathbf{u} = \mathbf{u}_{xxyy}$. After verifying consistency, the operators $\{\mathcal{T},\mathcal{A}\}$ defining the resulting PIE are then returned as `opvar2d` objects, `Top,Aop`, for which operations such as addition (`+`), multiplication ($\star$), and transpose (`'`) have been overloaded to allow for easy computation

of the sum, composition, and adjoint of PI operators. Moreover, given these operators, an LPI such as the program in Cor. 19 can be declared as an `lpiprogram` structure, using a programming structure similar to SOSTOOLS [37]. This program structure can be modified to add indefinite and positive semidefinite PI operator decision variables using `lpivar` and `poslpivar`, and declare equality and inequality constraints using `lpi_eq` and `lpi_ineq`, after which an SDP solver such as SeDuMi [45] or Mosek [5] can be used to solve the program by calling `lpisolve`. For example, to declare and solve the optimization program in Cor. 19 with $d = 4$, $\varepsilon = 10^{-1}$, and $k = 1$ for the heat equation in (7.1), we can call

```
>> d = 4;  eppos = 1e-1;  k = 1;
>> prog = lpiprogram([x;y],dom);
>> [prog,Pop] = lpivar(prog,Top.dim,d);
>> prog = lpi_eq(prog,Pop'*Top-Top'*Pop);
>> prog = lpi_ineq(prog,Top'*Pop-eppos^2*Top'*Top);
>> Qop = Pop'*Aop + Aop'*Pop + 2*k*Pop'*Top;
>> prog = lpi_ineq(prog,-Qop);
>> prog = lpisolve(prog);
>> Pop_sol = getsol_lpivar(prog,Pop);
```

A more optimized version of this LPI is also included as a script `PIETOOLS_stability`. We refer to the PIETOOLS manual [43] for more details on how to declare PDEs, PIEs, and LPIs using PIETOOLS, as well as for other applications of the software such as controller design and simulation. Note, however, that PIETOOLS does not currently support PDE models of spatial dimension greater than two. Therefore, the application of this software in the following subsection will be limited to analysis of heat, wave and plate equations in 2D.

7.2. *Numerical Examples*

In this subsection, we use the PIETOOLS software suite to verify exponential PIE to PDE stability of 2D heat, wave, and plate equations and to compute a maximal lower bound on the associated exponential rate of decay. For the wave and plate equations, second-order temporal derivatives are eliminated through introduction of auxiliary states so as to ensure compatibility with the first-order PDE representation in Eqn. (4.5). All numerical tests in this section were performed on a computer with Intel(R) Core(TM) i7-5960x CPU @3.00 GHz, with 128 GB RAM. For each example, stability was verified by using the scripts in Subsection 7.1 to solve the optimization problem in Cor. 19, with Mosek [5] being used to solve the underlying SDP. Maximal lower bounds on the decay rate were calculated via bisection on the parameter k in Cor. 19. All PDE simulations were performed using the PIESIM routines included in PIETOOLS, expanding the solution using a basis of of 17×17 Chebyshev polynomials along the spatial directions, and using the backward Euler method for time integration.

7.2.1. Reaction-Diffusion Equation

First, consider again the reaction-diffusion equation from Ex. 4,

$$\begin{aligned} \mathbf{u}_t(t,x,y) &= \mathbf{u}_{xx}(t,x,y) + \mathbf{u}_{yy}(t,x,y) + r\mathbf{u}(t,x,y), && (x,y) \in [0,1]^2, \\ \mathbf{u}(t,0,y) &= \mathbf{u}(t,1,y) = 0, \quad \mathbf{u}(t,x,0) = \mathbf{u}_y(t,x,1) = 0, && t \geq 0, \end{aligned} \tag{7.2}$$

where $\mathbf{u}(t) \in S_2^{(2,2)}[[0,1]^2]$. A PIE representation of this PDE is presented in Ex. 3. This PDE can be shown to be exponentially PIE to PDE stable if and only if $r \leq 1\frac{1}{4}\pi^2 \approx 12.337$, with rate $k = 1\frac{1}{4}\pi^2 - r -$

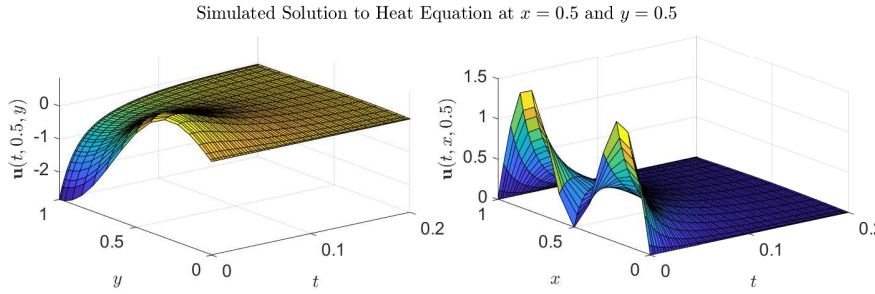


FIG. 1. Simulated solution $\mathbf{u}(t,x,y)$ to the reaction-diffusion equation in (7.2) at $x = 0.5$ (left) and $y = 0.5$ (right), using $r = 12$ and starting with initial state $\mathbf{u}(0,x,y) = \sqrt{2}\sin(\pi x)\sin(\frac{3\pi}{2}y) + \sqrt{2}\sin(3\pi x)\sin(\frac{\pi}{2}y)$.

see Appx. C.1. Using PIETOOLS to verify the stability LPI via Cor. 19 with $d = 1$, $k = 0$ and $\varepsilon = 10^{-1}$, and performing bisection on the value of r, stability of the PDE can be verified for any $r \leq 1\frac{1}{4}\pi^2 - 10^{-3}$. Fixing r and performing bisection on the value of k, exponential PIE to PDE stability can be verified with maximal rates k_{max} as in Table 1, for several values of r, and for several values of the maximal monomial degree d. The mean computation time t_{mean} needed to parse the stability LPI and solve the resulting semidefinite program for each value of d is also displayed in the table. A simulated solution to the reaction-diffusion equation for $r = 12$ is displayed in Fig. 1, at $x = 0.5$ and at $y = 0.5$, for initial state $\mathbf{u}(0,x,y) = \sqrt{2}\sin(\pi x)\sin(\frac{3\pi}{2}x) + \sqrt{2}\sin(3\pi x)\sin(\frac{\pi}{2}x)$.

	$d \setminus r$	0	2	4	6	8	10	12	12.3	t_{mean} (s)
k_{max}	0	12.336	10.336	8.3362	6.3364	4.3340	2.3340	0.33578	0.03613	43.44
	1	12.336	10.336	8.3363	6.3367	4.3367	2.3328	0.33666	0.03685	55.74
	2	12.337	10.337	8.3368	6.3369	4.3369	2.3369	0.33700	0.03700	149.1
	3	12.337	10.337	8.3368	6.3369	4.3369	2.3369	0.33700	0.03700	437.4
k_{true}		12.337	10.337	8.3370	6.3370	4.3370	2.3370	0.33700	0.03700	

TABLE 1 *Largest exponential decay rate k_{max} for which stability of the reaction-diffusion equation in (7.2) was verified for several values of the reaction parameter $r > 0$, solving the optimization problem from Cor. 19 with $\varepsilon = 10^{-1}$, and monomial degree d, for $d \in \{0:3\}$. The average time t_{mean} required to numerically parse and solve the optimization program for each value of d is displayed in the right-most column. The analytic rate of decay $k_{\text{true}} = \frac{1}{4}\pi^2 - r$ for each value r is also given.*

7.2.2. Wave Equation

Now consider the 2D wave equation with stabilizing feedback,

$$\boldsymbol{\phi}_{tt}(t,x,y) = \boldsymbol{\phi}_{xx}(t,x,y) + \boldsymbol{\phi}_{yy}(t,x,y) - 2\kappa\boldsymbol{\phi}_t(t,x,y) - \kappa^2\boldsymbol{\phi}(t,x,y), \qquad (x,y) \in [0,1]^2, \qquad (7.3)$$
$$\boldsymbol{\phi}(t,0,y) = \boldsymbol{\phi}_x(t,1,y) = 0, \qquad \boldsymbol{\phi}(t,x,0) = \boldsymbol{\phi}_y(t,x,1) = 0, \qquad t \geq 0,$$

where $\boldsymbol{\phi}(t) \in S_2^{(2,2)}[[0,1]^2]$, and $\kappa \in \mathbb{R}$. Using the augmented state $\mathbf{u}(t) = (\mathbf{u}_1(t), \mathbf{u}_2(t)) = (\boldsymbol{\phi}(t), \boldsymbol{\phi}_t(t)) \in S_2^{(2,2),2}$ to obtain a first-order form, we represent this PDE in the first-order form of Eqn. (3.1) as

$$\partial_t \begin{bmatrix} \mathbf{u}_1(t,x,y) \\ \mathbf{u}_2(t,x,y) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ \partial_x^2 + \partial_y^2 - \kappa^2 & -2\kappa \end{bmatrix} \begin{bmatrix} \mathbf{u}_1(t,x,y) \\ \mathbf{u}_2(t,x,y) \end{bmatrix}, \qquad (x,y) \in [0,1]^2, \tag{7.4}$$
$$\mathbf{u}(t,0,y) = \partial_x \mathbf{u}(t,1,y) = 0, \qquad \mathbf{u}(t,x,0) = \partial_y \mathbf{u}(t,x,1) = 0, \qquad t \geq 0.$$

For any given $\kappa \geq 0$, this PDE (7.4) can be shown to be exponentially PIE to PDE stable in the sense of Defn. 8 with rate of decay κ – see Appx. C.2.

Although PIETOOLS can be used to find a PIE representation of this PDE directly, we may provide some additional insight into the structure of the PIE by observing that $\mathbf{u}_1$ and $\mathbf{u}_2$ both satisfy the same regularity and boundary constraints. Thus, defining the operator $\mathcal{T} = \mathcal{T}_1\mathcal{T}_2$ for both $\mathbf{u}_1, \mathbf{u}_2$ as in Thm. 8, we see that if $\mathbf{u}(t)$ satisfies this PDE, then we have $\mathbf{u}_i(t) = \mathcal{T}\partial_x^2\partial_y^2\mathbf{u}_i(t)$, $\partial_x^2\mathbf{u}_i(t) = \mathcal{T}_1\partial_x^2\partial_y^2\mathbf{u}_i(t)$, and $\partial_y^2\mathbf{u}_i(t) = \mathcal{T}_2\partial_x^2\partial_y^2\mathbf{u}_i(t)$, for each $i \in \{1,2\}$, where for $\mathbf{v} \in L_2[[0,1]^2]$,

$$(\mathcal{T}_1\mathbf{v})(x,y) := -\int_0^y \eta\mathbf{v}(x,\eta)d\eta - \int_y^1 y\mathbf{v}(x,\eta)d\eta, \quad (\mathcal{T}_2\mathbf{v})(x,y) := -\int_0^x \theta\mathbf{v}(\theta,y)d\theta - \int_x^1 x\mathbf{v}(\theta,y)d\theta.$$

Introducing then the fundamental state variables $\mathbf{v}_i(t) = \partial_x^2\partial_y^2\mathbf{u}_i(t)$ for $i \in \{1,2\}$, it follows that $\mathbf{u}(t) = (\mathbf{u}_1(t), \mathbf{u}_2(t))$ satisfies the PDE (7.4) if and only if $\mathbf{v}(t) = (\mathbf{v}_1(t), \mathbf{v}_2(t))$ satisfies the PIE

$$\partial_t \overbrace{\begin{bmatrix} \mathcal{T} & 0 \\ 0 & \mathcal{T} \end{bmatrix}}^{\hat{\mathcal{T}}} \overbrace{\begin{bmatrix} \mathbf{v}_1(t) \\ \mathbf{v}_2(t) \end{bmatrix}}^{\mathbf{v}(t)} = \overbrace{\begin{bmatrix} 0 & \mathcal{T} \\ \mathcal{T}_1 + \mathcal{T}_2 - \kappa^2\mathcal{T} & -2\kappa\mathcal{T} \end{bmatrix}}^{\hat{\mathcal{A}}} \begin{bmatrix} \mathbf{v}_1(t) \\ \mathbf{v}_2(t) \end{bmatrix} = \hat{\mathcal{A}}\mathbf{v}(t), \qquad t \geq 0.$$

This PIE was simulated with PIESIM for $\kappa = 1$, and starting with initial values $\mathbf{u}_1(0,x,y) = \sin(\frac{\pi}{2}x)\sin(\frac{5\pi}{2}y)$ and $\mathbf{u}_2(0,x,y) = 0$. The resulting solution at $x = 0.5$ is plotted in Fig. 2. The figure shows that the values of both $\mathbf{u}_1(t,x,y) = \boldsymbol{\phi}(t,x,y)$ and $\mathbf{u}_2(t,x,y) = \boldsymbol{\phi}_t(t,x,y)$ converge to zero. This can be verified with PIETOOLS using Cor. 19 with operators $\{\hat{\mathcal{T}}, \hat{\mathcal{A}}\}$. Furthermore, using $\varepsilon = 10^{-1}$, $d = 1$, maximal lower bounds on the exponential decay rate, $k_{\max}$, are provided in Tab. 2, in each case tightly lower-bounding the true value of $k_{\text{true}} = \kappa$. The mean computation time to parse the stability LPI and solve the resulting semidefinite program for each fixed value of κ and $k_{\max}$ was 720 seconds, with the mean CPU time for solving the program being 477 seconds.

$\kappa = k_{\text{true}}$	1	2	3	4	5	6	7	8	9	10
$k_{\max}$	0.9999	1.9998	2.9874	3.9973	4.9980	5.9906	6.9645	7.9875	8.9859	9.9375

TABLE 2 *Largest verifiable lower bound on exponential decay rate, $k_{\max}$, for which exponential PIE to PDE stability of the wave equation in (7.4) was verified with PIETOOLS, for several values of the feedback gain parameter $\kappa > 0$ (which is also the true decay rate, k_{true}). The values of $k_{\max}$ were computed by solving Cor. 19 with, $d = 1$, and $\varepsilon = 10^{-1}$.*

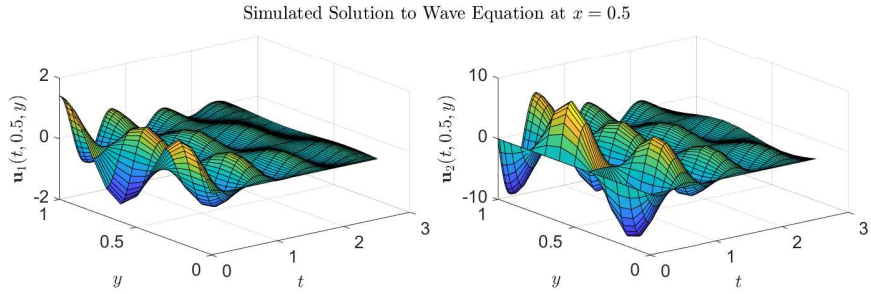


FIG. 2. Simulated solution $(\mathbf{u}_1(t,x,y), \mathbf{u}_2(t,x,y)) = (\boldsymbol{\phi}(t,x,y), \boldsymbol{\phi}_t(t,x,y))$ to the damped wave equation in (7.4) at $x = 0.5$, using $\kappa = 1$ and starting with initial values $\mathbf{u}_1(0,x,y) := \sin(\frac{\pi}{2}x)\sin(\frac{5\pi}{2}y)$ and $\mathbf{u}_2(0,x,y) = 0$.

7.2.3. Kirchhoff Plate Equation

As an example of a higher-order 2D PDE, consider the following Kirchhoff plate equation with structural damping, used to model the vertical deflection $\mathbf{w}$ of a clamped plate,

$$\begin{aligned} \mathbf{w}_{tt}(t) &= -\mathbf{w}_{xxxx}(t) - 2\mathbf{w}_{xxyy}(t) - \mathbf{w}_{yyyy}(t) + \alpha_0(\mathbf{w}_{txx}(t) + \mathbf{w}_{tyy}(t)), \qquad t \geq 0, \\ \mathbf{w}(t,0,y) &= \mathbf{w}_x(t,0,y) = \mathbf{w}(t,1,y) = \mathbf{w}_x(t,1,y) = 0, \\ \mathbf{w}(t,x,0) &= \mathbf{w}_y(t,x,0) = \mathbf{w}(t,x,1) = \mathbf{w}_y(t,x,1) = 0, \end{aligned} \tag{7.5}$$

where $\mathbf{w}(t) \in S_2^{(4,4)}[[0,1]^2]$, $(x,y) \in [0,1]^2$ and where $\alpha_0 \geq 0$ is the damping coefficient (see e.g. [16]). To represent this PDE in the form of Eqn. (3.1), we introduce the state variables $\mathbf{u}_1(t) = \mathbf{w}(t) \in S_2^{(4,4)}[[0,1]^2]$ and $\mathbf{u}_2(t) = \mathbf{w}_t(t) \in S_2^{(4,4)}[[0,1]^2]$ so that Eqn. (7.5) becomes

$$\begin{aligned} \partial_t \begin{bmatrix} \mathbf{u}_1(t) \\ \mathbf{u}_2(t) \end{bmatrix} &= \begin{bmatrix} 0 & 1 \\ -D^{(4,0)} - 2D^{(2,2)} - D^{(0,4)} & \alpha_0\left(D^{(2,0)} + D^{(0,2)}\right) \end{bmatrix} \begin{bmatrix} \mathbf{u}_1(t) \\ \mathbf{u}_2(t) \end{bmatrix}, \qquad t \geq 0, \\ \mathbf{u}_1(t,0,y) &= \partial_x \mathbf{u}_1(t,0,y) = \mathbf{u}_1(t,1,y) = \partial_x \mathbf{u}_1(t,1,y) = 0, \\ \mathbf{u}_1(t,x,0) &= \partial_y \mathbf{u}_1(t,x,0) = \mathbf{u}_1(t,x,1) = \partial_y \mathbf{u}_1(t,x,1) = 0, \\ \mathbf{u}_2(t,0,y) &= \partial_x \mathbf{u}_2(t,0,y) = \mathbf{u}_2(t,1,y) = \partial_x \mathbf{u}_2(t,1,y) = 0, \\ \mathbf{u}_2(t,x,0) &= \partial_y \mathbf{u}_2(t,x,0) = \mathbf{u}_2(t,x,1) = \partial_y \mathbf{u}_2(t,x,1) = 0. \end{aligned} \tag{7.6}$$

As for the wave equation, we use Thm. 8 to define PI operators $\mathcal{T}, \mathcal{R}_i, \mathcal{Q}_i$ such that for $\mathbf{v}_1(t) = D^{(4,4)}\mathbf{u}_1(t)$ and $\mathbf{v}_2(t) = D^{(4,4)}\mathbf{u}_2(t)$, we have

$$\partial_t \hat{\mathcal{T}}\mathbf{v}(t) = \partial_t \overbrace{\begin{bmatrix} \mathcal{T} & 0 \\ 0 & \mathcal{T} \end{bmatrix}}^{\hat{\mathcal{T}}} \begin{bmatrix} \mathbf{v}_1(t) \\ \mathbf{v}_2(t) \end{bmatrix} = \overbrace{\begin{bmatrix} 0 & \mathcal{T} \\ -\mathcal{Q}_1 - \mathcal{Q}_2 - 2\mathcal{Q}_3 & \alpha_0\left(\mathcal{R}_1 + \mathcal{R}_2\right) \end{bmatrix}}^{\hat{\mathcal{A}}} \begin{bmatrix} \mathbf{v}_1(t) \\ \mathbf{v}_2(t) \end{bmatrix} = \hat{\mathcal{A}}\mathbf{v}(t), \qquad t \geq 0.$$

Given this PIE representation, exponential stability of the PDE (7.6) can be verified with Cor. 19, using operators $\{\hat{\mathcal{T}}, \hat{\mathcal{A}}\}$. In particular, using degree $d = 0$ and $\varepsilon = 10^{-1}$, and performing bisection on the value of k, the PDE (7.6) with $\alpha_0 = 0.2$ was found to be exponentially PIE to PDE stable with rate at least $k = 3.6328$. Here, although stability properties of plate equations such as that in (7.5) have

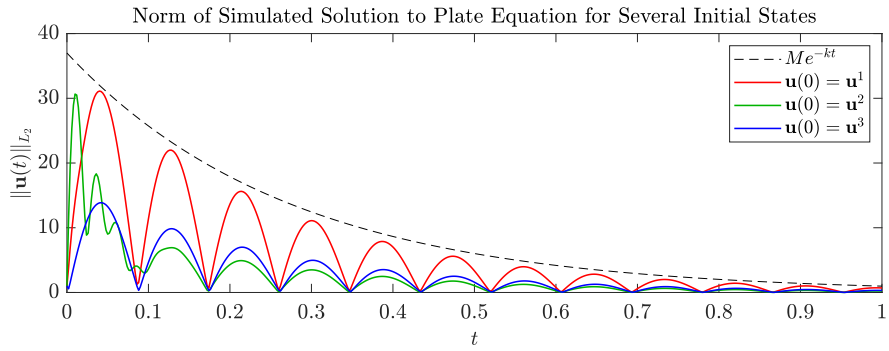


FIG. 3. Evolution of the norm $\|\mathbf{u}(t)\|_{L_2}$ of simulated solutions to the Kirchhoff plate equation in (7.6) for $\alpha_0 = 0.2$ and for normalized initial states $\mathbf{u}^i := \hat{\mathbf{u}}^i / \|\hat{\mathbf{u}}^i\|_{L_2}$, with $\hat{\mathbf{u}}^i$ as in (7.7) for $i \in \{1,2,3\}$. An exponential bound Me^{-kt} for $M = 37$ is also plotted, with rate $k = 3.6328$ corresponding to the largest rate for which exponential stability was verified using PIETOOLS.

been studied in the literature, including in e.g. [11, 16, 30], these results commonly prove exponential decay of a different energy functional $E(\mathbf{u}(t))$, rather than of the norm $\|\mathbf{u}(t)\|_{L_2}$. Therefore, to verify that the PDE is indeed exponentially stable with rate at least $k = 3.6328$, solutions to the PDE were also simulated using the PIESIM software included in PIETOOLS. The obtained evolution of the norm $\|\mathbf{u}(t)\|_{L_2} = \left\| \begin{bmatrix} \mathbf{w}(t) \\ \mathbf{w}_t(t) \end{bmatrix} \right\|_{L_2}$ for the simulated solutions is displayed in Fig. 3, for three initial conditions $\mathbf{u}(0) = \mathbf{u}^i := \hat{\mathbf{u}}^i / \|\hat{\mathbf{u}}^i\|_{L_2}$, where $\hat{\mathbf{u}}^i = (\hat{\mathbf{u}}_1^i, \hat{\mathbf{u}}_2^i)$ for

$$\begin{aligned}
\hat{\mathbf{u}}_1^1(x,y) &:= \Big(\cos(2\pi x) - 1\Big)\Big(\cos(2\pi y) - 1\Big), & \hat{\mathbf{u}}_2^1(x,y) &:= 0, \\
\hat{\mathbf{u}}_1^2(x,y) &:= \frac{1}{2}\Big(\cos(2\pi x) - 1\Big)\Big(\cos(4\pi y) - 1\Big), & \hat{\mathbf{u}}_2^2(x,y) &:= \Big(\cos(4\pi x) - 1\Big)\Big(\cos(2\pi y) - 1\Big), \\
\hat{\mathbf{u}}_1^3(x,y) &:= \frac{1}{2}x^2(1-x)^2y^2(1-y^2), & \hat{\mathbf{u}}_2^3(x,y) &:= x^2(1-x)^2y^2(1-y^2).
\end{aligned} \tag{7.7}$$

These numerical simulations verify that the norm of the solution for each initial condition decays exponentially with a rate greater than the computed lower bound of $k = 3.6328$.

8. Conclusion

We have shown how a broad class of coupled, linear PDEs on a hyper-rectangular spatial domain can be equivalently represented as Partial Integral Equations (PIEs). This representation was constructed inductively, based on a representation of the multivariate domain (including boundary conditions and Sobolev regularity constraints) as the intersection of lifted 1D domains. For such a domain decomposition, we proposed an algebraic consistency condition which was shown to be necessary and sufficient for invertibility of the multivariate spatial differential operator, D^δ, on the domain of the PDE – thus providing a bijection between the PDE domain and the Hilbert space L_2. Furthermore, an expression for this inverse was constructed inductively as the composition of univariate integral operators with polynomial semi-separable kernels. This inverse operator was then shown to be embedded in a *-algebra of multivariate Partial Integral (PI) operators, allowing for the construction of an equivalent PIE representation of the evolution of the PDE. Given this representation, a notion of

exponential PIE to PDE stability was defined and a sufficient condition for such stability was formulated as feasibility of a set of linear PI operator inequalities. Feasibility of such operator inequalities was then verified by constructing a basis for the PI operators and using positive matrices to enforce positivity. Software implementation was proposed which automated construction of the PIE representation and verification of the associated stability conditions. Finally, accuracy of the resulting stability analysis was verified by applying the stability test to 2D heat, wave, and plate equations, obtaining accurate bounds on the associated rate of decay.

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References

1. U. G. Abdulla. Generalized Newton–Leibniz formula and the embedding of the Sobolev functions with dominating mixed smoothness into Hölder spaces. *AIMS Mathematics*, 8(9):20700–20717, 2023.
2. M. Ahmadi, G. Valmorbida, and A. Papachristodoulou. Dissipation inequalities for the analysis of a class of PDEs. *Automatica*, 66:163–171, 2016.
3. M. Ahmadi, G. Valmorbida, D. Gayme, and A. Papachristodoulou. A framework for input–output analysis of wall-bounded shear flows. *Journal of Fluid Mechanics*, 873:742–785, 2019.
4. M. S. Andersen, J. Dahl, and L. Vandenberghe. *Cvxopt documentation*, 2021.
5. M. ApS. *The MOSEK optimization toolbox for MATLAB manual. Version 10.0.34*, 2023. URL http://docs.mosek.com/latest/toolbox/index.html.
6. A. Bensoussan, G. Da Prato, M. C. Delfour, and S. K. Mitter. *Representation and control of infinite dimensional systems*, volume 2. Springer, 2007.
7. S. I. Chernyshenko, P. Goulart, D. Huang, and A. Papachristodoulou. Polynomial sum of squares in fluid dynamics: a review with a look ahead. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 372(2020):20130350, 2014.
8. R. F. Curtain and H. Zwart. *An Introduction to Infinite-Dimensional Linear Systems Theory*. Springer-Verlag, Berlin, Heidelberg, 1995. ISBN 0387944753.
9. L. C. Evans. *Partial differential equations*, volume 19. American Mathematical Society, 2022.
10. G. Fantuzzi and I. Tobasco. Sharpness and non-sharpness of occupation measure bounds for integral variational problems. *arXiv eprint:2207.13570*, 2022.
11. E. Fridman, S. Nicaise, and J. Valein. Stabilization of second order evolution equations with unbounded feedback with time-dependent delay. *SIAM Journal on Control and Optimization*, 48(8):5028–5052, 2010.
12. F. Fuentes, D. Goluskin, and S. Chernyshenko. Global stability of fluid flows despite transient growth of energy. *Physical Review Letters*, 128(20):204502, 2022.
13. D. F. Gayme, B. J. McKeon, B. Bamieh, A. Papachristodoulou, and J. C. Doyle. Amplification and nonlinear mechanisms in plane Couette flow. *Physics of Fluids*, 23(6):065108, 2011.
14. I. Gohberg, S. Goldberg, and M. A. Kaashoek. *Classes of Linear Operators*, volume 1. Springer Basel AG, 1990. ISBN 9783034875110.
15. P. J. Goulart and S. Chernyshenko. Global stability analysis of fluid flows using sum-of-squares. *Physica D: Nonlinear Phenomena*, 241(6):692–704, 2012.
16. Z. Hajjej and H. Zhang. Exponential stability of a Kirchhoff plate equation with structural damping and internal time delay. *Symmetry*, 16(11):1427, 2024.
17. D. Henrion, M. Infusino, S. Kuhlmann, and V. Vinnikov. Infinite-dimensional moment-SOS hierarchy for nonlinear partial differential equations. *arXiv eprint:2305.18768*, 2023.
18. D. S. Jagt and M. M. Peet. A PIE representation of coupled linear 2D PDEs and stability analysis using LPIs. In *2022 American Control Conference (ACC)*, pages 1659–1666. IEEE, 2022.

19. M. R. Jovanović. From bypass transition to flow control and data-driven turbulence modeling: an input–output viewpoint. *Annual Review of Fluid Mechanics*, 53:311–345, 2021.
20. M. R. Jovanović and B. Bamieh. Componentwise energy amplification in channel flows. *Journal of Fluid Mechanics*, 534:145–183, 2005.
21. W. Kang and E. Fridman. Sampled-data control of 2-D Kuramoto-Sivashinsky equation. *IEEE Transactions on Automatic Control*, 67(3):1314–1326, 2021.
22. W. Kang, E. Fridman, and S. Zhuk. Sampled-data observer for 2D Navier-Stokes equation. In *2019 IEEE 58th Conference on Decision and Control (CDC)*, pages 1201–1206, 2019.
23. M. Korda, D. Henrion, and J. B. Lasserre. Moments and convex optimization for analysis and control of nonlinear PDEs. In *Handbook of Numerical Analysis*, volume 23, pages 339–366. Elsevier, 2022.
24. M. Krstic and A. Smyshlyaev. Backstepping boundary control – a tutorial. In *2007 American Control Conference*, pages 870–875. IEEE, 2007.
25. I. Lasiecka and R. Triggiani. Exact controllability and uniform stabilization of Kirchoff plates with boundary control only on $\Delta w|_{\Sigma}$ and homogeneous boundary displacement. *Journal of Differential Equations*, 93(1): 62–101, 1991.
26. I. Lasiecka and R. Triggiani. *Control theory for partial differential equations: Continuous and approximation theories*, volume 1. Cambridge University Press, 2000.
27. B. K. Lieu, M. R. Jovanović, and S. Kumar. Worst-case amplification of disturbances in inertialess Couette flow of viscoelastic fluids. *Journal of Fluid Mechanics*, 723:232–263, 2013.
28. C. Liu and D. F. Gayme. Input-output inspired method for permissible perturbation amplitude of transitional wall-bounded shear flows. *Physical Review E*, 102(6):063108, 2020.
29. C. Liu and D. F. Gayme. Structured input–output analysis of transitional wall-bounded flows. *Journal of Fluid Mechanics*, 927:A25, 2021.
30. Z. Liu and S. Zheng. Exponential stability of the Kirchhoff plate with thermal or viscoelastic damping. *Quarterly of Applied Mathematics*, 55(3):551–564, 1997.
31. J. Lofberg. YALMIP: A toolbox for modeling and optimization in MATLAB. In *2004 IEEE international conference on robotics and automation (IEEE Cat. No. 04CH37508)*, pages 284–289, 2004.
32. V. Magron and C. Prieur. Optimal control of linear pdes using occupation measures and SDP relaxations. *IMA Journal of Mathematical Control and Information*, 37(1):159–174, 2020.
33. R. Mahawattege and R. Triggiani. Fluid–structure interaction with Kelvin–Voigt damping: Analyticity, spectral analysis, exponential decay. *Applied Mathematics & Optimization*, 84(Suppl 2):1821–1863, 2021.
34. E. Meyer and M. M. Peet. Stability analysis of parabolic linear PDEs with two spatial dimensions using lyapunov method and SOS. In *2015 54th IEEE Conference on Decision and Control (CDC)*, pages 1884–1890, 2015.
35. A. A. Moghadam, I. Aksikas, S. Dubljevic, and J. F. Forbes. Boundary optimal (LQ) control of coupled hyperbolic PDEs and ODEs. *Automatica*, 49(2):526–533, 2013.
36. A. Papachristodoulou and M. M. Peet. On the analysis of systems described by classes of partial differential equations. In *2006 45th IEEE Conference on Decision and Control (CDC)*, pages 747–752. IEEE, 2006.
37. A. Papachristodoulou, J. Anderson, G. Valmorbida, S. Prajna, P. Seiler, P. Parrilo, M. M. Peet, and D. Jagt. SOSTOOLS version 4.00 sum of squares optimization toolbox for MATLAB, 2021.
38. P. A. Parrilo. *Structured semidefinite programs and semialgebraic geometry methods in robustness and optimization*. California Institute of Technology, 2000.
39. M. Riesmeier and F. Woittennek. Late lumping of transformation-based feedback laws for boundary control systems. *arXiv eprint:2211.01238*, 2022.
40. S. Shivakumar. *Analysis, Estimation, and Control of Partial Differential Equations Using Partial Integral Equation Representation*. PhD thesis, Arizona State University, 2024.
41. S. Shivakumar, A. Das, S. Weiland, and M. Peet. Extension of the partial integral equation representation to GPDE input-output systems. *arxiv eprint:2205.03735*, 2022.
42. S. Shivakumar, A. Das, S. Weiland, and M. Peet. Extension of the partial integral equation representation to GPDE input-output systems. *IEEE Transactions on Automatic Control*, pages 1–16, 2024.

43. S. Shivakumar, D. Jagt, D. Braghini, A. Das, Y. Peet, and M. Peet. PIETOOLS 2024: User manual, 2025.
44. A. Smyshlyaev and M. Krstic. Backstepping observers for a class of parabolic PDEs. *Systems & Control Letters*, 54(7):613–625, 2005.
45. J. F. Sturm. Using sedumi 1.02, a MATLAB toolbox for optimization over symmetric cones. *Optimization methods and software*, 11(1-4):625–653, 1999.
46. G. Valmorbida, M. Ahmadi, and A. Papachristodoulou. Convex solutions to integral inequalities in two-dimensional domains. In *2015 54th IEEE Conference on Decision and Control (CDC)*, pages 7268–7273, 2015.
47. R. Vazquez and M. Krstic. Boundary control and estimation of reaction–diffusion equations on the sphere under revolution symmetry conditions. *International Journal of Control*, 92(1):2–11, 2019.
48. R. Vázquez, E. Trélat, and J.-M. Coron. Control for fast and stable laminar-to-high-Reynolds-numbers transfer in a 2D Navier-Stokes channel flow. *Discrete and Continuous Dynamical Systems-Series B*, 10(4):925–956, 2008.
49. R. Vazquez, J. Auriol, F. Bribiesca-Argomedo, and M. Krstic. Backstepping for Partial Differential Equations. *arXiv eprint:2410.15146*, 2024.
50. J. Wang and L. Hu. Solving low-rank semidefinite programs via manifold optimization. *Journal of Scientific Computing*, 104(1):33, 2025.
51. C. Xu, E. Schuster, R. Vazquez, and M. Krstic. Stabilization of linearized 2D magnetohydrodynamic channel flow by backstepping boundary control. *Systems & control letters*, 57(10):805–812, 2008.
52. H. Yu, K. Kashima, and J. I. Imura. Stability analysis of 2-dimensional fluid flow based on sum of squares relaxation. In *2008 SICE Annual Conference*, pages 3321–3326. IEEE, 2008.
53. M. Zayats, E. Fridman, and S. Zhuk. Global observer design for Navier-Stokes equations in 2D. In *2021 60th IEEE Conference on Decision and Control (CDC)*, pages 1862–1867, 2021.
54. Y. Zheng, G. Fantuzzi, A. Papachristodoulou, P. Goulart, and A. Wynn. Chordal decomposition in operator-splitting methods for sparse semidefinite programs. *Mathematical Programming*, 180(1):489–532, 2020.
55. S. Zhuk, M. Zayats, and E. Fridman. Detectability and global observer design for 2D Navier-Stokes equations with uncertain inputs. *Automatica*, 153:111043, 2023.

A. A Bijection Between the PDE Domain and Fundamental State Space

In Section 4, an explicit expression was derived for the inverse of the multivariate differential operator $D^{\boldsymbol{\delta}} : \mathcal{D} \to L_2[\Omega]$, on the PDE domain $\mathcal{D} \subseteq S_2^{\boldsymbol{\delta}}[\Omega]$. Specifically, it was shown that if we can decompose $\mathcal{D} = \bigcap_{i=1}^N \mathcal{D}_i$, for lifted univariate domains $\mathcal{D}_i$, then the inverse to $D^{\boldsymbol{\delta}} : \mathcal{D} \to L_2[\Omega]$ can be expressed as the product $\mathcal{T} = \prod_{i=1}^N \mathcal{T}_i : L_2[\Omega] \to \mathcal{D}$ of the inverse $\mathcal{T}_i$ to each $\partial_{s_i}^{\delta_i} : \mathcal{D}_i \to L_2[\Omega]$. The proof of this result, however, relies on the commutability of the operators $\mathcal{T}_i, \partial_{s_j}^{\delta_j}$, and of the operators $\mathcal{T}_i, \mathcal{T}_j$, for distinct i, j. In this appendix, we show that these commutative properties are indeed satisfied under the conditions presented in Section 4 – in particular when the domains $\mathcal{D}_i$ are consistent – providing some of the proofs and details omitted from that section.

To start, consider a univariate domain $\mathcal{D} \subseteq S_2^{d,n}[a,b]$ as in (4.2), lifted to the multivariate setting as $\mathcal{D} \mapsto \mathcal{D}_i = \hat{\mathcal{D}}[i] \subseteq S_2^{\delta_i \cdot \mathbf{e}_i, n}[\Omega]$ using $d \mapsto \delta_i$ and $[a,b] \mapsto [a_i, b_i]$. Recall from Cor. 3 and Cor. 3 that if $\mathcal{D}$ is admissible as per Defn. 1, we can define the inverse to $\partial_{s_i}^{\delta_i} : \mathcal{D}_i \to L_2^n[\Omega]$ as an integral operator $\mathcal{T}_i : L_2^n[\Omega] \to \mathcal{D}_i$ with polynomial semi-separable kernel. As such, for any distinct $i, j \in \{1 : N\}$, we can define a right-inverse to $\partial_{s_i}^{\delta_i} \partial_{s_j}^{\delta_j} : \mathcal{D}_i \cap \mathcal{D}_j \to L_2^n[\Omega]$ as $\mathcal{T}_j \mathcal{T}_i$, satisfying $\partial_{s_i}^{\delta_i} \partial_{s_j}^{\delta_j} \mathcal{T}_j \mathcal{T}_i \mathbf{v} = \partial_{s_i}^{\delta_i} \mathcal{T}_i \mathbf{v} = \mathbf{v}$ for all $\mathbf{v} \in L_2^n[\Omega]$. In order to show that this operator also satisfies $\mathcal{T}_j \mathcal{T}_i \partial_{s_i}^{\delta_i} \partial_{s_j}^{\delta_j} \mathbf{u} = \mathbf{u}$ for $\mathbf{u} \in \mathcal{D}_i \cap \mathcal{D}_j$, however, we first need to prove that $\partial_{s_j}^{\delta_j} \mathbf{u} \in \mathcal{D}_i$, for which we have the following lemma.

Lemma 20 *Let $N \in \mathbb{N}$, $\boldsymbol{\delta} \in \mathbb{N}_0^N$, and $\Omega = \prod_{i=1}^N [a_i, b_i]$. For given $B_{j,k}^i, C_{j,k}^i$, let associated domains $\mathcal{D}_i$ be of the form in (4.2) (letting $B_{j,k} = B_{j,k}^i$, $C_{j,k} = C_{j,k}^i$, $[a,b] = [a_i, b_i]$ and $d = \delta_i$), and lifted to the multivariate setting as defined in Subsection 2.1 (so that $\mathcal{D}_i = \hat{\mathcal{D}}[i] \subseteq S_2^{\delta_i \cdot \mathbf{e}_i, n}[\Omega]$) for each $i \in \{1 : N\}$. If $\mathbf{u} \in \mathcal{D} := \mathcal{D}_1 \cap \cdots \cap \mathcal{D}_N$, then $\partial_{s_j}^{\alpha_j} \cdots \partial_{s_N}^{\alpha_N} \mathbf{u} \in \mathcal{D}_1 \cap \cdots \cap \mathcal{D}_{j-1}$ for all $j \in \{1 : N\}$ and $\vec{0} \leq \boldsymbol{\alpha} \leq \boldsymbol{\delta}$.*

Proof Fix arbitrary $\vec{0} \leq \boldsymbol{\alpha} \leq \boldsymbol{\delta}$. We prove the result by induction on j, ranging from N to 1. For the base case, $j = N$, fix arbitrary $\mathbf{u} \in \mathcal{D}$. Then $\mathbf{u} \in S_2^{\delta_i \cdot \mathbf{e}_i, n}[\Omega]$ for all $i \in \{1 : N\}$, and therefore $\partial_{s_N}^{\alpha_N} \mathbf{u} \in S_2^{\delta_i \cdot \mathbf{e}_i, n}[\Omega]$ for all $i \in \{1 : N-1\}$. In addition, by definition of the domains $\mathcal{D}_i$, we have

$$\sum_{k=0}^{\delta_i - 1} \left[B_{\ell,k}^i (\partial_{s_i}^k \mathbf{u})(s)|_{s_i = a_i} + C_{\ell,k}^i (\partial_{s_i}^k \mathbf{u})(s)|_{s_i = b_i} \right] = 0, \qquad \forall \ell \in \{0 : \delta_i - 1\},$$

for all $i \in \{1 : N-1\}$. It follows that also

$$\begin{aligned} &\sum_{k=0}^{\delta_i - 1} \left[B_{\ell,k}^i (\partial_{s_i}^k \partial_{s_N}^{\alpha_N} \mathbf{u})(s)|_{s_i = a_i} + C_{\ell,k}^i (\partial_{s_i}^k \partial_{s_N}^{\alpha_N} \mathbf{u})(s)|_{s_i = b_i} \right] \\ &\qquad = \partial_{s_N}^{\alpha_N} \sum_{k=0}^{\delta_i - 1} \left[B_{\ell,k}^i (\partial_{s_i}^k \mathbf{u})(s)|_{s_i = a_i} + C_{\ell,k}^i (\partial_{s_i}^k \mathbf{u})(s)|_{s_i = b_i} \right] = 0, \end{aligned}$$

for every $\ell \in \{0 : \delta_i - 1\}$, and therefore $\partial_{s_N}^{\alpha_N} \mathbf{u} \in \mathcal{D}_i$, for all $i \in \{1 : N-1\}$. Thus, $\partial_{s_N}^{\alpha_N} \mathbf{u} \in \mathcal{D}_1 \cap \cdots \cap \mathcal{D}_{N-1}$.

Now, suppose for induction the lemma statement holds for all $j \in \{J : N\}$, for some $J \in \{1 : N\}$, and let $j = J - 1$. Then, for any $\mathbf{u} \in \mathcal{D}$, we have $\partial_{s_{j+1}}^{\alpha_{j+1}} \cdots \partial_{s_N}^{\alpha_N} \mathbf{u} \in \mathcal{D}_1 \cap \cdots \cap \mathcal{D}_j \subseteq S_2^{\delta_1, n}[\Omega] \cap \cdots \cap S_2^{\delta_j \cdot \mathbf{e}_j, n}[\Omega]$,

by the induction hypothesis. It follows that $\partial_{s_j}^{\alpha_j}\cdots\partial_{s_N}^{\alpha_N}\mathbf{u}\in S_2^{\delta_i\cdot e_i,n}[\Omega]$ for all $i\in\{1:j-1\}$. In addition, by definition of the domains $\mathcal{D}_i$, we have

$$\sum_{k=0}^{\delta_i-1}\Big[B^i_{\ell,k}\big(\partial^k_{s_i}(\partial^{\alpha_{j+1}}_{s_{j+1}}\cdots\partial^{\alpha_N}_{s_N}\mathbf{u})\big)(s)|_{s_i=a_i}+C^i_{\ell,k}\big(\partial^k_{s_i}(\partial^{\alpha_{j+1}}_{s_{j+1}}\cdots\partial^{\alpha_N}_{s_N}\mathbf{u})\big)(s)|_{s_i=b_i}\Big]=0,\qquad\forall\ell\in\{0:\delta_i-1\},$$

for all $i\in\{1:j\}$. It follows that also

$$\begin{aligned}&\sum_{k=0}^{\delta_i-1}\Big[B^i_{\ell,k}\big(\partial^k_{s_i}(\partial^{\alpha_j}_{s_j}\cdots\partial^{\alpha_N}_{s_N}\mathbf{u})\big)(s)|_{s_i=a_i}+C^i_{\ell,k}\big(\partial^k_{s_i}(\partial^{\alpha_j}_{s_j}\cdots\partial^{\alpha_N}_{s_N}\mathbf{u})\big)(s)|_{s_i=b_i}\Big]\\&\quad=\partial^{\alpha_j}_{s_j}\sum_{k=0}^{\delta_i-1}\Big[B^i_{\ell,k}\big(\partial^k_{s_i}(\partial^{\alpha_{j+1}}_{s_{j+1}}\cdots\partial^{\alpha_N}_{s_N}\mathbf{u})\big)(s)|_{s_i=a_i}+C^i_{\ell,k}\big(\partial^k_{s_i}(\partial^{\alpha_{j+1}}_{s_{j+1}}\cdots\partial^{\alpha_N}_{s_N}\mathbf{u})\big)(s)|_{s_i=b_i}\Big]=0,\end{aligned}$$

for all $\ell\in\{0:\delta_j\}$, and therefore $\partial^{\alpha_j}_{s_j}\cdots\partial^{\alpha_N}_{s_N}\mathbf{u}\in\mathcal{D}_i$, for all $i\in\{1:j-1\}$. We find that $\partial^{\alpha_j}_{s_j}\cdots\partial^{\alpha_N}_{s_N}\mathbf{u}\in\mathcal{D}_1\cap\cdots\cap\mathcal{D}_{j-1}$, whence the result holds by induction. □

Lem. 20 proves that, for any $\mathbf{u}\in\mathcal{D}_1\cap\cdots\cap\mathcal{D}_N$, we have $\partial^{\delta_j}_{s_j}\cdots\partial^{\delta_N}_{s_N}\mathbf{u}\in\mathcal{D}_1\cap\cdots\cap\mathcal{D}_{j-1}$ for all $j\in\{1:N\}$. As such, if both $\mathcal{D}_i$ and $\mathcal{D}_j$ are admissible as per Defn. 1, then for any $\mathbf{u}\in\mathcal{D}_i\cap\mathcal{D}_j$ we have by Cor. 2 and Cor. 3 that $\partial^{\delta_j}_{s_j}\mathbf{u}=\mathcal{T}_i\partial^{\delta_i}_{s_i}(\partial^{\delta_j}_{s_j}\mathbf{u})$, and thus $\mathcal{T}_j\mathcal{T}_i\partial^{\delta_i}_{s_i}\partial^{\delta_j}_{s_j}\mathbf{u}=\mathbf{u}$. More generally, the following corollary proves that $\mathcal{T}:=\mathcal{T}_N\cdots\mathcal{T}_1$ defines a left-inverse to $D^\delta:\mathcal{D}\to L^n_2$, as well as a right-inverse to the extension of D^δ to the range of $\mathcal{T}$.

Corollary 21 *Let $\mathcal{T}:=\mathcal{T}_N\cdots\mathcal{T}_1$ and $\mathcal{D}:=\mathcal{D}_1\cap\cdots\cap\mathcal{D}_N$, for $\mathcal{T}_i$ as in Cor. 3. Then the following statements hold:*

1. $\mathcal{T}D^\delta\mathbf{u}=\mathbf{u}$ *for any* $\mathbf{u}\in\mathcal{D}$.
2. $D^\delta\mathcal{T}\mathbf{v}=\mathbf{v}$ *for any* $\mathbf{v}\in L^n_2$.

Proof For the first statement, we note that, by Lem. 20, we have $\partial^{\delta_{j+1}}_{s_{j+1}}\cdots\partial^{\delta_N}_{s_N}\mathbf{u}\in\mathcal{D}_1\cap\cdots\cap\mathcal{D}_j\subseteq\mathcal{D}_j$ for all $\mathbf{u}\in\mathcal{D}$ and $j\in\{1:N\}$. Since, by Cor. 3, $\mathcal{T}_j\partial^{\delta_j}_{s_j}\mathbf{u}=\mathbf{u}$ for all $\mathbf{u}\in\mathcal{D}_j$, it follows that for every $\mathbf{u}\in\mathcal{D}$,

$$\begin{aligned}\mathcal{T}D^\delta\mathbf{u}&=\mathcal{T}_N\cdots\mathcal{T}_2(\mathcal{T}_1\partial^{\delta_1}_{s_1})(\partial^{\delta_2}_{s_2}\cdots\partial^{\delta_N}_{s_N}\mathbf{u})\\&=\mathcal{T}_N\cdots\mathcal{T}_3(\mathcal{T}_2\partial^{\delta_2}_{s_2})(\partial^{\delta_3}_{s_3}\cdots\partial^{\delta_N}_{s_N}\mathbf{u})=\ \cdots\ =\mathcal{T}_N\partial^{\delta_N}_{s_N}\mathbf{u}=\mathbf{u}.\end{aligned}$$

Now, for the second statement, note that $\mathcal{T}_{j-1}\cdots\mathcal{T}_1\mathbf{v}\in L^n_2$ for all $\mathbf{v}\in L^n_2$ and $j\in\{1:N\}$. Since, by Cor. 3, $\partial^{\delta_j}_{s_j}\mathcal{T}_j\mathbf{v}=\mathbf{v}$ for all $\mathbf{v}\in L^n_2$, it follows that

$$\begin{aligned}D^\delta\mathcal{T}\mathbf{v}&=\partial^{\delta_1}_{s_1}\cdots\partial^{\delta_{N-1}}_{s_{N-1}}(\partial^{\delta_N}_{s_N}\mathcal{T}_N)(\mathcal{T}_{N-1}\cdots\mathcal{T}_1\mathbf{v})\\&=\partial^{\delta_1}_{s_1}\cdots\partial^{\delta_{N-2}}_{s_{N-2}}(\partial^{\delta_{N-1}}_{s_{N-1}}\mathcal{T}_{N-1})(\mathcal{T}_{N-2}\cdots\mathcal{T}_1\mathbf{v})=\ \cdots\ =\partial^{\delta_1}_{s_1}\mathcal{T}_1\mathbf{v}=\mathbf{v}.\end{aligned}$$

□

Cor. 21 shows that $\mathcal{T} := \mathcal{T}_N \cdots \mathcal{T}_1$ defines both a left- and right-inverse to $D^\delta : \mathcal{D} \to L_2^n$ on the image of D^δ. In order for $\mathcal{T}$ to define an inverse to $D^\delta : \mathcal{D} \to L_2^n$, then, it remains to prove that the image of D^δ is indeed L_2^n, or equivalently, that $\mathcal{T}\mathbf{v} \in D^\delta$ for all $\mathbf{v} \in L_2^n$. Unfortunately, admissibility of the domains $\mathcal{D}_i$ is insufficient to guarantee that this is indeed the case. Indeed, as shown in Subsection 4.2 for $\Omega = [0,1]^2$ and $\delta = (d,d)$, a unique inverse to $\partial_x^d \partial_y^d : \mathcal{D}_1 \cap \mathcal{D}_2 \to L_2^n[[0,1]^2]$ may not exist if the boundary conditions defining $\mathcal{D}_1$ and $\mathcal{D}_2$ impose conflicting constraints at any of the corners of the domain, i.e. $(x,y) \in \{(0,0),(1,0),(0,1),(1,1)\}$. A necessary condition on the parameters $\{B^1_{\ell,k}, C^1_{\ell,k}\}$ and $\{B^2_{\ell,k}, C^2_{\ell,k}\}$ to avoid such conflicting conditions was presented in Lem. 5, which we restate and proof here.

Lemma 22 *Suppose $B^i_{j,k}, C^i_{j,k}$ as in Eqn. (4.2) define admissible domains $\mathcal{D} \subseteq S_2^{d,n}[0,1]$ as per Defn. 1 (using $B_{j,k} = B^i_{j,k}$, $C_{j,k} = C^i_{j,k}$, $[a,b] = [0,1]$), lifted to the multivariate setting as $\mathcal{D} \mapsto \mathcal{D}_i$, and let $[H^i_a]_{j,k} = B^i_{j,k}$, $[H^i_b]_{j,k} = C^i_{j,k}$. Define associated $K^i := \left(H^i_a + H^i_b \boldsymbol{Q}(b_i - a_i)\right)^{-1} H^i_b \in \mathbb{R}^{nd \times nd}$ for $\boldsymbol{Q}$ as in Defn. 1, and decompose the matrix K^i into blocks as*

$$K^i = \begin{bmatrix} K^i_{1,1} & \cdots & K^i_{1,d} \\ \vdots & \ddots & \vdots \\ K^i_{d,1} & \cdots & K^i_{d,d} \end{bmatrix}, \qquad \text{where} \qquad K^i_{k,\ell} \in \mathbb{R}^{n \times n},\ \forall k,\ell \in \{1:d\}. \tag{A.1}$$

If there exists an operator $\mathcal{T} : L_2^n[[0,1]^2] \to \mathcal{D} := \mathcal{D}_i \cap \mathcal{D}_j$ such that $\partial_x^d \partial_y^d \mathcal{T}\mathbf{v} = \mathbf{v}$ for all $\mathbf{v} \in L_2^n[[0,1]^2]$, then

$$K^1_{k,p} K^2_{\ell,q} = K^2_{\ell,q} K^1_{k,p}, \qquad \forall k,p,\ell,q \in \{1:d\}. \tag{A.2}$$

Proof Suppose there exists an operator $\mathcal{T} : L_2^n[[0,1]^2] \to \mathcal{D}$ such that $\partial_x^d \partial_y^d \mathcal{T}\mathbf{v} = \mathbf{v}$ for all $\mathbf{v} \in L_2^n[[0,1]^2]$. For any $\mathbf{v} \in L_2^n[\Omega]$, let $\mathbf{u} = \mathcal{T}\mathbf{v}$. Then, $\mathbf{v} = \partial_x^d \partial_y^d \mathbf{u}$ and $\mathbf{u} \in \mathcal{D}$. By Lem. 20, it follows that also $\partial_y^d \mathbf{u} \in \mathcal{D}_1$ and $\partial_x^d \mathbf{u} \in \mathcal{D}_2$.

Now, fix arbitrary $k,\ell \in \{1:d\}$. To prove that (A.2) holds, we will use the fact that $\mathbf{u} \in \mathcal{D}_2$ and $\partial_y^d \mathbf{u} \in \mathcal{D}_1$, and the fact that $\mathbf{u} \in \mathcal{D}_1$ and $\partial_x^d \mathbf{u} \in \mathcal{D}_2$, to derive two distinct expressions for the same corner value, $(\partial_x^{k-1} \partial_y^{\ell-1} \mathbf{u})(0,0)$, in terms of $\mathbf{v} = \partial_x^d \partial_y^d \mathbf{u}$. First, given that $\mathbf{u} \in \mathcal{D}_2$, we remark that by the proof of Cor. 2, $\mathbf{u}$ must satisfy

$$\begin{aligned} 0 &= H^2_a \left(\partial_y^{(0:d-1)} \mathbf{u}\right)(x,0) + H^2_b \left(\partial_y^{(0:d-1)} \mathbf{u}\right)(x,1) \\ &= \left(H^2_a + H^2_b \boldsymbol{Q}(1)\right) \left(\partial_y^{(0:d-1)} \mathbf{u}\right)(x,0) + \int_0^1 H^2_b \mathbf{e}_d(1-y) \partial_y^d \mathbf{u}(x,y)\, dy. \end{aligned}$$

where $\boldsymbol{Q}$ is as in Defn. 1, and where

$$\mathbf{e}_d(z) := \begin{bmatrix} \frac{z^{d-1}}{(d-1)!} \\ \vdots \\ \frac{1}{2} z^2 \\ z \\ 1 \end{bmatrix} \otimes I_n, \qquad \partial_y^{(0:d-1)} \mathbf{u} := \begin{bmatrix} \mathbf{u} \\ \partial_y \mathbf{u} \\ \vdots \\ \partial_y^{d-1} \mathbf{u} \end{bmatrix}.$$

Since, by assumption, the domains $\mathcal{D}_1$ and $\mathcal{D}_2$ are admissible, it follows by Defn. 1 that the matrix $(H_a^2 + H_{j,k}^2 \boldsymbol{Q}(1)) \in \mathbb{R}^{nd \times nd}$ is invertible, and thus $\mathbf{u}$ must satisfy

$$\left(\partial_y^{(0:d-1)}\mathbf{u}\right)(x,0) = -\int_0^1 \left(H_a^2 + H_b^2 \boldsymbol{Q}(1)\right)^{-1} H_b^2 \mathbf{e}_d (1-y) \partial_y^d \mathbf{u}(x,y)\, dy = -\int_0^1 K^2 \mathbf{e}_d (1-y)\, \partial_y^d \mathbf{u}(x,y)\, dy,$$

Taking the derivative ∂_x^{k-1} of both sides of this identity, and remarking that $\partial_y^{\ell-1}\mathbf{u} = (\mathrm{e}_\ell \otimes I_n)^T \partial_y^{(0:d-1)}\mathbf{u}$ for $\mathrm{e}_\ell \in \mathbb{R}^d$ the ℓth standard Euclidean basis vector, it follows that the derivative $\partial_x^{k-1}\partial_y^{\ell-1}\mathbf{u}$ must satisfy

$$\left(\partial_x^{k-1}\partial_y^{\ell-1}\mathbf{u}\right)(x,0) = (\mathrm{e}_\ell \otimes I_n)^T \partial_x^{k-1}\left(\partial_y^{(0:d-1)}\mathbf{u}\right)(x,0) = -\int_0^1 K^2_{(\ell,:)} \mathbf{e}_d (1-y)\, \partial_x^{k-1}\partial_y^d \mathbf{u}(x,y)\, dy.$$

where now $K^2_{(\ell,:)} := (\mathrm{e}_\ell \otimes I_n)^T K^2$. Here, since $\partial_y^d \mathbf{u} \in \mathcal{D}_1$, by Cor. 2 and Cor. 3 we can express $\partial_x^{k-1}(\partial_y^d \mathbf{u}) = \mathcal{R}_1^{k-1}\partial_x^d(\partial_y^d \mathbf{u}) = \mathcal{R}_1^{k-1}\mathbf{v}$, where the operator $\mathcal{R}_1^k$ is defined by

$$(\mathcal{R}_1^{k-1}\mathbf{v})(x,y) := \int_0^1 \boldsymbol{G}_{k-1}(x,\theta)\mathbf{v}(\theta,y)d\theta, \qquad \boldsymbol{G}_k(x,\theta) := \begin{cases} \mathbf{c}_{k-1}(x)^T(I_{nd} - K^1)\mathbf{e}_d(1-\theta) & \theta \le x, \\ -\mathbf{c}_{k-1}(x)^T K^1 \mathbf{e}_d(1-\theta) & x < \theta, \end{cases}$$

where

$$\mathbf{e}_1(z) := \begin{bmatrix} 1 \\ z \\ \frac{1}{2}z^2 \\ \vdots \\ \frac{z^{d-1}}{(d-1)!} \end{bmatrix} \otimes I_n, \qquad \mathbf{c}_{k-1}(z) := \partial_z^{k-1}\mathbf{e}_1(z) = \begin{bmatrix} 0_{k-1} \\ 1 \\ z \\ \vdots \\ \frac{z^{d-k}}{(d-k)!} \end{bmatrix} \otimes I_n.$$

It follows that

$$\left(\partial_x^{k-1}\partial_y^{\ell-1}\mathbf{u}\right)(x,0) = -\int_0^1 K^2_{(\ell,:)} \mathbf{e}_d (1-y) \left(\mathcal{R}_1^{k-1}\mathbf{v}\right)(x,y)\, dy.$$

Evaluating this expression at $x = 0$, we note that, by definition of $\mathcal{R}_1^{k-1}$, we have

$$\left(\mathcal{R}_1^{k-1}\mathbf{v}\right)(0,y) = -\int_0^1 \mathbf{c}_{k-1}(0-0)^T K^1 \mathbf{e}_d (1-x)\, \mathbf{v}(x,y)\, dx = -\int_0^1 K^1_{(k,:)} \mathbf{e}_d (1-x)\, \mathbf{v}(x,y)\, dx.$$

Using this expression it follows that

$$\begin{aligned} \left(\partial_x^{k-1}\partial_y^{\ell-1}\mathbf{u}\right)(0,0) &= -\int_0^1 K^2_{(\ell,:)} \mathbf{e}_d (1-y) \left(\mathcal{R}_1^{k-1}\mathbf{v}\right)(0,y)\, dy \\ &= \int_0^1 K^2_{(\ell,:)} \mathbf{e}_d (1-y) \left(\int_0^1 K^1_{(k,:)} \mathbf{e}_d (1-x) \mathbf{v}(x,y)\, dx \right) dy \\ &= \int_0^1 \int_0^1 \left(K^2_{(\ell,:)} \mathbf{e}_d (1-y)\right)\left(K^1_{(k,:)} \mathbf{e}_d (1-x)\right) \mathbf{v}(x,y)\, dy\, dx. \end{aligned}$$

Here, to derive this expression for $\partial_x^{k-1}\partial_y^{\ell-1}\mathbf{u}(0,0)$, we only used the fact that $\mathbf{u} \in \mathcal{D}_2$ and $\partial_y^d \mathbf{u} \in \mathcal{D}_1$. Conversely, then, using the fact that $\mathbf{u} \in \mathcal{D}_1$ and $\partial_x^d \mathbf{u} \in \mathcal{D}_2$, we can apply similar reasoning to find that

also

$$\big(\partial_x^{k-1}\partial_y^{\ell-1}\mathbf{u}\big)(0,0) = \int_0^1\int_0^1 \big(K^1_{(k,:)}\mathbf{e}_d(1-x)\big)\big(K^2_{(\ell,:)}\mathbf{e}_d(1-y)\big)\mathbf{v}(x,y)\,dy\,dx.$$

Since $\mathbf{u}\in\mathcal{D}\subseteq S_2^{(d,d),n}[\Omega]$ and $k,\ell\le d$, the derivative $\partial_x^{k-1}\partial_y^{\ell-1}\mathbf{u}$ is continuous, and therefore we must have

$$\begin{aligned}\int_0^1\int_0^1 \big(K^2_{(\ell,:)}\mathbf{e}_d(1-y)\big)\big(K^1_{(k,:)}\mathbf{e}_d(1-x)\big)\mathbf{v}(x,y)\,dy\,dx&\\ = \big(\partial_x^{k-1}\partial_y^{\ell-1}\mathbf{u}\big)(0,0) = \int_0^1\int_0^1 \big(K^1_{(k,:)}\mathbf{e}_d(1-x)\big)\big(K^2_{(\ell,:)}\mathbf{e}_d(1-y)\big)\mathbf{v}(x,y)\,dy\,dx.&\end{aligned}$$

Since this holds for arbitrary $\mathbf{v}\in L_2^n[\Omega]$, it follows that in fact

$$\big(K^2_{(\ell,:)}\mathbf{e}_d(y)\big)\big(K^1_{(k,:)}\mathbf{e}_d(x)\big) - \big(K^1_{(k,:)}\mathbf{e}_d(x)\big)\big(K^2_{(\ell,:)}\mathbf{e}_d(y)\big) = 0 \qquad \text{a.e. } x,y\in[0,1]. \tag{$\star$}$$

Finally, to see that this implies (A.2), we decompose K^i and K^j as in (A.1), and invoke the definition of $\mathbf{e}_d$ to find

$$K^1_{(k,:)}\mathbf{e}_d(z) = \begin{bmatrix}K^1_{k,1} & K^1_{k,2} & \dots & K^1_{k,d}\end{bmatrix}\begin{bmatrix}\frac{z^{d-1}}{(d-1)!}I_n\\ \frac{z^{d-2}}{(d-2)!}I_n\\ \vdots\\ I_n\end{bmatrix} = \sum_{p=1}^{d}\frac{1}{(d-p)!}K^1_{k,p}z^{d-p}.$$

Similarly, we can express $K^2_{(\ell,:)}\mathbf{e}_d(z) = \sum_{q=1}^d \frac{1}{(d-q)!}K^2_{\ell,q}z^{d-q}$, for all $z\in\mathbb{R}$. It follows that

$$\begin{aligned}\big(K^2_{(\ell,:)}\mathbf{e}_d(y)\big)\big(K^1_{(k,:)}\mathbf{e}_d(x)\big) - \big(K^1_{(k,:)}\mathbf{e}_d(x)\big)\big(K^2_{(\ell,:)}\mathbf{e}_d(y)\big)&\\ = \sum_{p,q=1}^{d}\frac{1}{(d-p)!(d-q)!}\Big[K^1_{k,p}K^2_{\ell,q} - K^2_{\ell,q}K^1_{k,p}\Big]x^{d-p}y^{d-q},&\end{aligned}$$

for all $x,y\in[0,1]$. By ($\star$), it follows that $K^i_{k,p}K^j_{\ell,q} = K^j_{\ell,q}K^i_{k,p}$ for all $p,q\in\{1:d\}$. □

Lem. 22 proves that the differential operator $\partial_x^d\partial_y^d : \mathcal{D}_1\cap\mathcal{D}_2 \to L_2^n[[0,1]^2]$ admits a right-inverse only if a suitable commutative condition is satisfied, ensuring that the boundary conditions defining $\mathcal{D}_1$ and $\mathcal{D}_2$ can be simultaneously satisfied at the corner $(x,y)=(0,0)$. We refer to this commutative property as consistency of the domains $\mathcal{D}_1$ and $\mathcal{D}_2$ – see also Defn. 2. In order to prove that consistency of the PDE domains is also sufficient for $\partial_x^d\partial_y^d : \mathcal{D}_1\cap\mathcal{D}_2 \to L_2^n[[0,1]^2]$ to be right-invertible, we remark that the operators $\mathcal{T} = \mathcal{T}_2\mathcal{T}_1$ and $\tilde{\mathcal{T}} = \mathcal{T}_1\mathcal{T}_2$ both already satisfy $\partial_x^d\partial_y^d\mathcal{T} = \partial_x^d\partial_y^d\tilde{\mathcal{T}} = I_n$, though $\mathcal{T}\mathbf{v}\notin\mathcal{D}_1$ and $\tilde{\mathcal{T}}\mathbf{v}\notin\mathcal{D}_2$ for general $\mathbf{v}\in L_2^n[[0,1]^2]$. As such, a right-inverse to $\partial_x^d\partial_y^d : \mathcal{D}_1\cap\mathcal{D}_2\to L_2^n[[0,1]^2]$ exists only if the operators $\mathcal{T}_1$ and $\mathcal{T}_2$ commute. The following lemmas, generalizing Lem. 6 in Sec. 4.2, prove that this condition is equivalent to consistency of the domains $\mathcal{D}_1$ and $\mathcal{D}_2$.

Lemma 23 *For $i,j\in\{1:N\}$, let $\boldsymbol{T}_i(x,\theta) := -\mathbf{e}_1(x-a_i)^T K\mathbf{e}_{\delta_i}(b_i-\theta)$ for some $K\in\mathbb{R}^{n\delta_i\times n\delta_i}$, and where $\mathbf{e}_1,\mathbf{e}_{\delta_i}$ are as in Cor. 2 (using $d=\delta_i$). Then $\boldsymbol{T}_i\boldsymbol{T}_j = \boldsymbol{T}_j\boldsymbol{T}_i$ if and only if $K^i_{k,p}K^j_{\ell,q} = K^j_{\ell,q}K^i_{k,p}$ for all $k,p\in\{1:\delta_i\}$ and $\ell,q\in\{1:\delta_j\}$.*

Proof By definition, we have

$$\begin{aligned}\boldsymbol{T}_i(x+a_i,b_i-\theta) &:= -\mathbf{e}_1(x)K^i\mathbf{e}_{\delta_i}(\theta)\\ &= \begin{bmatrix} I_n \\ xI_n \\ \vdots \\ \frac{x^{\delta_i-1}}{(\delta_i-1)!}I_n \end{bmatrix}^T \begin{bmatrix} K^i_{1,1} & K^i_{1,2} & \cdots & K^i_{1,\delta_i} \\ K^i_{2,1} & K^i_{2,2} & \cdots & K^i_{2,\delta_i} \\ \vdots & \vdots & \ddots & \vdots \\ K^i_{\delta_i,1} & K^i_{\delta_i,2} & \cdots & K^i_{\delta_i,\delta_i} \end{bmatrix} \begin{bmatrix} \frac{\theta^{\delta_i-1}}{(\delta_i-1)!}I_n \\ \frac{\theta^{\delta_i-2}}{(\delta_i-2)!}I_n \\ \vdots \\ I_n \end{bmatrix}\\ &= \sum_{k,p=1}^{\delta_i} \frac{1}{(k-1)!(\delta_i-p)!}K^i_{k,p}x^{k-1}\theta^{\delta_i-p},\end{aligned}$$

for all $x,\theta\in[0,b_i-a_i]$. It follows that

$$\begin{aligned}&\boldsymbol{T}_i(x+a_i,b_i-\theta)\boldsymbol{T}_j(y+a_j,b_j-\eta)-\boldsymbol{T}_j(y+a_j,b_j-\eta)\boldsymbol{T}_i(x+a_i,b_i-\theta)\\ &\quad=\sum_{k,p=1}^{\delta_i}\sum_{\ell,q=1}^{\delta_j}\frac{1}{(k-1)!(\delta_i-p)!}\frac{1}{(\ell-1)!(\delta_j-q)!}\left[K^i_{k,p}K^j_{\ell,q}-K^j_{\ell,q}K^i_{k,p}\right]x^{k-1}\theta^{\delta_i-p}y^{\ell-1}\eta^{\delta_j-q},\end{aligned}$$

for all $x,\theta\in[0,b_i-a_i]$ and $y,\eta\in[0,b_j-a_j]$. It follows that $\boldsymbol{T}_i\boldsymbol{T}_j=\boldsymbol{T}_j\boldsymbol{T}_i$ if and only if $K^i_{k,p}K^j_{\ell,q}=K^j_{\ell,q}K^i_{k,p}$ for all k,p,ℓ,q. □

Lemma 24 *For $\delta\in\mathbb{N}_0^N$ and $\Omega=\prod_{i=1}^N[a_i,b_i]$, let $\mathcal{D}_i\subseteq S_2^{\delta_i\cdot\mathrm{e}_i,n}[\Omega]$ for each $i\in\{1:N\}$ be admissible, and define associated operators $\mathcal{T}_i$ as in Cor. 2 and Cor. 3 and matrices K^i as in Defn. 2 Then, for any $i,j\in\{1:N\}$ such that $i\neq j$, we have*

$$\mathcal{T}_i\circ\mathcal{T}_j=\mathcal{T}_j\circ\mathcal{T}_i \qquad\Leftrightarrow\qquad K^i_{k,p}K^j_{\ell,q}=K^j_{\ell,q}K^i_{k,p},\quad \forall k,p\in\{1:\delta_i\},\ell,q\in\{1:\delta_j\},$$

where $K^i_{k,p}:=(\mathrm{e}_k\otimes I_n)^TK^i(\mathrm{e}_p\otimes I_n)$ for $\mathrm{e}_k\in\mathbb{R}^{\delta_i}$ the kth standard Euclidean basis vector.

Proof Fix arbitrary $i,j\in\{1:N\}$ such that $i\neq j$. If either $\delta_i=0$ or $\delta_j=0$, then $\mathcal{T}_i=I_n$ or $\mathcal{T}_j=I_n$, and the result holds trivially. Otherwise, let parameters K^i be as in Defn. 2, and let $\boldsymbol{T}_i(x,\theta):=-\mathbf{e}_1(x-a_i)^TK^i\mathbf{e}_{\delta_i}(b_i-\theta)$ for $\mathbf{e}_1,\mathbf{e}_d$ as in Cor. 2. Then, by definition of $\mathcal{T}_i$ in Cor. 3, we can express $\mathcal{T}_i=\mathcal{Q}_i+\mathcal{R}_i$, where for any $\mathbf{v}\in L_2^n[\Omega]$,

$$(\mathcal{Q}_i\mathbf{v})(s):=\int_{a_i}^{b_i}\boldsymbol{T}_i(s_i,\theta_i)\mathbf{v}(s)|_{s_i=\theta_i}d\theta_i,\qquad\text{and}\qquad(\mathcal{R}_i\mathbf{v})(s):=\int_{a_i}^{s_i}\frac{(s_i-\theta_i)^{\delta_i-1}}{(\delta_i-1)!}\mathbf{v}(s)|_{s_i=\theta_i}d\theta_i,$$

for all $s=(s_1,\ldots,s_n)\in\Omega$. Here, we note by inspection that for any $\boldsymbol{T}_i$, we have $\mathcal{Q}_i\mathcal{R}_j=\mathcal{R}_j\mathcal{Q}_i$ and $\mathcal{R}_i\mathcal{R}_j=\mathcal{R}_j\mathcal{R}_i$. Hence the $\mathcal{T}_i$ will commute if and only if the $\mathcal{Q}_i$ commute. By definition of $\mathcal{Q}_i$, we have

for every $\mathbf{v} \in L_2^n[\Omega]$,

$$\begin{aligned}\Big(\big(\mathcal{Q}_i\mathcal{Q}_j - \mathcal{Q}_j\mathcal{Q}_i\big)\mathbf{v}\Big)(s) &= \int_{a_i}^{b_i} \boldsymbol{T}_i(s_i,\theta_i)\left[\int_{a_j}^{b_j} \boldsymbol{T}_j(s_j,\theta_j)\mathbf{v}(s)|_{s_i=\theta_i,s_j=\theta_j}\,d\theta_j\right]d\theta_i \\ &\quad - \int_{a_j}^{b_j} \boldsymbol{T}_j(s_j,\theta_j)\left[\int_{a_i}^{b_i} \boldsymbol{T}_i(s_i,\theta_i)\mathbf{v}(s)|_{s_i=\theta_i,s_j=\theta_j}\,d\theta_i\right]d\theta_j \\ &= \int_{a_i}^{b_i}\int_{a_j}^{b_j}\Big[\boldsymbol{T}_i(s_i,\theta_i)\boldsymbol{T}_j(s_j,\theta_j) - \boldsymbol{T}_j(s_j,\theta_j)\boldsymbol{T}_i(s_i,\theta_i)\Big]\mathbf{v}(s)|_{s_i=\theta_i,s_j=\theta_j}\,d\theta_j d\theta_i.\end{aligned}$$

Therefore, $\mathcal{T}_i\circ\mathcal{T}_j = \mathcal{T}_j\circ\mathcal{T}_i$ if and only if $\boldsymbol{T}_i\boldsymbol{T}_j = \boldsymbol{T}_j\boldsymbol{T}_i$ except on a set of measure zero. However, since the $\boldsymbol{T}_i$ are polynomial, by Lemma 23, $\boldsymbol{T}_i\boldsymbol{T}_j = \boldsymbol{T}_j\boldsymbol{T}_i$ if and only if $K^i_{k,p}K^j_{\ell,q} = K^j_{\ell,q}K^i_{k,p}$ for all k,p,ℓ,q, concluding the proof. □

Lem. 24 proves that for any distinct $i,j \in \{1:N\}$, the operators $\mathcal{T}_i$ and $\mathcal{T}_j$ commute if and only if the $n\times n$ sub-blocks of the matrices K^i and K^j commute – i.e. if the domains $\mathcal{D}_i$ are consistent as per Defn. 2. Assuming admissible and consistent domains, then, not only can we define an inverse $\mathcal{T}_i : L_2^n[\Omega] \to \mathcal{D}_i$ to each $\partial_{s_i}^{\delta_i} : \mathcal{D}_i \to L_2^n[\Omega]$, but these operators $\mathcal{T}_i$ and $\mathcal{T}_j$ also commute for $i \neq j$. Using this fact, Thm. 8 proves that for admissible and consistent $\mathcal{D}_i$, the inverse of the differential operator $D^\delta : \bigcap_{i=1}^N \mathcal{D}_i \to L_2^n[\Omega]$ takes the form $\mathcal{T} = \prod_{i=1}^N \mathcal{T}_i : L_2^n[\Omega] \to \mathcal{D}$. In proving this result, we also relied on the fact that for $i \neq j$, the integral operator $\mathcal{T}_i$ commutes with the differential operator $\partial_{s_j}^{\delta_j}$, as proven in the following lemma.

Lemma 25 *For $N \in \mathbb{N}$ and $\boldsymbol{\delta} \in \mathbb{N}_0^N$, let $\mathcal{D} := \bigcap_{i=1}^N \mathcal{D}_i \subset S_2^{\delta,n}$ with $\mathcal{D}_i \subset S_2^{\delta_i,n}$ admissible and consistent.*

Let $\mathcal{T}_i, \mathcal{A}_{i,j}$ be defined as in Cor. 2 and extended to multivariate space using Cor. 3 (letting $\mathcal{T} \mapsto \mathcal{T}_i$ and $\mathcal{A} \mapsto \mathcal{A}_{i,j}$), for $d = \boldsymbol{\delta}_i$, $\mathcal{D} = \mathcal{D}_i$ and $\{\boldsymbol{H}_k\}$ where $\boldsymbol{H}_k := \begin{cases} I_n, & k = j, \\ 0, & k \neq j, \end{cases}$ for each $k \in \{0:d\}$. Then, for any $i,j \in \{1:N\}$ such that $i \neq j$ and all $d \in \{0:\delta_i\}$ and $k \in \{0:\delta_j\}$, we have

$$\partial_{s_i}^d\mathcal{T}_j\mathbf{u} = \mathcal{T}_j\partial_{s_i}^d\mathbf{u}, \qquad \text{and} \qquad \partial_{s_i}^d\mathcal{A}_{j,k}\mathbf{u} = \mathcal{A}_{j,k}\partial_{s_i}^d\mathbf{u},$$

for all $\mathbf{u} \in S_{2,i}^{d,n}[\Omega] := \big\{\mathbf{u} \in L_2^n[\Omega] \mid \partial_{s_i}^\ell\mathbf{u} \in L_2^n,\ \forall 0 \le \ell \le d\big\}$.

Proof We prove the result only for the operators $\mathcal{A}_{j,k}$ (not for $\mathcal{T}_j$), noting that $\mathcal{T}_j = \mathcal{A}_{j,0}$.

Fix arbitrary $i,j \in \{1:N\}$ such that $i \neq j$, and let $d \in \{0:\delta_i\}$ and $k \in \{0:\delta_j\}$. Fix further arbitrary $\mathbf{u} \in S_{2,i}^{d,n}[\Omega]$. If $d = 0$ or $k = d$, we have $\partial_s^d = I_n$ or $\mathcal{A}_{j,k} = I_n$ (by the definition in Cor. 2), and therefore $\partial_{s_i}^d\mathcal{A}_{j,k}\mathbf{u} = \mathcal{A}_{j,k}\partial_{s_i}^d\mathbf{u}$ holds trivially. Otherwise, if $k \neq d$ and $d \neq 0$, then by definition of the operator $\mathcal{A}_{j,k}$ in Cor. 2,

$$\begin{aligned}\big(\mathcal{A}_{j,k}\mathbf{v}\big)(s) &= \int_{a_j}^{b_j} \boldsymbol{R}_{j,k}(s_j,\theta_j)\,\mathbf{v}(s_1,\ldots,s_{j-1},\theta_j,s_{j+1},\ldots,s_N)\,d\theta_j \\ &\quad + \int_{a_j}^{s_j} \frac{(s_j-\theta_j)^{k-1}}{(k-1)!}\,\mathbf{v}(s_1,\ldots,s_{j-1},\theta_j,s_{j+1},\ldots,s_N)\,d\theta_j,\end{aligned}$$

where $\boldsymbol{R}_{j,k}(x,y) := -\mathbf{c}_k(x-a_j)K\mathbf{e}_d(b_j-y)$ for K as defined in Cor. 2, and where

$$\mathbf{e}_d(z) = \begin{bmatrix} \frac{z^{d-1}}{(d-1)!} \\ \vdots \\ \frac{1}{2}z^2 \\ z \\ 1 \end{bmatrix} \otimes I_n, \qquad \mathbf{c}_k(z) = \begin{bmatrix} 0_k \\ 1 \\ z \\ \vdots \\ \frac{z^{d-k-1}}{(d-1)!} \end{bmatrix} \otimes I_n.$$

It follows that

$$\begin{aligned}
\left(\mathcal{A}_{j,k}\partial_{s_i}^d \mathbf{u}\right)(s) &= \int_{a_j}^{b_j} \boldsymbol{R}_{j,k}(s_j,\theta_j)\,\partial_{s_i}^d \mathbf{u}(s_1,\ldots,s_{j-1},\theta_j,s_{j+1},\ldots,s_N)\,d\theta_j \\
&\quad + \int_{a_j}^{s_j} \frac{(s_j-\theta_j)^{k-1}}{(k-1)!}\,\partial_{s_i}^d \mathbf{u}(s_1,\ldots,s_{j-1},\theta_j,s_{j+1},\ldots,s_N)\,d\theta_j \\
&= \partial_{s_i}^d \left(\int_{a_j}^{b_j} \boldsymbol{R}_{j,k}(s_j,\theta_j)\mathbf{u}(s_1,\ldots,s_{j-1},\theta_j,s_{j+1},\ldots,s_N)\,d\theta_j \right) \\
&\quad + \partial_{s_i}^d \left(\int_{a_j}^{s_j} \frac{(s_j-\theta_j)^{k-1}}{(k-1)!}\mathbf{u}(s_1,\ldots,s_{j-1},\theta_j,s_{j+1},\ldots,s_N)\,d\theta_j \right) = \left(\partial_{s_i}^d \mathcal{A}_{j,k}\mathbf{u}\right)(s).
\end{aligned}$$

□

B. The $*$-Algebra of ND 3-PI Operators

In Section 5, a class of multivariate 3-PI operators was defined – $\boldsymbol{\Pi}_N[\Omega]$ – generalizing the definition of univariate 3-PI operators to the ND setting. In this appendix, we show that the class of multivariate 3-PI operators includes a particular class of integral operators, namely that of integral operators with polynomial semi-separable kernels. In addition, we provide some of the proofs omitted from Subsection 5.2, proving that the class of multivaraite 3-PI operators is indeed closed under linear combinations, as well as composition and adjoint operations – thus defining a $*$-algebra.

To start, recall that by Defn. 6, the class of ND 3-PI operators, $\boldsymbol{\Pi}_N[\Omega]$, is defined by taking sums of products of 3-PI operators along each dimension $i \in \{1:N\}$. Here, we recall that any univariate 3-PI operator is given by the sum of a multiplier operator defined by a polynomial, and an integral operator with polynomial semi-separable kernel, so that $\mathcal{R} \in \boldsymbol{\Pi}_1[a,b]$ if $\mathcal{R}\mathbf{v}(s) = \boldsymbol{R}(s)\mathbf{v}(s) + \int_a^b \boldsymbol{K}(s,\theta)\mathbf{v}(\theta)d\theta$ for some $\boldsymbol{M} \in \mathbb{R}[s]$ and some $\boldsymbol{K}$ of the form

$$\boldsymbol{K}(s,\theta) = \begin{cases} \boldsymbol{K}_1(s,\theta), & \theta \le s, \\ \boldsymbol{K}_2(s,\theta), & \theta > s, \end{cases}$$

for $\boldsymbol{K}_1, \boldsymbol{K}_2$ polynomial. Taking sums of compositions of such 3-PI operators along different intervals $[a_i,b_i]$, then, the proposed class of ND 3-PI operators will include (but is not limited to) any operator of

the form

$$(\mathcal{R}\mathbf{v})(s) = \sum_{j=1}^{M}\left(\prod_{i=1}^{N}\boldsymbol{R}_{i,j}(s_i)\right)\mathbf{v}(s), \qquad \text{and} \qquad (\mathcal{K}\mathbf{v})(s) = \sum_{j=1}^{M}\left(\prod_{i=1}^{N}\int_{a_i}^{b_i}\boldsymbol{K}_{i,j}(s_i,\theta_i)\right)\mathbf{v}(\theta)d\theta,$$

where $\boldsymbol{M}_{i,j}$ are polynomial, and $\boldsymbol{K}_{i,j}$ are polynomial semi-separable. Note here that, any multivariate polynomial can be expressed as $\boldsymbol{R}(s) = \sum_{j=1}^{M}\prod_{i=1}^{N}\boldsymbol{R}_{i,j}(s_i)$ for some univariate polynomials $\boldsymbol{R}_{i,j}$. As such, the class $\boldsymbol{\Pi}_N[\Omega]$ will include any multiplier operator defined by a polynomial. In addition, defining multivariate polynomial semi-separable functions as in Defn. 7, any such polynomial semi-separable function can be expressed as $\sum_{j=1}^{M}\prod_{i=1}^{N}\boldsymbol{K}_{i,j}$ for some univariate polynomial semi-separable functions. Consequently, any integral operator with polynomial semi-separable kernel will be a multivariate 3-PI operator, yielding the following result.

Lemma 26 *Suppose that*

$$(\mathcal{K}\mathbf{v})(s) := \int_{\Omega}\boldsymbol{K}(s,\theta)\mathbf{v}(\theta)d\theta$$

where $\boldsymbol{K}(s,\theta) \in L_2[\Omega]$ *is multivariate polynomial semi-separable. Then* $\mathcal{K} \in \boldsymbol{\Pi}_N[\Omega]$.

Proof Suppose that $\boldsymbol{K}$ is multivaraite polynomia semi-separable. Then there exist polynomials $\boldsymbol{K}_\alpha \in \mathbb{R}[s,\theta]$ for $\alpha \in \{-1,1\}^N$ such that

$$\boldsymbol{K}(s,\theta) = \boldsymbol{K}_\alpha(s,\theta) \qquad \forall (s,\theta) \in \{(s,\theta) \in \Omega \mid \alpha_i(s_i-\theta_i) \le 0,\ \forall i \in \{1:N\}\}.$$

Given these polynomials, we can then express

$$(\mathcal{K}\mathbf{v})(s) = \int_{\Omega}\boldsymbol{K}(s,\theta)\mathbf{v}(\theta)d\theta = \sum_{\alpha\in\{-1,1\}^N}\int_{\Omega}\mathbf{I}_\alpha(s,\theta)\boldsymbol{K}_\alpha(s,\theta)\mathbf{v}(\theta)d\theta$$

where we define the indicator function

$$\mathbf{I}_\alpha(s,\theta) := \prod_{i=1}^{N}\mathbf{I}_{\alpha_i}(s_i,\theta_i) \qquad \text{where} \qquad \mathbf{I}_\alpha(s,\theta) := \begin{cases} 1, & \alpha_i\cdot(s_i-\theta_i) \le 0,\ \forall i \in \{1:N\}, \\ 0, & \text{else.} \end{cases}$$

Now, for each $\alpha \in \{-1,1\}^N$, since $\boldsymbol{K}_\alpha$ is polynomial, we can define $\boldsymbol{R}_{\alpha,i,j} \in \mathbb{R}[s_i,\theta_i]$ such that $\boldsymbol{K}_\alpha = \sum_{j=1}^{M}\prod_{i=1}^{N}\boldsymbol{R}_{\alpha,i,j}$ for some $M \in \mathbb{N}$ sufficiently large (e.g. expanding $\boldsymbol{K}_\alpha$ as a sum of monomials). Given these univariate polynomials, $\boldsymbol{R}_{\alpha,i,j}$, define the 3-PI parameters $\mathbf{R}^{\beta,\gamma}_{\alpha,i,j} \in \boldsymbol{\Gamma}_1[a_i,b_i]$ by

$$\mathbf{R}_{\alpha,i,j} := \begin{cases} \{0,\boldsymbol{R}_{\alpha,i,j},0\}, & \alpha_i = -1, \\ \{0,0,\boldsymbol{R}_{\alpha,i,j}\}, & \alpha_i = 1, \end{cases}$$

Then, by definition

$$\begin{aligned}\int_{\Omega}\mathbf{I}_\alpha(s,\theta)\boldsymbol{K}_\alpha(s,\theta)\mathbf{v}(\theta)d\theta &= \int_{\Omega}\left(\sum_{j=1}^{M}\prod_{i=1}^{N}\mathbf{I}_{\alpha_i}(s_i,\theta_i)\boldsymbol{R}_{\alpha,i,j}(s_i,\theta_i)\right)\mathbf{v}(\theta)d\theta \\ &= \sum_{j=1}^{M}\left[\left(\prod_{i=1}^{N}\Pi_1\left[\mathbf{R}_{\alpha,i,j}\right]\right)\mathbf{v}\right](s).\end{aligned}$$

Since $\prod_{i=1}^{N}\Pi_1\left[\mathbf{R}_{\alpha,i,j}\right] \in \boldsymbol{\Pi}_N[\Omega]$ for all α and j, it follows that also $\mathcal{K} \in \boldsymbol{\Pi}_N[\Omega]$. $\square$

Lem 26 shows that the class of ND PI operators includes a particular class of integral operators, of which the kernel is polynomial semi-separable. By construction, the operator $\mathcal{T}$ in Thm. 8, defining the inverse to $D^{\delta}:\mathcal{D}\to L_2^n[\Omega]$, will be of this form. However, the class of ND 3-PI operators also allows for mixed multiplier and integral operators along different spatial directions, e.g. $\int_a^x \boldsymbol{R}(x,y,\theta)\mathbf{v}(\eta,y)d\eta$ and $\int_y^d \boldsymbol{R}(x,y,\eta)\mathbf{v}(x,\zeta)d\zeta$. This ensures that the operator $\mathcal{A}$ defining the PIE representation in Thm. 9 is also an ND PI operator, as proven in the following lemma.

Lemma 27 *For $N\in\mathbb{N}$ and $\delta\in\mathbb{N}_0^N$, let $\mathcal{T}$ and $\mathcal{A}$ be as defined in Thm. 8. Then $\mathcal{T},\mathcal{A}\in\mathbf{\Pi}_N^{n\times n}[\Omega]$.*

Proof For the case $n=1$, the result follows by inspection. In particular, by definition, $\mathcal{T}=\prod_{i=1}^N\mathcal{T}_i$ and $\mathcal{A}=\sum_{\vec{0}\leq\alpha\leq\delta}\mathrm{M}[\boldsymbol{A}_\alpha]\prod_{i=1}^N\mathcal{A}_{i,\alpha_i}$, where the operators $\mathcal{T}_i$ and $\mathcal{A}_{i,\alpha_i}$ are of the form of the operators $\mathcal{T}$ and $\mathcal{A}$ in Cor. 2, respectively. By definition, then, we find that $\mathcal{T}_i$ and $\mathcal{A}_{i,\alpha_i}$ are univariate 3-PI operators, wherefore $\mathcal{T}$ is a multivariate 3-PI operator. Furthermore, since $\mathrm{M}[\boldsymbol{A}_\alpha]\in\mathbf{\Pi}_N$ for all $\boldsymbol{A}_\alpha\in\mathbb{R}[s]$ and $\mathbf{\Pi}_N$ is closed under compositions (by Prop. 14), it follows that also $\mathcal{A}$ is a multivariate 3-PI operator.

For the case $n>1$, we note that for each $k,\ell\in\{1,n\}$, we can express

$$[\mathcal{T}]_{k,\ell}=\left[\prod_{i=1}^N\mathcal{T}_i\right]_{k,\ell}=\sum_{j_1=1}^n\sum_{j_2=1}^n\cdots\sum_{j_{N-1}=1}^n[\mathcal{T}_1]_{k,j_1}[\mathcal{T}_2]_{j_1,j_2}\cdots[\mathcal{T}_N]_{j_{N-1},\ell}.$$

Since, as established for the case $n=1$, each $[\mathcal{T}_i]_{j_{i-1},j_i}$ is a univariate 3-PI operator, it follows that each $[\mathcal{T}]_{k,\ell}$ is a multivariate 3-PI operator, and therefore $\mathcal{T}\in\mathbf{\Pi}_N^{n\times n}[\Omega]$. By similar reasoning, it follows that also $\mathcal{A}\in\mathbf{\Pi}_N^{n\times n}[\Omega]$. □

By Lem. 11, given a multivariate PDE as in (4.5), the operators $\mathcal{T}$ and $\mathcal{A}$ defining the associated PIE representation as per Thm. 8 are in fact ND 3-PI operators. Having thus shown that the class $\mathbf{\Pi}_N[\Omega]$ is suitable for parameterization of ND PIEs, we now prove that this class in fact defines a $*$-algebra. To start, we show that $\mathbf{\Pi}_N[\Omega]$ is a vector space, for which we have the following lemma.

Lemma 28 *For $N\in\mathbb{N}$ and $\Omega:=\prod_{i=1}^N$, if $\mathcal{Q},\mathcal{R}\in\mathbf{\Pi}_N[\Omega]$, then $\lambda\mathcal{Q}+\mu\mathcal{R}\in\mathbf{\Pi}_N[\Omega]$ for all $\lambda,\mu\in\mathbb{R}$.*

Proof Fix arbitrary $\mathcal{Q}=\sum_{j=1}^M\Pi_N[\mathbf{Q}_j]\in\mathbf{\Pi}_N[\Omega]$, where for each $j\in\{1:M\}$, we have $\mathbf{Q}_j:=\{\mathbf{Q}_{j,i}\}_{i=1}^N$ for $\mathbf{Q}_{j,i}\in\mathbf{\Gamma}_1[a_i,b_i]$. Suppose that $\mathbf{Q}_{j,1}=\{\mathbf{Q}_{j,1}^0,\mathbf{Q}_{j,1}^1,\mathbf{Q}_{j,2}^2\}\in\mathbf{\Gamma}_1[a_1,b_1]$, so that $\mathbf{Q}_{j,1}^0\in\mathbb{R}[s_1]$ and $\mathbf{Q}_{j,1}^1,\mathbf{Q}_{j,1}^2\in\mathbb{R}[s_1,\theta_1]$. For $\lambda\in\mathbb{R}$, define $\lambda\mathbf{Q}_{j,1}:=\{\lambda\mathbf{Q}_{j,1}^0,\lambda\mathbf{Q}_{j,1}^1,\lambda\mathbf{Q}_{j,2}^2\}\in\mathbf{\Pi}_1[a_1,b_1]$. Then, by definition of univariate 3-PI operators (Defn. 5) and linearity of multiplier and integral operators, $\lambda\Pi_1[\mathbf{Q}_{j,1}]\mathbf{v}=\Pi_1[\lambda\mathbf{Q}_{j,1}]\mathbf{v}$ for all $\mathbf{v}\in L_2$. Define further $\lambda\mathbf{Q}_j:=\{\lambda\mathbf{Q}_{j,1},\mathbf{Q}_{j,2},\ldots,\mathbf{Q}_{j,N}\}\in\mathbf{\Pi}_N[\Omega]$. Then, for each $j\in\{1:M\}$

$$\lambda\mathcal{Q}=\lambda\sum_{j=1}^M\Pi_N[\mathbf{Q}_j]=\sum_{j=1}^M\lambda\prod_{i=1}^N\Pi_1[\mathbf{Q}_{j,i}]=\sum_{j=1}^M\Pi_1[\lambda\mathbf{Q}_{j,1}]\Pi_1[\mathbf{Q}_{j,2}]\cdots\Pi_1[\mathbf{Q}_{j,N}]=\sum_{j=1}^M\Pi_N[\lambda\mathbf{Q}_j]\in\mathbf{\Pi}_N[\Omega].$$

Finally, fix arbitrary $\mathcal{R}=\sum_{k=1}^K\Pi_N[\mathbf{R}_k]\in\mathbf{\Pi}_N[\Omega]$. Then, for all $\lambda,\mu\in\mathbb{R}$,

$$\lambda\mathcal{Q}+\mu\mathcal{R}=\sum_{j=1}^M\Pi_N[\lambda\mathbf{Q}_j]+\sum_{k=1}^K\Pi_N[\mu\mathbf{R}_k]\in\mathbf{\Pi}_N[\Omega].$$

□

Lem. 28 shows that the class of multivariate 3-PI operators is closed under linear combinations, and is thus a vector space. Next, we show that this vector space is also closed under compositions. Here, the composition of two multivariate 3-PI operators can be defined inductively, using the composition rules for unviariate 3-PI operators. In particular, if $\mathcal{Q} = \prod_{i=1}^{N} \Pi_1[\boldsymbol{Q}_i]$ and $\mathcal{R} = \prod_{i=1}^{N} \Pi_1[\boldsymbol{R}_i]$ for $\boldsymbol{Q}_i, \boldsymbol{R}_i \in \boldsymbol{\Gamma}_1[a_i, b_i]$, then $\mathcal{Q} \circ \mathcal{R} = \prod_{i=1}^{N} (\Pi_1[\boldsymbol{Q}_i] \circ \Pi_1[\boldsymbol{R}_i])$. In order for this to hold, however, the different operators $\Pi_1[\boldsymbol{Q}_i]$ and $\Pi_1[\boldsymbol{R}_j]$ for $i \neq j$ must commute. The following lemma proves that this is indeed the case.

Lemma 29 *Let $N \in \mathbb{N}$ and $\Omega = \prod_{i=1}^{N}[a_i, b_i]$. For $\mathbf{R} \in \boldsymbol{\Gamma}_1[a_i, b_i]$ and $\mathbf{Q} \in \boldsymbol{\Gamma}_1[a_j, b_j]$ let $\mathcal{R}$ and $\mathcal{Q}$ be the multivariate extensions of $\Pi_1[\mathbf{R}]$ and $\Pi_1[\mathbf{Q}]$, respectively, so that e.g. $(\mathcal{R}\mathbf{v})(s) := (\Pi_1 \mathbf{v}(s_1, \ldots, s_{i-1}, \bullet, s_{i+1}, \ldots, s_N))(s_i)$ for $\mathbf{v} \in L_2[\Omega]$. If $i \neq j$, then $(\mathcal{R}\mathcal{Q})\mathbf{v} = (\mathcal{Q}\mathcal{R})\mathbf{v}$ for all $\mathbf{v} \in L_2[\Omega]$.*

Proof Let $\mathbf{R} = \{\boldsymbol{R}_0, \boldsymbol{R}_1, \boldsymbol{R}_2\} \in \boldsymbol{\Gamma}[a_i, b_i]$ and $\mathbf{Q} = \{\boldsymbol{Q}_0, \boldsymbol{Q}_1, \boldsymbol{Q}_2\} \in \boldsymbol{\Gamma}[a_j, b_j]$. To prove the result, we first note that by Defn. 5, for any $\mathbf{v} \in L_2[\Omega]$, we can express $\mathcal{R}\mathbf{v} = \mathcal{R}_0\mathbf{v} + \mathcal{R}_1\mathbf{v} + \mathcal{R}_2\mathbf{v}$ and $\mathcal{Q}\mathbf{v} = \mathcal{Q}_0\mathbf{v} + \mathcal{Q}_1\mathbf{v} + \mathcal{Q}_2\mathbf{v}$, where we define

$$
\begin{aligned}
(\mathcal{R}_0\mathbf{v})(s) &:= \boldsymbol{R}_0(s_i)\mathbf{v}(s), & (\mathcal{Q}_0\mathbf{v})(s) &:= \boldsymbol{Q}_0(s_j)\mathbf{v}(s),\\
(\mathcal{R}_1\mathbf{v})(s) &:= \int_{a_i}^{s_i} \boldsymbol{R}_1(s_i, \theta_i)\mathbf{v}(s)|_{s_i=\theta_i} d\theta_i, & (\mathcal{Q}_1\mathbf{v})(s) &:= \int_{a_j}^{s_j} \boldsymbol{Q}_1(s_j, \theta_j)\mathbf{v}(s)|_{s_j=\theta_j} d\theta_j,\\
(\mathcal{R}_2\mathbf{v})(s) &:= \int_{s_i}^{b_i} \boldsymbol{R}_2(s_i, \theta_i)\mathbf{v}(s)|_{s_i=\theta_i} d\theta_i, & (\mathcal{Q}_2\mathbf{v})(s) &:= \int_{s_j}^{b_j} \boldsymbol{Q}_2(s_j, \theta_j)\mathbf{v}(s)|_{s_j=\theta_j} d\theta_j,
\end{aligned}
$$

for all $s = (s_1, \ldots, s_N) \in \Omega$. Here, we remark that

$$
\big(\mathcal{R}_0\mathcal{Q}_0\mathbf{v}\big)(s) = \boldsymbol{R}_0(s_i)\boldsymbol{Q}_0(s_j)\mathbf{v}(s) = \boldsymbol{Q}_0(s_j)\boldsymbol{R}_0(s_i)\mathbf{v}(s) = \big(\mathcal{Q}_0\mathcal{R}_0\mathbf{v}\big)(s),
$$

as well as

$$
\begin{aligned}
\big(\mathcal{R}_0\mathcal{Q}_1\mathbf{v}\big)(s) &= \boldsymbol{R}_0(s_i) \int_{a_j}^{s_j} \boldsymbol{Q}_1(s_j, \theta_j)\mathbf{v}(s)|_{s_j=\theta_j} d\theta_j\\
&= \int_{a_j}^{s_j} \boldsymbol{Q}_1(s_j, \theta_j)\boldsymbol{R}_0(s_i)\mathbf{v}(s)|_{s_j=\theta_j} d\theta_j = \big(\mathcal{Q}_1\mathcal{R}_0\mathbf{v}\big)(s),
\end{aligned}
$$

and

$$
\begin{aligned}
\big(\mathcal{R}_1\mathcal{Q}_1\mathbf{v}\big)(s) &= \int_{a_i}^{s_i} \boldsymbol{R}_1(s_i, \theta_i) \left[\int_{a_j}^{s_j} \boldsymbol{Q}_1(s_j, \theta_j)\mathbf{v}(s)|_{s_i=\theta_i, s_j=\theta_j} d\theta_j \right] d\theta_i\\
&= \int_{a_i}^{s_i} \int_{a_j}^{s_j} \left[\boldsymbol{R}_1(s_i, \theta_i)\boldsymbol{Q}_1(s_j, \theta_j)\mathbf{v}(s)|_{s_i=\theta_i, s_j=\theta_j} \right] d\theta_j d\theta_i\\
&= \int_{a_j}^{s_j} \boldsymbol{Q}_1(s_j, \theta_j) \left[\int_{a_i}^{s_i} \boldsymbol{R}_1(s_i, \theta_i)\mathbf{v}(s)|_{s_i=\theta_i, s_j=\theta_j} d\theta_i \right] d\theta_j = \big(\mathcal{Q}_1\mathcal{R}_1\mathbf{v}\big)(s),
\end{aligned}
$$

By similar reasoning, we find that $\mathcal{R}_\ell\mathcal{Q}_k\mathbf{v} = \mathcal{Q}_k\mathcal{R}_\ell\mathbf{v}$, for all $k, \ell \in \{0, 1, 2\}$. It follows that

$$
\begin{aligned}
\mathcal{R}\mathcal{Q}\mathbf{v} &= \big(\mathcal{R}_0 + \mathcal{R}_1 + \mathcal{R}_2\big)\big(\mathcal{Q}_0 + \mathcal{Q}_1 + \mathcal{Q}_2\big)\mathbf{v}\\
&= \big(\mathcal{Q}_0 + \mathcal{Q}_1 + \mathcal{Q}_2\big)\big(\mathcal{R}_0 + \mathcal{R}_1 + \mathcal{R}_2\big)\mathbf{v} = \mathcal{Q}\mathcal{R}\mathbf{v}.
\end{aligned}
$$

□

Using Lem. 29, the following corollary proves that the composition of two ND PI operators can indeed be expressed using the composition of the defining 1D PI operators, as used in the proof of Prop. 14.

Corollary 30 *For given $N \in \mathbb{N}$ and $\Omega = \prod_{i=1}^{N}[a_i, b_i]$, let $\mathbf{Q} = \{\mathbf{Q}_i\}_{i=1}^{N} \in \mathbf{\Gamma}_N[\Omega]$ and $\mathbf{R} = \{\mathbf{R}_i\}_{i=1}^{N} \in \mathbf{\Gamma}_N[\Omega]$, where $\mathbf{Q}_i \in \mathbf{\Gamma}_N[a_i, b_i]$ and $\mathbf{R}_i \in \mathbf{\Gamma}_N[a_i, b_i]$ for each $i \in \{1:N\}$. Then*

$$\left(\prod_{i=1}^{N} \Pi_1[\mathbf{Q}_i]\right) \circ \left(\prod_{j=1}^{N} \Pi_1[\mathbf{R}_j]\right) = \prod_{i=1}^{N} \left(\Pi_1[\mathbf{Q}_i] \circ \Pi[\mathbf{R}_i]\right).$$

Proof By Lem. 29 we have that for all $\mathbf{u} \in L_2[\Omega]$,

$$\begin{aligned}\left(\prod_{i=1}^{N} \Pi_1[\mathbf{Q}_i]\right)\left(\prod_{j=1}^{N} \Pi_1[\mathbf{R}_j]\right)\mathbf{u} &= \Big(\Pi_1[\mathbf{Q}_1]\cdots\Pi_1[\mathbf{Q}_N]\Pi_1[\mathbf{R}_1]\cdots\Pi_1[\mathbf{R}_N]\Big)\mathbf{u} \\ &= \big(\Pi_1[\mathbf{Q}_1]\Pi_1[\mathbf{R}_1]\big)\cdots\big(\Pi_1[\mathbf{Q}_N]\Pi_1[\mathbf{R}_N]\big)\mathbf{u} = \left(\prod_{i=1}^{N}\big(\Pi_1[\mathbf{Q}_i]\Pi_1[\mathbf{R}_i]\big)\right)\mathbf{u}.\end{aligned}$$

□

Using Cor. 30, Prop. 14 proves that the composition of two multivariate 3-PI operators is again a multivariate 3-PI operator, so that $\mathbf{\Pi}_N$ forms an algebra. Finally, to prove that $\mathbf{\Pi}_N$ in fact forms a $*$-algebra, we show that this set is also closed under the adjoint operation. Specifically, we have the following result, corresponding to Lem. 16 in Subsection 5.2

Lemma 31 *For $N \in \mathbb{N}$ and $\Omega := \prod_{i=1}^{N}[a_i, b_i] \subseteq \mathbb{R}^N$, if $\mathcal{R} \in \mathbf{\Pi}_N[\Omega]$, then $\mathcal{R}^* \in \mathbf{\Pi}_N[\Omega]$. In particular, if $\mathcal{R} = \sum_{j=1}^{M} \Pi_N[\mathbf{R}_j]$, where $\mathbf{R}_j := \{\mathbf{R}_{j,i}\}_{i=1}^{N} \in \mathbf{\Gamma}_N[\Omega]$, then $\mathcal{R}^* = \sum_{j=1}^{M} \Pi_N[\mathbf{R}_j^*]$, where*

$$\mathbf{R}_j^* := \{\mathbf{R}_{j,i}^*\}_{i=1}^{N},$$

where the univariate adjoint parameters $\mathbf{R}_{j,i}^* \in \mathbf{\Gamma}_1[a_i, b_i]$ are as defined in Lem. 15.

Proof To prove the result we first remark that, by Lem. 15, for each $i \in \{1:N\}$ and all $\mathbf{R} \in \mathbf{\Gamma}_1[a_i, b_i]$ we have

$$\int_{a_i}^{b_i} \mathbf{u}(s)\left(\Pi_1[\mathbf{R}]\mathbf{v}\right)(s)\,ds_i = \int_{a_i}^{b_i} \left(\Pi_1[\mathbf{R}^*]\mathbf{u}\right)(s)\,\mathbf{v}(s)\,ds_i, \qquad \forall \mathbf{u}, \mathbf{v} \in L_2[\Omega],$$

where we define $(\Pi_1[\mathbf{R}]\mathbf{v})(s) := \big(\Pi_1[\mathbf{R}]\mathbf{v}(s_1,\dots,s_{i-1},\bullet,s_{i+1},\dots,s_N)\big)(s_i)$. Now, let $\mathcal{R} = \sum_{j=1}^{M}\Pi_N[\mathbf{R}_j]$, where $\mathbf{R}_j := \{\mathbf{R}_{j,i}\}_{i=1}^{N} \in \mathbf{\Gamma}_N[\Omega]$. Then, for all $j \in \{1:M\}$,

$$
\begin{aligned}
&\big\langle \mathbf{u}, \Pi_N[\mathbf{R}_j]\mathbf{v}\big\rangle_{L_2[\Omega]} = \int_\Omega \Big[\mathbf{u}(s)\big(\Pi_N[\mathbf{R}_j]\mathbf{v}\big)(s)\Big]\,ds \\
&= \int_{a_1}^{b_1}\cdots\int_{a_N}^{b_N}\Big[\mathbf{u}(s_1,\dots,s_N)\big(\Pi_1[\mathbf{R}_{j,1}]\cdots\Pi_1[\mathbf{R}_{j,N}]\mathbf{v}\big)(s_1,\dots,s_N)\Big]\,ds_N\cdots ds_1 \\
&= \int_{a_1}^{b_1}\cdots\int_{a_N}^{b_N}\Big[\big(\Pi_1[\mathbf{R}^*_{j,1}]\mathbf{u}\big)(s_1,\dots,s_N)\,\big(\Pi_1[\mathbf{R}_{j,2}]\cdots\Pi_1[\mathbf{R}_{j,N}]\mathbf{v}\big)(s_1,\dots,s_N)\Big]\,ds_N\cdots ds_1 \\
&= \int_{a_1}^{b_1}\cdots\int_{a_N}^{b_N}\Big[\big(\Pi_1[\mathbf{R}^*_{j,2}]\Pi_1[\mathbf{R}^*_{j,1}]\mathbf{u}\big)(s_1,\dots,s_N)\,\big(\Pi_1[\mathbf{R}_{j,3}]\cdots\Pi_1[\mathbf{R}_{j,N}]\mathbf{v}\big)(s_1,\dots,s_N)\Big]\,ds_N\cdots ds_1 \\
&\vdots \\
&= \int_{a_1}^{b_1}\cdots\int_{a_N}^{b_N}\Big[\big(\Pi_1[\mathbf{R}^*_{j,N}]\cdots\Pi_1[\mathbf{R}^*_{j,1}]\mathbf{u}\big)(s_1,\dots,s_N)\,\mathbf{v}(s_1,\dots,s_N)\Big]\,ds_N\cdots ds_1 \\
&= \int_{a_1}^{b_1}\cdots\int_{a_N}^{b_N}\Big[\big(\Pi_1[\mathbf{R}^*_{j,1}]\cdots\Pi_1[\mathbf{R}^*_{j,N}]\mathbf{u}\big)(s_1,\dots,s_N)\,\mathbf{v}(s_1,\dots,s_N)\Big]\,ds_N\cdots ds_1 \\
&= \int_\Omega\Big(\big(\Pi_N[\mathbf{R}^*_j]\mathbf{u}\big)(s)\mathbf{v}(s)\Big)\,ds \\
&= \big\langle \Pi_N[\mathbf{R}^*_j]\mathbf{u},\mathbf{v}\big\rangle_{L_2[\Omega]},
\end{aligned}
$$

where we remark that $\Pi_1[\mathbf{R}^*_{j,N}]\cdots\Pi_1[\mathbf{R}^*_{j,1}] = \Pi_1[\mathbf{R}^*_{j,1}]\cdots\Pi_1[\mathbf{R}^*_{j,N}]$ by Lem. 29. By linearity of the inner product, it follows that if $\mathcal{R} := \sum_{i=1}^{M}\Pi_N[\mathbf{R}_j]$, then

$$
\langle \mathbf{u}, \mathcal{R}\mathbf{v}\rangle_{L_2} = \sum_{i=1}^{M}\big\langle \mathbf{u}, \Pi_N[\mathbf{R}_j]\mathbf{v}\big\rangle_{L_2} = \sum_{i=1}^{M}\big\langle \Pi_N[\mathbf{R}^*_j]\mathbf{u},\mathbf{v}\big\rangle_{L_2} = \langle \mathcal{R}^*\mathbf{u},\mathbf{v}\rangle_{L_2}.
$$

□

C. Exponential PIE to PDE Stability of Classical PDE Examples

In this appendix, we verify exponential stability of the reaction-diffusion equation from Subsec. 7.2.1 and the wave equation from Subsec. 7.2.2. For each example, an explicit solution is constructed using separation of variables, and it is proven that the norm of this solution is bounded by an exponentially decaying function, providing an explicit value of the decay rate. Since the reaction-diffusion and wave equations are classical examples of PDEs, for which existence and uniqueness of solutions have been well-studied in the literature, several details in the derivation of the solutions will be omitted, and it will not be shown that the obtained solutions are indeed unique. We refer to standard textbooks on PDEs such as [9] for more details and proofs.

C.1. *Exponential PIE to PDE Stability of the Reaction-Diffusion Equation*

Consider the following reaction-diffusion equation, with Dirichlet-Neumann boundary conditions

$$\begin{aligned} &\mathbf{u}_t(t,x,y) = \nu[\mathbf{u}_{xx}(t,x,y) + \mathbf{u}_{yy}(t,x,y)] + r\mathbf{u}(t,x,y), && (x,y) \in [0,L_x] \times [0,L_y], \\ &\mathbf{u}(t,0,y) = \mathbf{u}(t,L_x,y) = 0, \qquad \mathbf{u}(t,x,0) = \mathbf{u}_y(t,x,L_y) = 0. \end{aligned} \tag{C.1}$$

We solve this PDE using separation of variables. In particular, suppose $\mathbf{u}(t,x,y) = T(t)U(x,y)$ for some function T that depends only on time, and a function U that varies only in space. Substituting the relation $\mathbf{u}(t,x,y) = T(t)U(x,y)$ into the PDE, we find that it must satisfy

$$T'(t)U(x,y) = \nu T(t)[\partial_x^2 U(x,y) + \partial_y^2 U(x,y)] + rT(t)U(x,y).$$

Dividing both sides by $T(t)U(x,y)$, and rearranging the terms, this yields

$$\frac{T'(t)}{T(t)} - r = \nu \frac{\partial_x^2 U(x,y)}{U(x,y)} + \nu \frac{\partial_y^2 U(x,y)}{U(x,y)},$$

where now the left-hand side depends only on t, and the right-hand side depends only on x,y. It follows that both sides must be constant in time and space, whence solutions must satisfy

$$\frac{T'(t)}{T(t)} - r = -\lambda \qquad \text{and} \qquad \nu \frac{\partial_x^2 U(x,y)}{U(x,y)} + \nu \frac{\partial_y^2 U(x,y)}{U(x,y)} = -\lambda,$$

for some constant $\lambda \in \mathbb{R}$. Solving the first of these equations, we obtain

$$T(t) = T(0)e^{[r-\lambda]t}.$$

For the second equation, suppose further that $U(x,y) = X(x)Y(y)$ for some functions $X : [0,L_x] \to \mathbb{R}$ and $Y : [0,L_y] \to \mathbb{R}$. Substituting this identity into the equation, it follows that these functions must satisfy

$$\nu \frac{X''(x)}{X(x)} = -\nu \frac{Y''(y)}{Y(y)} - \lambda.$$

Here, the left-hand side depends only on x, and the right-hand side depends only on y, implying that both sides must be constant in x and y. Thus, the functions X and Y must satisfy

$$\nu \frac{X''(x)}{X(x)} = -\mu, \qquad \text{and} \qquad -\nu \frac{Y''(y)}{Y(y)} - \lambda = -\mu,$$

for some constant $\mu \in \mathbb{R}$. To solve these problems, recall that the function $u(t,x,y) = T(t)X(x)Y(y)$ must further satisfy the boundary conditions

$$\begin{aligned} &T(t)X(0)Y(y) = u(t,0,y) = 0, && T(t)X(L_x)Y(y) = u(t,L_x,y) = 0, \\ &T(t)X(x)Y(0) = u(t,x,0) = 0, && T(t)X(x)Y'(L_y) = u_y(t,x,L_y) = 0. \end{aligned}$$

Excluding trivial solutions, these boundary conditions can only be satisfied if

$$X(0) = X(L_x) = 0, \qquad \text{and} \qquad Y(0) = Y'(L_y) = 0.$$

Thus, we obtain two boundary-value problems

$$X''(x) = -\frac{\mu}{\nu}X(x), \qquad \text{and} \qquad Y''(y) = -\frac{\lambda-\mu}{\nu}Y(y),$$
$$X(0) = X(L_x) = 0, \qquad Y(0) = Y'(L_y) = 0.$$

These are both standard 1D Sturm-Liouville problems, of which solutions are well-known. In particular, we note that non-trivial solutions exist only for $\mu = \mu_m := \nu\frac{m^2\pi^2}{L_x^2}$ and $\lambda - \mu = \lambda_{m,n} - \mu_n := \nu\frac{(n-1/2)^2\pi^2}{L_y^2}$ for $m,n \in \mathbb{N}$, taking the form

$$X_m(x) = \sin\left(\frac{m\pi}{L_x}x\right), \qquad Y_n(y) = \sin\left(\frac{(n-1/2)\pi}{L_y}y\right), \qquad \forall m,n \in \mathbb{N}.$$

Here, the functions $\mathbf{u}_{m,n}(x,y) := X_m(x)Y_n(y)$ form a basis for functions on $L_2[[0,1]^2]$ satisfying the imposed Dirichlet and Neumann boundary conditions, and any solution to the original PDE takes the form

$$\mathbf{u}(t,x,y) = \sum_{m=1}^{\infty}\sum_{n=1}^{\infty} a_{m,n}e^{[r-\lambda_{m,n}]t}\mathbf{u}_{m,n}(x,y) = \sum_{m=1}^{\infty}\sum_{n=1}^{\infty} a_{m,n}e^{[r-\lambda_{m,n}]t}\sin\left(\frac{m\pi}{L_x}x\right)\sin\left(\frac{(n-1/2)\pi}{L_y}y\right),$$

for some coefficients $a_{m,n}$ defined by the initial conditions, and where

$$\lambda_{m,n} = \mu_m + \nu\frac{(n-1/2)^2\pi^2}{L_y^2} = \nu\pi^2\left[\frac{m^2}{L_x^2} + \frac{(n-1/2)^2}{L_y^2}\right], \qquad \forall m,n \in \mathbb{N}.$$

It follows that

$$\mathbf{u}_{xxyy}(t,x,y) = \sum_{m=1}^{\infty}\sum_{n=1}^{\infty}\frac{m^2\pi^2}{L_x^2}\frac{(n-1/2)^2\pi^2}{L_y^2}a_{m,n}e^{[r-\lambda_{m,n}]t}\sin\left(\frac{m\pi}{L_x}x\right)\sin\left(\frac{(n-1/2)\pi}{L_y}y\right).$$

By orthogonality of the basis functions $\mathbf{u}_{m,n}$ with respect to the L_2 inner product, we find

$$\begin{aligned}\|\mathbf{u}(t)\|_{L_2}^2 &= \sum_{m=1}^{\infty}\sum_{n=1}^{\infty}\frac{L_xL_y}{4}a_{m,n}^2e^{2[r-\lambda_{m,n}]t} \\ &\leq \sum_{m=1}^{\infty}\sum_{n=1}^{\infty}\left(m^2\pi^2(n-1/2)^2\pi^2\right)^2\frac{L_xL_y}{4}a_{m,n}^2e^{2[r-\lambda_{m,n}]t} = L_x^2L_y^2e^{2[r-\lambda_{m,n}]t}\left\|\mathbf{u}_{xxyy}(0)\right\|_{L_2}^2.\end{aligned}$$

Thus, the PDE is exponentially PIE to PDE stable whenever $r \leq \lambda_{\min} := \min_{m,n\in\mathbb{N}}\lambda_{m,n} = \nu\pi^2\left[\frac{1}{L_x^2} + \frac{1}{4L_y^2}\right]$, with rate of decay $k = r - \lambda_{\min}$.

C.2. *Exponential PIE to PDE Stability of the Wave Equation*

Consider the following wave equation with feedback, with Dirichlet-Neumann boundary conditions

$$\mathbf{u}_{tt}(t,x,y) = \mathbf{u}_{xx}(t,x,y) + \mathbf{u}_{yy}(t,x,y) - 2\kappa\mathbf{u}_t(t,x,y) - \kappa^2\mathbf{u}(t,x,y), \qquad (x,y) \in [0,L_x]\times[0,L_y], \tag{C.2}$$
$$\mathbf{u}(t,0,y) = \mathbf{u}_x(t,L_x,y) = 0, \qquad \mathbf{u}(t,x,0) = \mathbf{u}_y(t,x,L_y) = 0,$$

where $\kappa \geq 0$. Suppose solutions take the form $\mathbf{u}(t,x,y) = T(t)U(x,y)$ for some function T that depends only on time, and a function U that varies only in space. Substituting the relation $\mathbf{u}(t,x,y) = T(t)U(x,y)$

into the PDE, we find that T and U must satisfy

$$T''(t)U(x,y) = T(t)[\partial_x^2 U(x,y) + \partial_y^2 U(x,y)] + 2\kappa T'(t)U(x,y) - \kappa^2 T(t)U(x,y).$$

Dividing both sides by $T(t)U(x,y)$, and re-arranging the terms, this yields

$$\frac{T''(t)+2\kappa T'(t)}{T(t)} + \kappa^2 = \frac{\partial_x^2 U(x,y)}{U(x,y)} + \frac{\partial_y^2 U(x,y)}{U(x,y)},$$

where now the left-hand side depends only on t, and the right-hand side depends only on x,y. It follows that both sides must be constant in time and space, whence solutions must satisfy

$$\frac{T''(t)+2\kappa T'(t)}{T(t)} + \kappa^2 = -\lambda, \qquad \text{and} \qquad \frac{\partial_x^2 U(x,y)}{U(x,y)} + \frac{\partial_y^2 U(x,y)}{U(x,y)} = -\lambda,$$

for some constant $\lambda \in \mathbb{R}$. Solving the first of these equations, we find that if $\lambda > 0$, solutions take the form

$$T(t) = \gamma e^{-\kappa t} \sin\left(t\sqrt{\lambda}\right) + \delta e^{-\kappa t} \cos\left(t\sqrt{\lambda}\right),$$

for some constants γ, δ depending on $T(0)$ and $T'(0)$. For the second equation, suppose further that $U(x,y) = X(x)Y(y)$ for some functions $X : [0,L_x] \to \mathbb{R}$ and $Y : [0,L_y] \to \mathbb{R}$. Substituting this identity into the equation, it follows that these functions must satisfy

$$\frac{X''(x)}{X(x)} = -\frac{Y''(y)}{Y(y)} - \lambda.$$

Here, the left-hand side depends only on x, and the right-hand side depends only on y, implying that both sides must be constant in x and y. Thus, the functions X and Y must satisfy

$$\frac{X''(x)}{X(x)} = -\mu, \qquad \text{and} \qquad -\frac{Y''(y)}{Y(y)} - \lambda = -\mu,$$

for some constant $\mu \in \mathbb{R}$. To solve these problems, recall that the function $u(t,x,y) = T(t)X(x)Y(y)$ must further satisfy the boundary conditions

$$\begin{aligned} T(t)X(0)Y(y) &= u(t,0,y) = 0, & T(t)X'(L_x)Y(y) &= u_x(t,L_x,y) = 0, \\ T(t)X(x)Y(0) &= u(t,x,0) = 0, & T(t)X(x)Y'(L_y) &= u_y(t,x,L_y) = 0. \end{aligned}$$

Excluding trivial solutions, these boundary conditions can only be satisfied if

$$X(0) = X'(L_x) = 0, \qquad \text{and} \qquad Y(0) = Y'(L_y) = 0.$$

Thus, we obtain the boundary-value problems

$$\begin{aligned} X''(x) &= -\mu X(x), & \text{and} \qquad & Y''(y) = -(\lambda-\mu)Y(y), \\ X(0) &= X'(L_x) = 0, & & Y(0) = Y'(L_y) = 0. \end{aligned}$$

These are again standard Sturm-Liouville problems, which we can solve to find that a solution exists only if $\mu = \mu_m = (m-\frac{1}{2})^2\frac{\pi^2}{L_x^2}$ and $\lambda_{m,n} - \mu_m = (n-\frac{1}{2})^2\frac{\pi^2}{L_y^2}$ for $m,n \in \mathbb{N}$, with solutions taking the form

$$X_m(x) = \alpha\sin(\sqrt{\mu_m}x), \qquad Y_n(y) = \beta\sin(\sqrt{\lambda_n - \mu_n}y), \qquad \forall m,n \in \mathbb{N},$$

for arbitrary $\alpha,\beta \in \mathbb{R}$. Thus, non-trivial solutions to the PDE exist only for

$$\lambda_{m,n} := \mu_m + \left(n-\frac{1}{2}\right)^2\frac{\pi^2}{L_y^2} = \pi^2\left[\frac{(m-1/2)^2}{L_x^2} + \frac{(n-1/2)^2}{L_y^2}\right], \qquad \forall m,n \in \mathbb{N}.$$

Note that the functions $\mathbf{u}_{m,n}(x,y) := X_m(x)Y_n(y)$ for $m,n \in \mathbb{N}$ define a basis for the space of functions satisfying the imposed Dirichlet and Neumann boundary conditions. Reintroducing the dependence on time, $T(t)$, we find that solutions to the wave equation take the form

$$\begin{aligned}\mathbf{u}(t,x,y) = &\sum_{m,n=1}^{\infty} a_{m,n}e^{-\kappa t}\sin\left(\sqrt{\lambda_{m,n}}t\right)\sin\left(\sqrt{\mu_m}x\right)\sin\left(\sqrt{\lambda_{m,n}-\mu_m}y\right)\\ &+\sum_{m,n=1}^{\infty} b_{m,n}e^{-\kappa t}\cos\left(\sqrt{\lambda_{m,n}}t\right)\sin\left(\sqrt{\mu_m}x\right)\sin\left(\sqrt{\lambda_{m,n}-\mu_m}y\right),\end{aligned}$$

for coefficients $a_{m,n}, b_{m,n}$ determined by $\mathbf{u}(0,x,y)$ and $\mathbf{u}_t(0,x,y)$. Given these coefficients, the temporal derivative of the solution is then given by

$$\begin{aligned}\mathbf{u}_t(t,x,y) &= -\kappa\mathbf{u}(t) + \sum_{m,n=1}^{\infty} a_{m,n}\sqrt{\lambda_{m,n}}e^{-\kappa t}\cos\left(\sqrt{\lambda_{m,n}}t\right)\sin\left(\sqrt{\mu_m}x\right)\sin\left(\sqrt{\lambda_{m,n}-\mu_m}y\right)\\ &\quad - \sum_{m,n=1}^{\infty} b_{m,n}\sqrt{\lambda_{m,n}}e^{-\kappa t}\sin\left(\sqrt{\lambda_{m,n}}t\right)\sin\left(\sqrt{\mu_m}x\right)\sin\left(\sqrt{\lambda_{m,n}-\mu_m}y\right)\\ &= -\sum_{m,n=1}^{\infty}\left[\kappa a_{m,n} + b_{m,n}\sqrt{\lambda_{m,n}}\right]e^{-\kappa t}\sin\left(\sqrt{\lambda_{m,n}}t\right)\sin\left(\sqrt{\mu_m}x\right)\sin\left(\sqrt{\lambda_{m,n}-\mu_m}y\right)\\ &\quad - \sum_{m,n=1}^{\infty}\left[\kappa b_{m,n} - a_{m,n}\sqrt{\lambda_{m,n}}\right]e^{-\kappa t}\cos\left(\sqrt{\lambda_{m,n}}t\right)\sin\left(\sqrt{\mu_m}x\right)\sin\left(\sqrt{\lambda_{m,n}-\mu_m}y\right).\end{aligned}$$

By orthogonality of the spatial basis functions with respect to the L_2-inner product, it follows that

$$\|\mathbf{u}(t)\|_{L_2}^2 = \frac{L_xL_y}{4}e^{-2\kappa t}\sum_{m,n=1}^{\infty}\left[a_{m,n}\sin\left(\sqrt{\lambda_{m,n}}t\right) + b_{m,n}\cos\left(\sqrt{\lambda_{m,n}}t\right)\right]^2,$$

$$\|\mathbf{u}_t(t)\|_{L_2}^2 = \frac{L_xL_y}{4}e^{-2\kappa t}\sum_{m,n=1}^{\infty}\left[\left(\kappa a_{m,n} + b_{m,n}\sqrt{\lambda_{m,n}}\right)\sin\left(\sqrt{\lambda_{m,n}}t\right) + \left(\kappa b_{m,n} - a_{m,n}\sqrt{\lambda_{m,n}}\right)\cos\left(\sqrt{\lambda_{m,n}}t\right)\right]^2.$$

Given these expressions, we note that solutions do not satisfy $\left\|\begin{bmatrix}\mathbf{u}(t)\\ \mathbf{u}_t(t)\end{bmatrix}\right\|_{L_2} \leq Me^{-kt}\left\|\begin{bmatrix}\mathbf{u}(0)\\ \mathbf{u}_t(0)\end{bmatrix}\right\|_{L_2}$ for any $M \geq 1$ and $k \geq 0$. Indeed, to illustrate, consider $L_x = L_y = 1$, with the initial condition $\mathbf{u}(0,x,y) =$

$\sin((j-\frac{1}{2})\pi x)\sin((j-\frac{1}{2})\pi y)$ for some $j \in \mathbb{N}$. Then $a_{m,n} = 0$ for all $m,n \in \mathbb{N}$, $b_{j,j} = 1$, and $b_{m,n} = 0$ for all other $m,n \in \mathbb{N}$. It follows that, the norm of the solution at each time t is given by

$$\|\mathbf{u}(t)\|_{L_2}^2 = \frac{L_x L_y}{4} e^{-2\kappa t} \cos\left(\sqrt{2}(j-1/2)\pi t\right),$$
$$\|\mathbf{u}_t(t)\|_{L_2}^2 = \frac{L_x L_y}{4} e^{-2\kappa t} \left[\sqrt{2}(j-1/2)\pi \sin\left(\sqrt{2}(j-1/2)\pi t\right) + \kappa \cos\left(\sqrt{2}(j-1/2)\pi t\right)\right]^2.$$

Therefore, at $t = 0$, we have

$$\left\| \begin{bmatrix} \mathbf{u}(0) \\ \mathbf{u}_t(0) \end{bmatrix} \right\|_{L_2}^2 = \|\mathbf{u}(0)\|_{L_2}^2 + \|\mathbf{u}_t(0)\|_{L_2}^2 = \frac{L_x L_y}{4} + \frac{L_x L_y}{4}\kappa^2 = \frac{L_x L_y}{4}(1+\kappa^2),$$

whereas at $t = T_j := \frac{1}{2\sqrt{2}(j-1/2)}$, we have

$$\left\| \begin{bmatrix} \mathbf{u}(T_j) \\ \mathbf{u}_t(T_j) \end{bmatrix} \right\|_{L_2}^2 = \left\|\mathbf{u}(T_j)\right\|_{L_2}^2 + \left\|\mathbf{u}_t(T_j)\right\|_{L_2}^2 = \frac{L_x L_y}{4}\sqrt{2}(j-1/2)\pi.$$

It follows that, for any $M \geq 1$, there exists $j \in \mathbb{N}$ such that $\left\| \begin{bmatrix} \mathbf{u}(T_j) \\ \mathbf{u}_t(T_j) \end{bmatrix} \right\|_{L_2}^2 > M \left\| \begin{bmatrix} \mathbf{u}(0) \\ \mathbf{u}_t(0) \end{bmatrix} \right\|_{L_2}^2$, and thus the wave equation cannot be exponentially stable in the classical sense. However, we can prove that the PDE is exponentially PIE to PDE stable, in the sense of Defn. 8. In particular, note that

$$\mathbf{u}_{xxyy}(0,x,y) = \sum_{m,n=1}^{\infty} b_{m,n}\mu_m(\lambda_{m,n}-\mu_m)\sin\left(\sqrt{\mu_m}x\right)\sin\left(\sqrt{\lambda_{m,n}-\mu_m}y\right),$$
$$\mathbf{u}_{txxyy}(0,x,y) = -\sum_{m,n=1}^{\infty}\left[\kappa b_{m,n} - a_{m,n}\sqrt{\lambda_{m,n}}\right]\mu_m(\lambda_{m,n}-\mu_m)\sin\left(\sqrt{\mu_m}x\right)\sin\left(\sqrt{\lambda_{m,n}-\mu_m}y\right).$$

By orthogonality of the spatial basis functions, it follows then that

$$\left\| \begin{bmatrix} \mathbf{u}_{xxyy}(0) \\ \mathbf{u}_{txxyy}(0) \end{bmatrix} \right\|_{L_2}^2 = \frac{L_x L_y}{4} \sum_{m,n=1}^{\infty}\left[b_{m,n}^2 + \left(\kappa b_{m,n} - a_{m,n}\sqrt{\lambda_{m,n}}\right)^2\right]\mu_m^2(\lambda_{m,n}-\mu_m)^2.$$

Thus, we can bound the norm of the solution to the wave equation at each time as

$$\begin{aligned}
\left\| \begin{bmatrix} \mathbf{u}(t) \\ \mathbf{u}_t(t) \end{bmatrix} \right\|_{L_2}^2 &= \frac{L_x L_y}{4} e^{-2\kappa t} \sum_{m,n=1}^{\infty}\left[a_{m,n}\sin\left(\sqrt{\lambda_{m,n}}t\right) + b_{m,n}\cos\left(\sqrt{\lambda_{m,n}}t\right)\right]^2 \\
&\quad + \frac{L_x L_y}{4} e^{-2\kappa t} \sum_{m,n=1}^{\infty}\left[\left(\kappa a_{m,n} + b_{m,n}\sqrt{\lambda_{m,n}}\right)\sin\left(\sqrt{\lambda_{m,n}}t\right) + \left(\kappa b_{m,n} - a_{m,n}\sqrt{\lambda_{m,n}}\right)\cos\left(\sqrt{\lambda_{m,n}}t\right)\right]^2 \\
&\leq \frac{L_x L_y}{4} e^{-2\kappa t} \sum_{m,n=1}^{\infty}\left[a_{m,n}^2 + b_{m,n}^2 + \left(\kappa a_{m,n} + b_{m,n}\sqrt{\lambda_{m,n}}\right)^2 + \left(\kappa b_{m,n} - a_{m,n}\sqrt{\lambda_{m,n}}\right)^2\right] \\
&\leq e^{-2\kappa t}\left\| \begin{bmatrix} \mathbf{u}_{xxyy}(0) \\ \mathbf{u}_{txxyy}(0) \end{bmatrix} \right\|_{L_2}^2 + \frac{L_x L_y}{4} e^{2\kappa t} \sum_{m,n=1}^{\infty}\left[a_{m,n}^2 + \left(\kappa a_{m,n} + b_{m,n}\sqrt{\lambda_{m,n}}\right)^2\right].
\end{aligned}$$

Here, we note that for any L_x, L_y, we can find a constant C such that $\lambda_{m,n} \leq C\mu_m^2(\lambda_{m,n} - \mu_m)^2$ for all $m,n \in \mathbb{N}$. It follows that

$$\begin{aligned}
\sum_{m,n=1}^{\infty} \left[a_{m,n}^2 + \left(\kappa a_{m,n} + b_{m,n}\sqrt{\lambda_{m,n}} \right)^2 \right] &\leq \sum_{m,n=1}^{\infty} \left[a_{m,n}^2 + \left(\kappa^2 + \lambda_{m,n} \right)\left(a_{m,n}^2 + b_{m,n}^2 \right) \right] \\
&\leq \sum_{m,n=1}^{\infty} \left[a_{m,n}^2 + \lambda_{m,n}\left(\kappa^2 + 1 \right)\left(a_{m,n}^2 + b_{m,n}^2 \right) \right] \\
&\leq \sum_{m,n=1}^{\infty} 2\lambda_{m,n}(1+\kappa^2)\left(a_{m,n}^2 + b_{m,n}^2 \right) \\
&\leq 2C(1+\kappa^2)^2 \sum_{m,n=1}^{\infty} \left[b_{m,n}^2 + \left(\kappa b_{m,n} - a_{m,n}\sqrt{\lambda_{m,n}} \right)^2 \right] \mu_m^2(\lambda_{m,n} - \mu_m)^2 \\
&= \frac{8C(1+\kappa^2)^2}{L_x L_y} \left\| \begin{bmatrix} \mathbf{u}_{xxyy}(0) \\ \mathbf{u}_{txxyy}(0) \end{bmatrix} \right\|_{L_2}^2 .
\end{aligned}$$

Defining, then, $M := \sqrt{1 + 8C(1+\kappa^2)^2}$, it follows that

$$\left\| \begin{bmatrix} \mathbf{u}(t) \\ \mathbf{u}_t(t) \end{bmatrix} \right\|_{L_2} \leq Me^{-2\kappa t} \left\| \begin{bmatrix} \mathbf{u}_{xxyy}(0) \\ \mathbf{u}_{txxyy}(0) \end{bmatrix} \right\|_{L_2} .$$